

Program overview

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Year 2023/2024
Organization Civil Engineering and Geosciences
Education Master Environmental Engineering

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
ENVE Master 2023							
ENVE Year 1 2023							
ENVE Faculty Core Module (MUDE)							
CEGM1000	Modelling, Uncertainty and Data for Engineers	12					
ENVE Programme: Principles of Environmental Engineering							
ENVM1000	Principles of Environmental Engineering	9					
ENVE Track Atmospheric Environment Engineering (AEE)							
ENVM1700	Fundamentals of AEE	15					
ENVM1800	Atmospheric Measurements and Modelling	9					
ENVM1900	Grand Challenges in AEE	15					
ENVE Track Water Resources Engineering (WRE)							
ENVM1401	Quality, Quantity and Design, Information and Advice	15					
ENVM1500	Water Quality and Principles of Environmental Engineering	9					
ENVM1501	Design and Modeling of Urban Water Infrastructure Systems	9					
ENVM1502	River Basin Hydrology and Water Management	9					
ENVM1600	Water Treatment Technologies	15					
ENVM1601	Operation, Control, Management and Adoption of Urban Water Infrastructure Systems	15					
ENVM1602	Regional Hydrology	15					
ENVM1603	Water Resources Engineering and Management	15					
ENVE Track Resource & Waste Engineering (R&WE)							
ENVM1100	Fundamentals of RW and E	15					
ENVM1200	Resource Engineering	9					
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ENVM1301	Reactive Resources and Wastes	15					
ENVE Year 2 2023							
ENVE Joint Interdisciplinary Project							
TUD4040	Joint Interdisciplinary Project	15					
ENVE Multidisciplinary Project							
CEGM3000	Multidisciplinary Project (MDP)	15					
ENVE Internship							
CIEM0120	Research Internship	10					
ENVM0121	Extension Research Internship	5					
ENVM3000	Internship Environmental Engineering	15					
ENVE Study abroad							
ENVE C-modules							
ENVM2100	Industry Water	5					
ENVM2102	Water and Health	5					
ENVM2104	Design Nature Based Solutions in Water Management Using Aquatic Ecology and Morphodynamic Principles	5					
ENVM2105	Water Law & Organisation	5					
ENVM2106	Engineering and Development	5					
ENVE D-Modules (Cross-overs)							
CEGM2000	Suspensions, Sludges and Soils	10					
CEGM2001	Sustainable Cities	10					
CEGM2002	Engineering for Global Development	10					
CEGM2003	Data Science and Artificial Intelligence for Engineers	10					
CEGM2004	Noise and Vibration: Generation, Propagation and Effect on Humans and Environment	10					
CEGM2005	Probabilistic Modelling of real-world phenomena through ObseRvations and Elicitation (MORE)	10					
CEGM2006	Subsurface Storage for Energy, Water and Climate Applications	10					
CEGM2007	Resilient Deltas	10					
CEGM2008	Monitoring of Structural Health and Geohazards	10					
ENVE Master Thesis Preparation and Portfolio							
ENVM4000	Master Thesis Preparation and Portfolio Environmental Engineering	5					
ENVE Master Thesis							
ENVM4010	Master Thesis Environmental Engineering	30					

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ENVE Faculty Core Module (MUDE)
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CEGM1000	Modelling, Uncertainty and Data for Engineers	12
Responsible Instructor	Dr.ir. R.C. Lanzafame	
Module Manager	Dr.ir. R.C. Lanzafame	
Instructor	Dr.ir. C.C.J.M. Tiberius	
Instructor	Dr.ir. A.A. Verhagen	
Instructor	Dr. A. Amiri Simkoei	
Instructor	F.P. van der Meer	
Instructor	D. Mehta	
Instructor	Dr.ir. G. Homem de Almeida Correia	
Instructor	Dr.ir. I. Barcelos Carneiro M Da R	
Instructor	Dr. R. Taormina	
Instructor	A. Jagadeesh	
Instructor	Dr. P.M.N. Mares Nasarre	
Instructor	T.R. van Woudenberg	
Instructor	J.C. Pittman	
Contact Hours / Week x/x/x/x	4HC, 4W, 4PJ	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Modelling, Uncertainty and Data for Engineers (MUDE) is a module shared by students in all three programs within the Civil Engineering and Geosciences Faculty (Civil Engineering, Environmental Engineering, Applied Earth Sciences). In a typical week you will be introduced to new theory, practice it on your own through a series of exercises and work together with teachers and classmates in a medium-sized classroom setting. At the end of each week you will apply these concepts in a project along with a group of fellow students. You will also have weekly coding assignments to build your programming skills in a way that will help you apply the methods learned in MUDE using real data sets and models, as well as prepare you for your future MSC thesis work.	
Study Goals	After successfully completing the MUDE module a student is able to 1. Apply a diverse set of models and analysis techniques to solve engineering problems through a combination of mathematical formulation and implementation in computer code. 2. Process and analyze data from different sources and formats to gain insight on the underlying systems and incorporate in modeling approaches. 3. Mathematically formulate, solve and discuss estimation, risk, reliability and optimization problems for systems to support decision making. 4. Analyze and reflect on the methodology, performance, and results of modeling approaches and analysis techniques, accounting for uncertainty. 5. Implement and carry out analysis using Python programming language following specific coding standards and best practices while facilitating collaboration. 7. Communicate and interpret results of analyses with appropriate documentation, metrics and visualization techniques.	
Education Method	The weekly format is: - Mon: lecture - Tue: open question hours (attendance optional) - Wed: classroom workshops (teacher-led activities) - Fri: supervised group projects (bi-weekly assignments) Practice material is provided in the form of a digital textbook with analytic and programming exercises and is the primary focus of the Mon-Wed activities. These concepts are applied on Friday during the project sessions. Weekly coding assignments build fundamental programming skills that are also applied in the Friday project sessions.	
Assessment	- Exam at end of Q1 (resit in Q2) (25%) - Exam at end of Q2 (resit in Q3) (25%) - Project portfolio (50%) The exams are paper-based. Project portfolio includes 7 bi-weekly projects (80%) as well as weekly coding assignments (20%). You will pass the module if you obtained at least a 5.8 for the three individual components. There will be a resit opportunity for the exams. If the grade for your project portfolio is below 5.8, you will get the opportunity to submit a revised version for which you will get a 6.0 if judged sufficient.	
Expected prior Knowledge	University units, or equivalent, in Calculus, Linear Algebra, Probability & Statistics, Numerical Analysis, Python programming	
Academic Skills	Analysis, coding, communication, visualization, team skills	
Literature & Study Materials	An online digital textbook consisting of text, practice material and videos, as well as related materials (e.g., Jupyter Notebooks).	
Judgement	- Exam at end of Q1 (resit in Q2) (25%) - Exam at end of Q2 (resit in Q3) (25%) - Project portfolio (50%) The exams are paper-based. Project portfolio includes 7 bi-weekly projects (80%) as well as periodic coding assignments (20%). You will pass the module if you obtained at least a 5.8 for the three individual components. There will be a resit opportunity for the exams. If the grade for your project portfolio is below 5.8, you will get the opportunity to submit a revised version for which you will get a 6.0 if judged sufficient.	
Permitted Materials during Exam	Calculator, pen/pencil. All formulas will be provided by the instructors.	
Colleggerama	Yes No	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Environmental Engineering

ENVE Programme: Principles of Environmental Engineering

ENVM1000	Principles of Environmental Engineering	9
Responsible Instructor	Dr.ir. M.W. Ertsen	
Instructor	Dr. E. Mostert	
Instructor	Dr. B.M. van Breukelen	
Instructor	Dr.ir. B.J.H. van de Wiel	
Instructor	Dr. J. Gebert	
Instructor	A.R. Gammon	
Contact Hours / Week x/x/x/x	14/x/x/x 8(seminar)/4 (practicum)/2 (project)/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Today's societies are facing multiple environmental changes of natural and anthropogenic origin that can both impair human and environmental health. The main priority of environmental engineers is to safeguard environmental quality, human livelihood, health, well-being, and safety. To design effective engineering and management solutions, environmental engineers need to understand both physico(bio)chemical processes in soil, water, and air as well as the societal processes that affect environmental legislation and courses of action. These two aspects are covered within the themes Environmental Quality, Cycles and Fluxes and Environment and Society. Environmental Quality, Cycles, and Fluxes Link to learning objectives of the module: ENV-PC: LO1, LO2, LO3, LO4, LO5 and LO8ab. Regarding physico(bio)chemistry, this module provides the student with quantitative insights into the interlinkages between the soil, water, and air compartments, specifically the processes distributing water (the hydrological cycle) and elements (for example C, N, P, S cycles). The module will also discuss how human activity influences these cycles and impairs soil, water, and air quality. It is explained how various environmental quality standards protect environmental and human health. Environment and Society Link to learning objectives of the module: ENV-PC: LO1, LO4, LO5, LO6, LO7 and LO8ab. Several different Environmental Engineering problems will be selected to show the variety of environmental engineering problems and how they could be analysed and solved. Problem ownership and responsibility will be discussed together with distribution of costs and benefits and ethical dilemmas. Students learn about the role of law and politics at different levels. With the gained understanding and skills, the student is able to identify integrated strategies to practically prevent or resolve environmental engineering problems. Other The students will start compiling their portfolio that will be assessed in Q6. An introduction will be given on the portfolio compilation, life-long learning skills and the ethics learning line.	
Study Goals	After completion of this module the students are able to: ENV-PC-LO1: Understand the natural cycles of water, carbon, nitrogen, phosphorus, and sulphur, and how human activities interfere and may cause environmental problems. (ILO: a) ENV-PC-LO2: Describe the key water, air, and soil quality parameters and their interrelations, explain possible causes of their variations, how these variations relate to natural processes and human interventions, and impact human health and the environment. (ILO: a) ENV-PC-LO3: Understand conceptually how water, solutes and gases are transported in the environment, how the respective fluxes can be quantified, and how contaminants affect the quality of water, soil, and air. (ILO: a) ENV-PC-LO4: Explain how to collect water, air, and soil samples; analyse those on relevant parameters; and provide a basic Interpretation of these data. (ILO: e) ENV-PC-LO5: Assess the role of stakeholders (individuals, private sector, government bodies) and the contribution of environmental standards, wider legislation, and politics in solving environmental problems. (ILO: h) ENV-PC-LO6: Identify and analyse ethical dilemmas concerning environmental problems and solution strategies. (ILO: l) ENV-PC-LO7: Design a draft strategy for solving environmental problems that integrates both understanding of the technical aspects of the problem and of the legal, political, and ethical constraints and possibilities. (ILO: a, l) EE-PC-LO8a: Perform effectively in a team of environmental engineering students that explore and analyze environmental engineering problems (ILO: k) EE-PC-LO8b: Prepare effective communication products to share the results of an environmental engineering project (ILO: i)	
Education Method	The efforts of students will strongly focus on the integration learning goals. Students teams work on real-life issues in student teams. In the teams, students apply aspects that are offered through a series of separate sessions on states/fluxes, environment&society and measurements to their project setting. This includes working on questions including: For whom is what the problem? What are related fluxes/stocks? Who is responsible for which part of the process? What data are used to act on this responsibility? How can these flows be measured? What are potential solutions? The group prepares a set of documents focusing on a feasible strategy for the given setting: a description of the origin, pathways, destinations and effects of the substance concerned, an identification of the stakeholders concerned, and selected actions that the group proposes plus a justification why to take these actions. In the last few weeks, students elaborate on their individual additions to the group work. Students start compiling their portfolio with information on ethics and lifelong learning as well as proof for fulfilling their ILOs. More information on the portfolio can be found in the Thesis preparation & Portfolio module.	
Assessment	During the sessions and project meetings, several formative formats will be applied: Quizzes on Brightspace and interactive discussions in class Discussion Forum on Brightspace	

	Short exercises with immediate feedback during class Peer feedback on draft chapters and reports on field and lab practicals Feedback from teachers on presentations and texts before handing in final versions. Practice exam and Q&A before the exam session Interviews on the student portfolio Students complete the module successfully when passing the 2 summative (individual) assessments: Written Exam fluxes/states (including measurements) = 40% Project (integrating fluxes/states and env/soc = 60% A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.
Expected prior Knowledge	NA
Academic Skills	Communication and responsiveness, scientific writing and presentation, time management and anticipation, application of theory to new situations academic reading, writing, referencing, information collection and evaluation, inquiry thinking, critical thinking, ethical reflection, and teamwork.
Literature & Study Materials	Short manual with links to study material prepared by the teaching teams, scientific papers, chapters/sections in open access e-books, and links to high quality information on internet.
Judgement	See Assessment
Permitted Materials during Exam	TBD
Collegerama	No

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Organization	Civil Engineering and Geosciences
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ENVE Track Atmospheric Environment Engineering (AEE)

ENVM1700	Fundamentals of AEE	15
Responsible Instructor	Dr.ir. B.J.H. van de Wiel	
Instructor	Prof.dr. A. Metrikine	
Instructor	Dr.ir. O.A.C. Hoes	
Instructor	Dr. G. Biskos	
Instructor	A. Tsouvalas	
Instructor	Dr.ir. B.J.H. van de Wiel	
Instructor	S. Basu	
Instructor	Dr. A. Cabboi	
Instructor	Dr. A. Cicirello	
Contact Hours / Week x/x/x/x	Urban climate physics: Q2: W1-7, HC(4), PT(4); W8, S(1) Atmospheric chemistry: Q2: W1-8, 4 HC (2x2) Air pollution meteorology & dispersion: Q3: W1-8, 4 HC (2x2) Physics of sound and human perception: Q3: W1-8, 4 HC (2x2)	
Education Period	2 3	
Start Education	2	
Exam Period	2 3 4	
Course Language	English	
Course Contents	This module is aimed to teach students the fundamentals of physics and chemistry in the urban atmospheric environment. They will learn to appreciate the inherent complexity of the urban environment, where due to complex building geometry the atmosphere is highly non-uniform and transient in space and time. By performing order-of-magnitude (scaling) analysis of various physical and chemical processes, the students will learn to simplify complexity into a limited number of key-processes and to describe the evolution of relevant physical/chemical quantities using the (simplified) governing equations. On an introductory level, the student will learn how those quantities can be probed by simple measurement principles. They will also learn basic tools of numerical modelling/conceptualization to analyze the impact of novel engineering designs on urban atmospheric environments. While both measurement and modelling techniques are introduced on a basic level here as to understand the key processes, an in-depth treatise of these techniques are introduced in the next module. Urban Climate Physics Link to learning objectives of the module: SS: LO1, LO2, LO3, LO4, LO5, LO6 and LO8. The module Fundamentals requires a basic introduction to urban microclimate. Here we introduce a range of physical concepts that are utilized in the remainder of the track modules. Apart from theory, some simple experiments are performed as an introduction to more rigorous explanation of experimental principles in later modules. The central theme of Urban Climate Physics is the scientific analysis of the urban climate. We aim at understanding the dominant physical processes that occur in urban environments as well as to understand the interactions between a city and the overlying atmosphere and how it impacts our living environment. Special emphasis is given to quantification of thermodynamic processes, such as radiative, convective and conductive heat transport and to airflow in complex geometries. The theoretical part connects with short practical's which illustrates the various physical principles through simple hands-on experiments. These practical's facilitate students to develop scientific intuition for the processes at hand and prepare them for the rest of the module (Modelling of Urban Flows, Urban & Indoor Air Quality, and Urban Extreme Weather). The following topics will be addressed: Introduction and basic concepts; Research Methodologies; Transport phenomena and urban air flows (part I); Transport phenomena and urban air flows (part II); Radiative processes; The urban energy balance (including moisture effects); Human thermal comfort in cities. Atmospheric chemistry Link to learning objectives of the module: SS: LO1, LO2, LO3, LO4, LO5, LO6 and LO8. In Atmospheric chemistry a range of fundamental concepts for understanding the key chemical processes that take place in the atmospheric environment will be introduced. These concepts will allow the students to identify, analyze, and explain certain atmospheric phenomena including physicochemical reactions, that will prove to be highly useful for the rest of the module. Apart from theory, simple experiments will be included during which the students will be able to get hands-on experience. The following topics will be addressed: Background, composition, and structure of the atmosphere Concepts of atmospheric chemistry (thermodynamics, kinetics, lifetime, photochemistry, etc); From stratospheric to tropospheric chemistry; Chemistry in urban and rural environments; Basic properties of atmospheric aerosol particles; Processes involving atmospheric aerosols*; Impacts of air pollution on human health, the environment and climate. Course Contents Continuation Air Pollution Meteorology & Dispersion Link to learning objectives of the module: SS: LO1, LO2, LO3, LO5 and LO6. In Air Pollution Meteorology & Dispersion the fundamental concepts related to turbulent diffusion and dispersion are introduced. The more advanced Urban & Indoor Air Quality, Integrated AEE Project will leverage on these concepts. Air Pollution Meteorology & Dispersion provides an in-depth understanding of how various atmospheric processes impact air pollutant transport, especially in urban environments. First, a number of fundamental concepts related to diffusion and dispersion (e.g., turbulence mixing, plume rise, deposition) are introduced. Then, different types of pollutant sources (e.g., point, line, area)	

and associated mathematical equations for diffusion are derived from the first principles. More advanced approaches, based on statistical and similarity theories, are discussed next. As complementary to the theoretical discussions, a state-of-the-art dispersion modelling software is utilized for further understanding of real-life pollutant transport problems. Lastly, accidental release of hazardous materials in mega-cities and related emergency response management will be briefly touched upon.

The following topics will be addressed:
 Introduction; Gaussian diffusion models;
 Plume rise, settling, and deposition;
 Statistical theories of puff and plume diffusion;
 Similarity theories of turbulence and diffusion;
 Modelling dispersion at city scale;
 Impacts of highways and buildings on urban air quality;
 Accidental releases and emergency response management.

Physics of Sound and Human Perception
 Link to learning objectives of the module:
 SS: LO2, LO3, LO5, LO6 and LO8.

In Physics of Sound and Human Perception the fundamental concepts related to physics of sound and its effect on people are introduced. The more advanced Noise Modelling & Mitigation, Integrated AEE Project will build on these concepts.

Firstly, the physics of sound are introduced with a combination of practical experimentation and mathematical analysis. Basic examples such as open and closed tubes, and moving sources are studied to provide a solid foundation of acoustical understanding in preparation for the rest of the module. Then, the way we hear and respond to sound, the way we quantify sounds, and the relationship between those measures and human responses to the sounds are introduced.

More specifically the following topics will be covered in a number of teaching sessions combining theory and (in-class) experiments related to sound and its perception:

- Derivation of the one-dimensional (1D) wave equation for a fluid-filled pipe based on first principles and explanation of the assumptions necessary to do so;
- Evaluation and analysis of free vibrations of a fluid-filled pipe: the physical phenomena of resonance and standing waves in a fluid-filled pipe;
- Derivation of the three-dimensional wave equation based on first principles and explanation the assumptions necessary to do so;
- Explanation of the physics of the wave equation in application to acoustic problems;
- Explanation of the meaning of common terms in wave mechanics, such as frequency, wavelength, wavenumber, wave speed, diffraction, reflection, dispersion and give examples of how they apply to sound;
- Acoustic sources;
- Speed of sound in a fluid medium and dependence on its physical properties;
- Demonstration (by means of a computer program where appropriate) of standing waves, interference fields, and other wave phenomena;
- Parts of the outer, middle and inner ear and their role in hearing;
- Hearing of an individual;
- Analysis of sound measurements;
- Conversion of narrowband data into third and octave band levels either with linear or A weighting.

Study Goals

After completion of this module the students are able to:

SS-LO1: Describe the overall physical and chemical structure and characteristics of the urban atmosphere and be able to express its difference with the rural surroundings. (ILO: a)

SS-LO2: Simplify the complexity of urban environments and to define the key physical and chemical concepts that describe the urban environment. (ILO: a, f)

SS-LO3: Quantify those physical and chemical concepts through an understanding of the underlying governing equations. (ILO: a, f)

SS-LO4: Perform simple hands-on experiments to measure various physical/chemical aspects of urban environments. (ILO: e)

SS-LO5: Analyze observational data of urban environments and be able to quantify its uncertainty. (ILO: e)

SS-LO6: Perform numerical modelling to quantify the effects of novel engineering designs (e.g., green roofs, parks, sound barrier) on urban environments. (ILO: d)

SS-LO7: Reflect on the societal impact of interventions and be able to discuss ethical aspects of those within the context of atmospheric environment engineering. (ILO: i)

SS-LO8. Describe how atmospheric environment pollution (e.g. related to aerosols, noise pollution) affects human health. (ILO: b)

Education Method

Teaching and learning that encourages students to take responsibility for their learning process:

Urban Climate Physics

Lectures. Prior to lectures corresponding book chapters/handouts should be studied. To engage the students, short mini exercises are given during the lectures. In addition, table-top experimental demonstrations are given during the lectures which require active student participation.

Hands-on experiments on campus. Students perform small practical's to learn basic principles of airflow & mass-conservations, radiative transport, atmospheric moisture, etc.

Article discussion in class. Students are encouraged to provide related articles from the news or from popular scientific reading, which then are discussed in class.

Instructions and theoretical exercises. On campus instructions are given, as to practice exercises. Feedback is provided by the teachers and teaching assistant during the instructions. Students are encouraged to work in small groups to stimulate discussions. To facilitate self-study, apart from direct feedback, written answers are also provided.

Prior to the exam a question/answer session is organized to better prepare for the exam.

Atmospheric chemistry

Form(s) of instruction, individual and/or group work: lectures and group work where they will carry out specific exercises and in which all the students should participate. Groups will contain 2-3 members.

Required preparation and required effort: The students are expected (i) to read background material in the form of key papers related to the topics covered each week and present the material in groups; (ii) carry out basic calculations related to the context of the lectures, and work through these with the class each week.

On campus and online activities: Online lectures and group interaction will apply only if restrictions are in place.

Air Pollution Meteorology & Dispersion

In-class lectures will be the primary mode of instruction. In one of the lectures, the students will be given a hands-on introduction to a dispersion modeling software. In addition, throughout the course, the students will be required to read relevant journal articles and critique certain aspects of the papers. They will form small discussion groups during lecture hours and will brainstorm to come up with a different scientific approach/solution.

Assessment	Physics of Sound and Human Perception In-class lectures will be the primary mode of instruction. Students will be guided through the theory combined with in-class experiments for demonstration purposes of the basic concepts of acoustics. Urban Climate Physics Exercises to actively understand and apply lecture material - are made under supervision of the instructors (this may include usage of a simple, interactive computer-program to visualize the result; no specific computer skills are required). All answers to exercises are provided after completion. Feedback to the experimental work by the students is given both on performance and on reporting. Atmospheric chemistry Feedback from presentation and group calculation work; formative multiple-choice questionnaire and short exercises provide understanding of distinct lectures. Air Pollution Meteorology & Dispersion Every week, there will be a muddiest point activity in class. The students will be asked to create a list of the unclear or confusing concepts from the previous lecture and ongoing assignment. Such written feedback from the students will be fully anonymous and the lecturer will address them during the following class. The lecturer might re-teach some of these difficult concepts with different illustrations in order to improve student learning. There will be three individual assignments which will empower the students with the necessary understanding and skill sets to execute a class project. The assignments will be comprised of theoretical problem solving, data analysis (e.g., analysis of air quality index), and small-scale model development (e.g., Gaussian plume model, mixed layer model). The students will receive formative feedback from the instructor. In addition, there will be a weekly in-class quiz (non-graded) activity. This activity will be interactive in nature and will make use of an online polling website or a cell phone app. Physics of Sound and Human Perception During the teaching period, a number of small assignments/exercises will be distributed as practicing material so that students get familiar with the theory of sound and its perception by humans. Formative feedback will be provided to students who decide to deliver assignments by both the instructor and the peers. Students complete the module successfully when passing the following 4 summative assessments: Urban climate physics: final written exam with closed and open-ended questions (33%) Atmospheric chemistry: written exam (27%) Air pollution, Meteorology and dispersion: individual assignment report (20%) Physics of Sound and Human Perception: written exam (20%) A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.
Expected prior Knowledge	The pre-requisites are inline with the admission criteria. The students should have prior knowledge of mathematics (calculus, linear algebra, differential equations, probability and statistics, numerical methods), fluid mechanics, and python programming language.
Connection to other modules	This foundational module will enable the students to learn more advanced materials in subsequent modules.
Academic Skills	Communication and responsiveness, scientific writing and presentation, time management and anticipation, application of theory to new situations using the relevant equations, academic reading, writing, referencing, information collection and evaluation, inquiry & critical thinking and teamwork.
Literature & Study Materials	Urban Climate Physics The book: Urban Climates by Oke, Mills, Christen and Voegt. Cambridge University Press, Un. Printing House, Cambridge CB2 8BS, U.K. Additional background material and/or papers will be provided via BrightSpace. Experimental material is provided in the practical's. Atmospheric chemistry One main textbook: Atmospheric Chemistry and Physics: From Air Pollution to Climate Change, J. H. Seinfeld and S. N. Pandis* Background material, papers and chapters from relevant literature will be provided via BrightSpace. *Ask Prof. Spyros Pandis to come as a guest lecturer to introduce PMCAMx; one of the main algorithms for predicting regional air pollution Air Pollution Meteorology & Dispersion Textbook: de Visscher, A. (2014), Air Dispersion Modeling: Foundations and Applications, John Wiley & Sons, Inc., 634 pp. Reference book: Arya, S. P. (1999), Air Pollution Meteorology and Dispersion, Oxford University Press, 310 pp. On a regular basis, the students will be provided with various journal articles related to the course topics. It is anticipated that they will peruse the articles before class and participate in class discussion. Physics of Sound and Human Perception Background material, papers and chapters from relevant literature will be provided via Brightspace. Material will be borrowed by the following books: https://www.taylorfrancis.com/books/fundamentals-sound-vibration-frank-fahy-david-thompson/e/10.1201/b18348 https://www.springer.com/gp/book/9783540240525 https://www.springer.com/gp/book/9783030447861 https://www.cambridge.org/core/books/fundamentals-of-noise-and-vibration-analysis-for-engineers/8768D986559E0B54ECDDF0A167C9CDF1 https://www.springer.com/gp/book/9783662053911
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	During open book exams, the students are permitted to bring any written materials.
Collegerama	Yes

ENVM1800	Atmospheric Measurements and Modelling	9
Responsible Instructor	Dr. S.R. de Roode	
Instructor	C.M.H. Unal	
Instructor	Dr.ir. J.A.E. ten Veldhuis	
Instructor	Dr. G. Biskos	
Instructor	Dr. S.R. de Roode	
Instructor	A. Tsouvalas	
Instructor	Dr. M.A. Schleiss	
Instructor	C. Garcia Sanchez	
Instructor	Prof.dr.ir. R. Uijlenhoet	
Contact Hours / Week x/x/x/x	6 (HC)/4 (PC)/2 (PT)/10 (PJ) /x/x/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This module provides in-depth educational training on the principles of measurement and modelling techniques with a focus on urban environments. Through various activities, the students will gain significant hands-on experience in conducting field experiments and performing numerical simulations. In collaboration with ethics experts from the TPM faculty, the instructors of the AEE track will organize a half-day long workshop on ethics. The theme of the workshop will be on ethical issues related to Open Science. The students will actively participate and prepare a reflection document as part of a portfolio. During Q6, this portfolio will be evaluated by ethics experts. Atmospheric Measurement Techniques for Urban Environments This module theme covers the fundamentals of observational techniques for atmospheric processes with an emphasis on urban environments and high spatial/temporal resolutions. The following topics will be addressed: Importance of collecting data Difficulties encountered in urban environments General overview of different types of measurement techniques including Doppler and polarimetry Basic concepts: wavelength, scattering, absorption, propagation Basic definitions and concepts: scale, coverage, temporal/spatial resolution, accuracy, precision etc. Rain gauges, Radar, Turbulent flux measurements of momentum, heat and moisture and chemical species, Lidar, Sodar, Radiometers, Accuracy, precision, calibration Noise measurements in the field, analysis of measured data and filtering Quality control, noise, bias, uncertainties and errors. Missing values Sensor placement, good practices etc. Citizen weather stations Microwave links, smartphones. User engagement and awareness. Modelling of Urban Flows Content module theme: The module theme focuses on the fundamentals of Computational Fluid Dynamics (CFD) tailored for urban environments. The course will introduce the governing equations first and elaborate on the mathematical methods applied to model the flow around buildings. Real world examples will be used to show direct application of CFD during computer labs, and students will have hands-on practice to learn and use CFD in urban areas. The following topics will be addressed: Governing equations Reynolds decomposition and averaging Discretization of governing equations (finite-difference, finite-volume, finite-element approaches) Time-marching methods Stability, accuracy, convergence Boundary conditions for urban environments Mesh and complex geometries generation Turbulence modelling (RANS/LES/DNS) Introduction to uncertainty quantification	
Study Goals	Measurements: LO2, LO3, LO4, LO6 and LO7. Modeling: LO1, LO2, LO3, LO4, LO5, LO6 and LO7.	
Education Method	Teaching and learning that encourages students to take responsibility for their learning process: Atmospheric Measurement Techniques for Urban Environments Lectures + in-class activities + group project List of topics for projects: Topic 1: noise measurements on campus. Topic 2: measurement of temperature, wind and rainfall on campus using Davis and Netatmo weather stations Topic 3: radiometer and pyranometer data analysis Topic 4: heat stress and urban heat island analysis Topic 5: rainfall estimations using radar Topic 6: sodar-based wind data analysis Topic 7: indoor/outdoor air quality measurements Topic 8: citizen weather data analysis and quality assessment Topic 9: rainfall estimations using microwave links Topic 10: wind estimations using radar Modelling of Urban Flows This theme includes lectures and computer labs in which students work in groups as well as individually. During lectures active learning activities are included to keep all the students on the same page, and enhance participation. Some examples include:	

	flipped classroom, where materials would be provided for the students to read before the class (for example when covering CFD domain size design we will use published literature); in-class, online quizzes after covering theoretical concepts; students will be asked to associate physical processes with various terms of the fluid dynamics equations; 5 minutes notes at the end of the theoretical lectures to collect the most difficult concepts of the session. During computer labs, students will perform CFD simulations. Several topics will be provided with solutions to train for the final group project. During computer labs, they will work on their own simulations and the teacher will be there for troubleshooting. Some potential modelling topics are: - wind flows around high-rise buildings; - dispersion of passive scalars; - wind flows around trees; - wind flows over heterogeneous roughness areas. On campus and online activities Classes will take place online if on campus is not possible
Assessment	Atmospheric Measurement Techniques for Urban Environments All in-class activities (including the practical's and homework) will be formative. Students will receive peer-feedback during the in-class activities and formative written feedback on the homework from the TA's. Modelling of Urban Flows Formative feedback during in-class activities, with interaction between peers. Formative assignments will be proposed to the students during laboratories, with feedback provided within 2 weeks. During the first week, their peers will have the opportunity to provide feedback. During the second week, the teacher will provide feedback. Students complete the module successfully when passing the 3 summative assessments: - Atmospheric Measurement Techniques for Urban Environments: written exam (30%), group project (30%) - Modelling of Urban Flows: group assignment (report (35%) and presentation (5%)). Groups will be composed by 2-3 students (depending on the group size and computational resources). We will use buddycheck to evaluate the group participants contribution. A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.
Connection to other modules	The AEE track will start with only one module A.
Academic Skills	Problem-based learning and solving, making inferences of physical situations by analyzing observational and modelled data, academic reading, scientific writing, referencing, inquiry thinking, critical thinking and teamwork.
Literature & Study Materials	Atmospheric Measurement Techniques for Urban Environments Introduction to Meteorological Instrumentation and Measurement, by Thomas P. deFelice. Peer-reviewed articles and review papers will be provided through Brightspace. Modelling of Urban Flows Slides will be provided for the students before every lecture. Additional materials as papers or content not included in the reference book will be provided for the students. Most lectures will be based in the reference book mentioned hereafter. Reference book: Computational Fluid Dynamics: A practical approach, Jiyuan Tu, Guan-Heng Yeoh, Chaoqun Liu. Additional cluster resources will be necessary to run CFD simulations (Delft cluster/INSY)
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations

ENVM1900	Grand Challenges in AEE	15
Responsible Instructor	Dr.ir. B.J.H. van de Wiel	
Instructor	C.M.H. Unal	
Instructor	Dr.ir. O.A.C. Hoes	
Instructor	Dr.ir. J.A.E. ten Veldhuis	
Instructor	Dr. G. Biskos	
Instructor	Dr. S.R. de Roode	
Instructor	Dr. A.A. Nunez Vicencio	
Instructor	A. Tsouvalas	
Instructor	Dr. K. Anupam	
Instructor	Dr. Z. Yang	
Instructor	Dr. Dr. H.O. Sertlek	
Instructor	Dr. M.A. Schleiss	
Instructor	C. Garcia Sanchez	
Instructor	Dr. A. Cicirello	
Contact Hours / Week x/x/x/x	14 (HC)/ 4 (PT)/ x/x/x/x	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The previous modules focused on understanding, conceptualization and technical aspects of Atmospheric Environment Engineering (AEE). This module takes a step beyond. Students first learn to identify and analyze grand challenges within AEE. Next, students learn how to create and evaluate engineering solutions for such challenges of contemporary and future societies. Rather than working on an isolated technical aspect, they learn how to integrate knowledge of different disciplines and learn to integrate different tools to tackle the problem at hand. To monitor and manage a research project and to work in an interdisciplinary team. To communicate their scientific and engineering solutions and to reflect on the ethical implications of their proposed solutions to the grand challenges. In collaboration with ethics experts from the TPM faculty, we will organize a half-day long workshop on ethics. The theme of the workshop will be on ethical issues related to Public Health. The students will actively participate and prepare a reflection document as part of a portfolio. During Q6, this portfolio will be evaluated by ethics experts. Urban and Indoor Air Quality LO1, LO3, LO4, LO7 and LO9. This course provides an in-depth understanding of how various sources of pollution can affect air quality in the urban outdoor and indoor environment. The course will start by covering the different types of air pollutants (gaseous pollutants, particulate matter, and bioaerosols), their sources, as well as the processes they may undergo to lead to secondary pollutants. Links with the associated health effects will be made. We will introduce simple observational and computational techniques for assessing and predicting the evolutions of air quality in the outdoor environment. We will draw upon the knowledge of module A in order to demonstrate how the concentration of different air pollutants can be monitored in the outdoor and indoor environment. Simple computational models will be introduced for the case of the indoor environment, showing the students how different ventilation approaches and other mitigation measures (humidification/purification) can improve air quality in buildings and/or vehicles. Topics of special interest on how different air pollutants, including particles containing viruses, can disperse/transport in the indoor environment will also be covered (guest lecture). Noise Modelling and Mitigation LO1, LO2, LO3, LO4, LO6 and LO9. We first introduce computational methods available to solve practical problems in acoustics and compared them with benchmark analytical solutions. Potential and limitations of analytical and numerical approaches are discussed. The module units introduces available methods but also stimulate to think beyond for those seeking to apply or extend them further. The second part of the module theme deals with noise control engineering, focusing primarily on noise mitigation and noise trapping, to reduce noise in practical situations. The noise control engineer needs to know how to set targets, how to characterize and quantify noise sources, and how to reduce noise either at source or in the transmission path. Suitable formulae are provided and explained for this. The main assessment consists of a design calculation/field study which makes use of the above to solve a practical problem	
Course Contents Continuation	Extreme Urban Weather LO1, LO2, LO3, LO4, LO6 and LO9. The first part focuses on rainfall extremes. Students learn how environmental factors such as temperature and relative humidity influence the spatial/temporal distribution of rainfall. They learn how to use radar and numerical weather prediction models to predict rainfall over urban scales. This enables them to identify dangerous rainfall patterns, assess risks and discuss the effectiveness of mitigation strategies. The second half of the module theme focuses on modelling the urban microclimate on the scale level of the street. This enables students to indicate problematic areas in relation to heat stress, as well as propose and assess the effectiveness of mitigation strategies. The following topics are addressed: Scaling of precipitation with temperature. Extreme precipitation in numerical weather prediction models; Nowcasting using radar Droughts & dry spell analysis; Influence of urban environment on heat stress and mitigation measures; Modelling urban heat using ENVI-met; Presentations from experts in the field of climate adaptation. Two assignments will run in parallel: a re-analysis of a heavy rain and a heat stress event (including analysis of potential mitigation) Integrated AEE Project LO1, LO2, LO3, LO4, LO5, LO6, LO7, LO8 and LO9. This module theme aims to integrate knowledge acquired during previous modules. Here we go beyond theoretical	

understanding and application of techniques by analyzing a complex research question as a whole. The student will break down the complex problem in sub-parts and will identify which methods/techniques are most appropriate to answer the research question. New ideas will be developed and new insights will be generated by combining or improving knowledge from different sources, which may include novel data analysis, modeling or theoretical analysis. The results are placed within a societal context and environmental and/or ethical issues are taken into consideration. Students will therefore condense the scientific outcome into a summarized form displaced in poster format. The posters will be presented to a broad audience and will strive to address the critical questions posed by the audience. Finally, the student will reflect on the full cycle of his/her research within his/her peer group as to learn how to improve his/her approach to this type of assignments in the future (master thesis).

Examples of projects:

- Understanding and prediction of intra-diurnal pollution level variations in a large city. As both human activity (rush hours) and atmospheric activity (turbulent mixing) strongly varies within a circadian cycle, this is also true for pollution levels exposed to the inhabitants.

- Harvesting the Urban Wind Island: a potential source of wind energy? The wind over city regions with low buildings may exceed wind over their rural counterpart that are less rough in general. This counterintuitive aspect results from the fact that the atmospheric boundary layer is generally deeper over cities.

- Greening the city: the role of vegetation in climate adaptation. Trees, shrubs and grass bring many benefits to cities including cooling by transpiration, filtering aerosols, absorbing noise and enhancing the general well-being of the urban population. Yet urban vegetation lives under stressful conditions. The project can take different angles. E. g by observing and modeling of temperature, radiation, noise, air quality modification. Or by looking into the design of urban greening by finding sweet spots in urban microclimate.

Study Goals

After completion of this module the students are able to:

B-LO1: Identify grand challenges in atmospheric environmental engineering related to sustainability of the built environment and to human health. (ILO: b)

B-LO2: To translate these problems into scientific research and/or engineering questions. (ILO: b)

B-LO3: Select the appropriate scientific tools (theoretical/ observational/ numerical) to address those questions and become cognizant of the limitations of the chosen tools. (ILO: a)

B-LO4: Apply various scientific tools in a rigorous and efficient manner. (ILO: e, f)

B-LO5: Plan, monitor and manage a project and to collaborate in an interdisciplinary team with people from diverse backgrounds. (ILO: j, k)

B-LO6: Develop novel ideas for interventions; existing knowledge is critically assessed, and alternatives are sought for by integrating multi-disciplinary approaches. (ILO: c)

B-LO7: Reflect on the societal impact of interventions and be able to discuss ethical aspects of those within the context of atmospheric environment engineering. (ILO: l)

B-LO8: Effectively communicate research findings to other researchers, engineers, as well as to the general public. (ILO: i)

B-LO9: Demonstrate understanding of the following UN sustainable development goals: (a) good health and well-being; (b) affordable and clean energy; (c) sustainable cities and communities; and (d) climate action. (ILO: h)

Education Method

Teaching and learning that encourages students to take responsibility for their learning process:

Urban and Indoor Air Quality

In-class lectures will primarily cover the necessary background knowledge so that the students will be capable to collect necessary data and use appropriate models to assess air quality in the indoor and outdoor environment. In a small number of the lectures the students will be given hands-on instruction on how to use specific sensors and models. They will get the opportunity to employ these tools at a later stage in order to carry out an assignment. In addition, throughout the course, the students will be required to read relevant journal articles and critique certain aspects of the papers in groups during the lecture hours.

Noise Modelling and Mitigation

Class lectures (on-campus) will be the primary mode of instruction to convey the theoretical concepts of the module. Students will be guided through the theory combined with in-class experiments.

Note: This theme will make use of collected noise measurement data from the theme Atmospheric Measurement Techniques for Urban Environments in which students will work on a field assignment which includes the measurement and analysis of recorded sound signals.

Extreme Urban Weather

In class lectures + practicals;
Assignments running in parallel with lectures.

Integrated AEE Project

Lectures: 2x 2 hours of lecture is given. The first lecture is to introduce and to give an overview on potential research topics, an outline of methodologies and a presentation of learning objectives. The second lecture is an introduction on available methods/guidelines for scientific poster presentation.

Training exercise: during an afternoon several examples of characteristic posters from conferences are discussed in class. Groups of students discuss e.g., highlights & pitfalls and propose a ranking of those examples.

Group work. The project work is done in groups of 2-5 students. This strategy is to stimulate interaction among persons with different backgrounds and different characters and competences. Each group is supervised by an instructor who coaches the research process.

Mini symposium. The results are displayed in a mini-symposium (potentially with peer-students and faculty members). If desired by the students, this symposium could be done in a quasi-formal setting of a mock conference (including drinks and formal clothing).

Assessment

Urban and Indoor Air Quality

Specific questions (which will require short calculations) related to the covered materials will be provided to the students in every lecture. The answers to these questions will be discussed either in following lectures or during specific sessions that can be arranged depending on demand. In addition, feedback will be provided in the framework of the assignments, both during the explanation/training but also during the presentation of the results.

Noise Modelling and Mitigation

Formative assessment of the work-in-progress will be provided to each group in several occasions in the form of oral feedback by both the teacher and the peers.

Extreme Urban Weather

All in-class activities (including the practical's and homework) will be formative. Students will receive peer-feedback during the practical's and formative written feedback on the homework from the TA's.

Integrated AEE Project

	Once a week the project group meets with their respective instructor/coach. Students are expected to be active: a brief update on progress is given by preparing and presenting a very short PowerPoint (few slides only). As all students should be engaged, this preparation is done on a rotational basis, with one student responsible for a specific week. This alternating strategy stimulates continuous engagement by all the students and also allows individual feedback at an early stage. During the poster presentation, direct feedback is given by the peer students (interview). Together with the teacher the students reflect on what could have been improved in the project and what are the recommendations for future work. Summative feedback. After the poster presentation, a project grade is given by the teacher team according to (rubric) criteria that take into account various aspects like e.g.: scientific approach, theoretical depth, organization and group work, end result, written (poster) presentation, contribution to discussion and oral presentation. Students complete the module successfully when passing the following 4 summative assessments: Urban and Indoor Air Quality: group assignment (20%) (including report 15% and presentation 5%) To avoid 'freeriding', a statement is made at the end of the report which indicates specific contributions by each student. Likewise, each student orally presents part of the presentation. Noise Modelling and Mitigation: group assignment (presentation of the case study (10%) and report on it (10%)) Both the report and the presentation are group components. Free riding will be prevented as follows: &#9679; All group members will be asked to present part of their group assignment. &#9679; After the presentation of the group assignment, the assessor will ask each individual member of the group specific questions regarding the report and the presentation. Questions can span the whole assignment regardless of which part is presented by the individual group member. This way individual contributions to the group work will be assessed and differentiation of the grades (between group members) will be possible based on the answers provided. Extreme Urban Weather: written exam with closed and open-ended questions (20%) Integrated AEE project: group project (40%) As the emphasis of the project is on integration, connecting and collaboration, assessment will take place on group level. A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.
Expected prior Knowledge	No specific prior knowledge
Academic Skills	Communication and responsiveness, scientific writing and presentation, time management and anticipation, application of theory to new situations using the relevant equations, presenting, academic reading, writing, referencing, information collection and evaluation, inquiry thinking, critical thinking and teamwork.
Literature & Study Materials	Urban and Indoor Air Quality Written background material, consisting of book chapters, scientific papers and in-house developed readers will be provided to the student in advance via Brightspace. Experimental material is directly provided in the practical's. Noise Modelling and Mitigation Background material, papers and chapters from relevant literature will be provided via Brightspace. Material will be borrowed by the following books: https://www.taylorfrancis.com/books/fundamentals-sound-vibration-frank-fahy-david-thompson/e/10.1201/b18348 https://www.springer.com/gp/book/9783540240525 https://www.springer.com/gp/book/9783030447861 https://www.cambridge.org/core/books/fundamentals-of-noise-and-vibration-analysis-for-engineers/8768D986559E0B54ECDDF0A167C9CDF1 https://www.springer.com/gp/book/9783662053911 https://www.springer.com/gp/book/9781475760118 https://www.taylorfrancis.com/books/environmental-methods-transport-noise-reduction-mats-nilsson-j%C3%B6rgen-bengtsson-ronny-klaeboe/e/10.1201/b17606 Extreme Urban Weather Lenzholzer, S. (2015). Weather in the City-How Design Shapes the Urban Climate. Nai 010 Uitgevers/Publishers. Willems, P. (2019): Impact of Climate Change on Rainfall Extremes and Urban Drainage Systems, IWA Publishing, Vol.11, Open access ebook at https://doi.org/10.2166/9781780401263 Additional background material and/or papers will be provided via Brightspace. Experimental material is provided in the practical's. Integrated AEE Project Book (selected chapters). Eloquent Science: A Practical Guide to Becoming a Better Writer, Speaker, and Atmospheric Scientist. By Schultz. Springer. ISBN 978-1-935704-03-4. Handouts on presenting a scientific poster. Journal papers for topic at hand, as collected by the students. Rubric for conference posters, American Meteorological Society
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	calculator
Collegerama	No

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Environmental Engineering

ENVE Track Water Resources Engineering (WRE)

ENVM1401	Quality, Quantity and Design, Information and Advice	15
Responsible Instructor	Dr.ir. A.M.J. Coenders	
Instructor	Dr.ir. M.W. Ertsen	
Instructor	Dr.ir. J.G. Langeveld	
Instructor	Dr. E. Mostert	
Instructor	Dr. R.R.P. van Nooijen	
Instructor	Dr.ir. A.M.J. Coenders	
Instructor	Dr.ir. O.A.C. Hoes	
Instructor	Prof.dr.ir. M. Bakker	
Instructor	Dr.ir. R.J. van der Ent	
Instructor	Dr.ir. G.H.W. Schoups	
Instructor	Dr. B.M. van Breukelen	
Instructor	M. Lauren	
Instructor	J.A. van der Werf	
Instructor	Prof.dr.ir. R. Uijlenhoet	
Contact Hours / Week x/x/x/x	x/8/variable/x (8 contact hours pw in Q2. Q3 varies)	
Education Period	2 3	
Start Education	2	
Exam Period	2 3 4	
Course Language	English	
Course Contents	Water resources engineers develop engineering interventions to protect society from droughts, flooding and water pollution and minimize the scarcity of potable water. The design of these interventions requires fundamental understanding of processes that influence water availability and water quality. Steppingstone part 1: Quality, quantity and design In this part of the module students will study water quality and biochemical processes, the natural hydrological cycle, and the man-made urban water cycle (drinking water, wastewater, etc.) through lectures and design exercises of a range of water resources engineering systems (irrigation, reservoirs, polders, drinking water supply, etc.). This module contains three different subthemes: Environmental Water Quality & Biogeochemical Processes, Environmental Water Quantity & Hydrological Processes, and Water resources engineering system design with distinct teaching approaches. Environmental Water Quality & Biogeochemical Processes Link to learning objectives of the module: SS: LO1 and LO4. This subtheme addresses environmental water quality and biogeochemistry aspects relevant to water resources management. Knowledge about the water quality and reaction processes and matter cycles taught in the ENV programme module will be used. Through cases students learn to describe relevant water quality and biogeochemical processes and make first order quantitative descriptions using hand calculations and open access software. Specific topics addressed include: Dissolution and precipitation of minerals; Thermodynamics and kinetics; pH/alkalinity Water-Air interaction and effects Sorption reactions: adsorption, ion-exchange Sorption reactions: surface complexation, pathogen filtration Redox reactions and microbial growth; kinetics Mass transport: Advection, dispersion, reaction, mixing Chemical modelling of water quality Interpreting and modelling of water quality data from environmental systems Environmental Water Quantity & Hydrological Processes Link to learning objectives of the module: SS: LO2 and LO4. This subtheme addresses environmental water quantity with a specific focus on hydrological and hydraulic processes, both in natural systems and man-made systems. It builds upon the knowledge on the hydrological cycle and water transport processes introduced in the ENV programme module. Through cases students learn to describe and quantify fluxes and stores in both the natural and the man-made water cycles. Topics that will be discussed are precipitation, evaporation, groundwater, drinking water, wastewater. Specific topics addressed include: Hydrological cycle in natural and man-made systems: precipitation, evaporation and vegetation, unsaturated zone, groundwater, human fluxes) Flow and discharge in rivers and open channels Flow and discharge in (pressurised) pipe systems and networks Flow at water control structures (weirs, orifices, pumps) Water resources engineering system design Link to learning objectives of the module: SS: LO3, LO4 and LO5. Process understanding is required to design water resources engineering systems. In this subtheme the functional dimensioning of several common water resources engineering systems will be discussed. Partly process understanding introduced in the abovementioned subthemes will be applied and partly new process understanding will be taught. Furthermore, first order design and dimensioning models (formulas) will also be part of the material.	
Course Contents Continuation	Specific topics addressed include: Multi-purpose reservoirs Irrigation systems Polder and drainage systems Urban drainage systems (piped and non-piped)	

Rainwater harvesting systems
Piped water distribution systems
Blue green infrastructure
Wastewater treatment systems

Steppingstone part 2: Information and Advice

Water resources engineers advise stakeholders on solution strategies for problems such as water pollution, water scarcity, floods and limited access to safe water. They build advice on extensive problem and data analysis and consider the ethical implications of their advice.

The second part of the steppingstone module Information and Advice will take place in Q3 and consists of a deepening of the data analysis skills from the MUDE specifically for water and of an advisory group project. In this project students will advise a stakeholder on a real-life water resources engineering project based on substantial data analysis. Several projects will be offered on various scales (local to catchment) and systems (urban, rural etc.). Steppingstone 2 will be taught as an intensive course of three weeks full time, including lectures, workshops and tutorial sessions. Information and advice are divided into the lecture- and exercise-base sub-theme "Data quality and data analysis" and the project-based sub-theme "Evidence-based advice".

Data quality and data analysis
Link to learning objectives of the module:
SS: LO6, LO7 and LO8.

This subtheme focusses on understanding and predicting the spatiotemporal behaviour of water resources engineering systems based on data. Students learn about typical monitoring methods and uncertainties associated and how to use statistical techniques for data analysis and prediction. This subtheme builds upon the faculty wide MUDE.

Specific topics include:
Geographical Information Systems
Raster and polygon data
Spatiotemporal patterns in man-dominated fluxes such as (drinking) water demand
Uncertainty and propagation of errors in hydrological time series
Correlation and persistence in precipitation and other timeseries
Frequency analysis and extreme value analysis of precipitation and discharge
Methods for determining extreme (design) precipitation flood and drought conditions
Time dependent groundwater flow
(groundwater) timeseries analysis
Spatiotemporal patterns in surface and groundwater quality
Leakage detection

Evidence-based advice
Link to learning objectives of the module:
SS: LO6, LO9, LO10 and LO11.

In this advisory group project students will work in a diverse team in terms of personalities, cultures and study background and advise a stakeholder on a real-life water resources engineering project based on substantial data analysis. In addition, they will reflect on the challenges and opportunities that working in a diverse team brings and on their (ethical) professional responsibility as an engineer in such an advisory project. This part builds upon data quality and data analysis and on the experience with teamwork and ethics that students gained in the ENV programme module.

Specific attention will be paid to data availability and how to handle limited data availability. Students will practice systematic procedure to make assumptions and how to assess the effect of these assumptions on their final advice.

Specific topics include:
Problem analysis for water resources engineering students
Data availability and data scarcity
Assumption development and assessment (sensitivity analysis)
Team roles and team behaviour
SCRUM
Professional ethics on the responsible use of data
Presentation skills

Study Goals

SS-LO1: Describe and quantify chemical and biological processes in environmental water systems, including transport by advection, diffusion and dispersion, (bio) chemical reactions and conversions and dissolution and precipitation of minerals. (ILO: a)

SS-LO2: Quantify stores and (mainly stationary) fluxes in environmental water systems, including precipitation, evaporation, surface water and groundwater flows, drinking water flows, wastewater flows etc. (ILO: a)

SS-LO3: Analyse water resources engineering problems. (ILO: b)

SS-LO4: Conceptualize relevant physical, chemical and/or biological processes towards a quantitative description of environmental water systems. (ILO: f)

SS-LO5: Explain the functioning of typical engineering solutions in the field of water resources management, water treatment and urban water infrastructure. (ILO: d)

SS-LO6: Identify and analyze complex water resources problems that may threaten the environment and human wellbeing. (ILO: b, d, e)

SS-LO7: Analyse and assess the quality of observations typical in water resources engineering. (ILO: g)

SS-LO8: Analyse temporal and spatial patterns in data in water resources engineering, including extreme values to gain system understanding. (ILO: a, g)

SS-LO9: Advise on solution and/or monitoring strategies for problems in the field of water resources engineering supported with quantitative data analysis considering (inter)national goals, norms and regulations for water resources management. (ILO: c, d, h)

SS-LO10: Assess data needs and required quality based on requirements set by stakeholders involved in the problem and solution strategies. (ILO: e, h)

SS-LO11: Work effectively in a team of diverse talents, skills, characters, and cultures to solve Water Resources Engineering problems. (ILO: j, k, l)

Education Method

Teaching and learning that encourages students to take responsibility for their learning process:

	Part 1 The teaching method is through cases, where students collaboratively work in small groups on several cases. Through the 7-jump method (1: clarifying concepts, 2: formulating the central case, 3: free brainstorm, 4: clustering, 5: learning objectives, 6: self-study, 7: report) students will actively construct (new) knowledge and skills to master the module content. This requires high-engagement and active participation of students. Furthermore, through the collaborative group process students can benefit from each others background. Each group has a tutor who ensures that the learning goals will be met. Study materials will be provided in different forms (text books, papers, videos, interactive lectures, (computer)practical (e.g., PHREEQC), etc.). Part 2 Data quality and data analysis Data quality and analysis will be taught via lectures, computer tutorials and computer exercises. Easy tutorials will be homework whereas more complex exercises will be done in class under supervision of the teaching team. Evidence-based advice Students practice the project management method SCRUM. SCRUM Teams will be formed based on diversity in personalities. Students will plan their project in three weekly sprints, organise daily stand-ups and assess weekly progress in review and retrospective meetings. Teachers facilitate this process through a series of workshops. In addition, specific guidance on problem analysis and assumption making will be offered.
Assessment	Q2: - 1 written exam 3 EC (20%) - 1 written exam 3 EC (20%) - 1 graded assignment of 3 EC (20%) Q3: - Individual computer test - Data quality and data analysis (15%) - Evidence-based advice (25%): The last review and retrospective meeting will be used to assess the students using a rubric that will be developed based on the learning objectives associated with this subtheme.
Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.
Academic Skills	Not applicable; however, having followed the Core Program Module - Environmental Engineering is highly recommended.
Literature & Study Materials	Critical thinking; Analytical thinking; Problem-based learning; Solving; Collaboration with peers; Application of theory to new situations using relevant equations; Data analysis; Advice stakeholders; Ethical reflection/consideration. Part 1: Study materials (text books, papers, videos, interactive lectures, (computer)practical (e.g., PHREEQC), etc.) will be provided via Brightspace Part 2: Data quality and data analysis For GIS students will use the free and open source program QGIS. They will be introduced via an updated version of https://ocw.tudelft.nl/course-lectures/introduction-to-gis/?course_id=9282 (accessed 2-4-2021). For the other topics dedicated lecture notes, background material, papers and chapters from relevant literature will be provided via Brightspace. Evidence-based advice Dedicated lecture notes, background material, papers and chapters from relevant literature will be provided via Brightspace, including the official Scrum guide (scrumguide.org).https://scrumguides.org/
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	None (or check Brightspace for updates)
Collegerama	Yes

ENVM1500	Water Quality and Principles of Environmental Engineering	9
Responsible Instructor	M. Laureni	
Instructor	Dr. B.M. van Breukelen	
Instructor	Dr.ir. R.E.F. Lindeboom	
Instructor	M. Laureni	
Instructor	S.J. Smith	
Instructor	K.M. Lompe	
Contact Hours / Week x/x/x/x	x/x/4/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	To define interventions for any future water resources challenge in a systematic manner, water technology students need to be able to apply the mass balance concept and calculate the time-based performance of a system unit in terms of biological and chemical reaction rates. By starting with physicochemical processes and then expanding to biochemical processes the complexity is gradually increased while using the same methodology. Additionally, these calculations can be combined with knowledge on pollutant removal and degradation processes/factors to analyse and adapt the operational conditions of a lab experiment with the aim to optimise process performance for both efficient conversions and/or pollutant removal. The ethical aspects of working as an engineer in the field of human health will be discussed in a workshop. Principles of physicochemical and biological processes Link to learning objectives of the module: A1: LO1, LO2 and LO3. Central topic of the module sub theme is learning to apply the principles of physicochemical and biochemical processes, reactor technologies and pollutant removal/degradation processes with the aim of using this know-how for designing and evaluating a treatment unit in module B1. It provides an environmental process engineering perspective and addresses the fundamentals on: Calculating equilibria of (bio-)chemical conversion processes in natural or engineered treatment units. Assessing biological and chemical reaction kinetics in dependence to applied reactor conditions and operational process characteristics using mathematical models. Assessing removal efficiency of pollutants by physicochemical and biochemical processes, by exploring, understanding and applying underlying concepts like mass balances, (bio-)chemical thermodynamics, mass transfer phenomena, dissolution, precipitation, degradation kinetics. Biological growth models (Monod kinetics), redox processes and reaction stoichiometry. Reactor technologies. Factors that impact removal, degradation or conversion of pollutants (e.g. chemicals of concern) that pose a risk to human health. Biological or chemical reactor performance analysis Link to learning objectives of the module: A1: LO1, LO2, LO3, LO4 and LO5. Central topic of the module sub theme is the execution of a biochemical lab experiment and mathematical modelling and performance analysis of this system. Through the lab experiment students can use know-how from module sub theme I to describe the chemical and/or biological conversion processes and biological growth rates and evaluate the removal efficiencies of contaminants (chemicals and pathogens that pose a risk to human health) in a single process unit. The following topics will be addressed: Using mathematical modelling in different software environments (e.g., phreeqC, matlab, python, aquasim) the experimental results can be described with chemical and biological conversion/growth models and evaluated for their potential to protect human health. Presenting, proposing and reporting about improved operational conditions for performance enhancement.	
Study Goals	A1-LO1: evaluate how physicochemical and biological processes in natural and engineered environments work and how they could be used to remove, degrade or recover pollutants (chemicals, pathogens, antibiotic resistance) that pose a risk to environmental and human health. (ILO: c, d) A1-LO2: Assess mass balances and (geo)chemical speciation reactions to calculate the equilibrium of (bio-)chemical conversion processes in natural or engineered treatment sub theme (batch/CSTR). (ILO: a) A1-LO3: Assess biological and chemical reaction kinetics in dependence to applied reactor conditions and operational process characteristics (Batch, fed-batch, porous media filters, CSTR, FBR/MBBR, PFR) using mathematical models. (ILO: a) A1-LO4: Execute a small lab experiment focused on a biological and/or chemical conversion technique, including a model-based prediction and verification of the results. (ILO: b, g) A1-LO5: Propose an improved approach for an environmental challenge based on laboratory results, external datasets, modelling output and scientific literature and convey one's advice through a presentation and report. (ILO: b, g)	
Education Method	Teaching and learning that encourages students to take responsibility for their learning process: Principles of physicochemical and biological processes Lectures (individual) Case studies (group work), Project and problem-based teaching, The students are expected to read background material in the form of research articles and chapters from several handbooks available in digital form in the TU Delft library. Students can consult knowledge clips from online sub themes on specific topics. The assignments need to be carried out during the lecture period. Online lectures and on campus workshops Biological or chemical reactor performance analysis Simulations (for hypothesis formation) by student (group work) Groups contain 2/3 members. Laboratory Experiment (group work) Data analysis and Modelling (group work) Presentation of results (group work) The students are expected to read background material in the form of research articles and chapters from several handbooks available in digital form in the TU Delft library. The assignments need to be completed during Q3.	

Assessment	Students complete the module successfully when passing the 3 summative assessments: Principles of physicochemical and biological processes Individual written exam (50%) Biological or chemical reactor performance analysis phreeqC assignments (25%) Written Report on lab experiment (25%) A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment. The results may be oral instead of written in case of less than 5 students.
Expected prior Knowledge	Completed MUDE and EE Stepping stone 1 and 2
Connection to other modules	This is a prerequisite module for B1 (in water technology track)
Academic Skills	Critical thinking, analytical thinking and problem solving are the main academic skills which will be trained. In addition, students will need to collaborate with their peers and thus organizational and interpersonal skills will be developed as well. Ethical reflections/considerations are included in a formative way.
Literature & Study Materials	The following books cover the main content of the module and relevant chapters will be offered to the students via Brightspace. Soil and water chemistry: an integrated approach. Michael E. Essington. CRC Press. ISBN 9781466573154 Water quality engineering Physical/chemical treatment processes. Mark Benjamin & Desmond Lawler. John Wiley & Sons. ISBN 978-1118169650 Biological Wastewater Treatment: Principles, Modeling and Design. Guanghao Chen, George A. Ekama, Mark C. M. van Loosdrecht, Damir Brdjanovic. IWA Publishing. ISBN (electronic): 9781789060362 Biological Wastewater Treatment: Principles, Modelling and Design: Examples & Exercises Additional background material and papers will be provided via BrightSpace.
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Open book exam (calculator and paper-printed material allowed)
Collegegama	No

ENVM1501	Design and Modeling of Urban Water Infrastructure Systems	9
Responsible Instructor	Prof.dr. Z. Kapelan	
Instructor	Dr.ir. J.G. Langeveld	
Instructor	E.J.M. Blokker	
Instructor	Dr.ir. I.W.M. Pothof	
Instructor	Prof.dr. G.J. Medema	
Instructor	Prof.dr.ir. J.P. van der Hoek	
Instructor	Dr. E. Abraham	
Instructor	J.A. van der Werf	
Instructor	Prof.dr. Z. Kapelan	
Contact Hours / Week x/x/x/x	14/x/x/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Urban water infrastructure systems sustain human life in towns and villages by supplying drinking water in sufficient quantity and quality, collecting and conveying wastewater and stormwater to a recipient whilst protecting human health, maintaining acceptable level of flood risk and protecting the natural environment. Modern urban water infrastructure is based increasingly on the concept of integrated water management ensuring that natural and other resources are (re)used in an efficient and sustainable manner. This module deals with concepts, systems, solutions, models, data and knowledge required for the effective design (and redesign / rehabilitation) of urban water infrastructure systems and their monitoring. A range of related issues are addressed covering water quantity, quality and other aspects. Both conventional, grid type systems (based on drinking water and sewer pipe networks) and latest off-grid type systems and solutions (such as sustainable drainage and other solutions) are considered including integrated water systems and solutions. Sub-theme "Design of urban water infrastructure systems" (link to learning objectives LO1, LO2 and LO3) This subtheme focusses on developing students understanding of concepts, system configurations, solutions, models, data and knowledge required for the effective design and redesign / rehabilitation of urban water infrastructure systems. Both drinking water and urban drainage systems are covered, including related modelling required for design. The focus of this subtheme is on water quantity related aspects of design. Sub-theme "Water quality processes in urban water infrastructure systems" (link to learning objectives LO3, LO4 and LO5). This subtheme focusses on developing a students understanding of water quality processes in urban water infrastructure systems and related monitoring. Water quality covers chemical parameters, microbiological parameters, physical parameters and the interaction between these parameters. Both drinking water and urban drainage systems are covered, including related water quality modelling.	
Study Goals	LO1: Design new and re-design existing urban water infrastructure systems. LO2: Design, dimension and analyse individual components of urban water infrastructure systems. LO3: Understand and apply relevant simulation and other models to analyse and plan new and suggest improvements to existing urban water infrastructure systems and their components. LO4: Design monitoring setup/networks for water quantity and water quality parameters taking specific challenges and dynamics of the range of urban water infrastructures into account. LO5: Study and understand water quality processes in urban water infrastructure systems using models and monitoring data and identify knowledge gaps in scientific literature on water quality processes.	
Education Method	Teaching and learning that encourages students to take responsibility for their learning process: "Design of urban water infrastructure systems" This subtheme will be taught via lectures, assignment and demonstration of software tools. Lectures will be delivered in a conventional format together with suitable questions / discussions taking place during the lecture. The assignment will focus on the design of a real urban drainage or drinking water system with weekly feedback sessions provided to support the students. Relevant software tool demonstrations will be conducted too, via half-daily workshops. These include Epanet software for drinking water distribution systems and SWMM for urban drainage / sewer systems. At these workshops students will have to use these tools to answer in situ provided problems. "Water quality processes in urban water infrastructure systems" This subtheme will be taught via lectures assignments and field experiments. Lectures will be delivered in a conventional format together with suitable questions / discussions taking place during the lecture. The assignment considers the analysis and evaluation of water quality in a sewer system.	
Assessment	Students complete the module by passing the following: 1. Oral exam, single exam covering material taught in both sub-themes, 50% of the overall module grade. Design of urban water infrastructure systems: 2. Assignment report on urban water infrastructure design (drinking water or urban drainage related, 30%) Water quality processes in urban water infrastructure systems: 3. Assignment report on water quality processes in urban water infrastructure (20%) Note: All of above must be passed to complete the module. A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.	
Expected prior Knowledge	Undergraduate hydraulics in civil engineering	
Connection to other modules	This module provides necessary input for the ENV1601 module.	

Academic Skills	Information collection and synthesis; Application of theory to situations in engineering practice; Making inferences of physical situations by analysing observations/data; Use of modelling tools; Ethical reflection/consideration.
Literature & Study Materials	Design of urban water infrastructure systems Butler, D., Digman, C.J, Makropoulos, C. and Davies, J.W., Urban Drainage, Blackwells, ISBN-13: 978-1498750585 Savic, D. and Banyard, J., Water Distribution Systems, Thomas Telford Publishing, ISBN-13: 978-0727741127 Epanet software for the modelling of drinking water systems (both quantity and quality), https://www.epa.gov/water-research/epanet SWMM software, for the modelling of urban drainage / sewer system (both quantity and quality), https://www.epa.gov/water-research/storm-water-management-model-swmm Lecture notes (via Brightspace) Assignment manual with requirements and data for the design of an urban drainage system in part of Delft (via Brightspace) Papers and chapters from relevant literature (via Brightspace) Water quality processes in urban water infrastructure systems Hvitved-Jacobsen, T., Vollertsen, J., Nielsen, A.H. Sewer Processes. Microbial and Chemical Process Engineering of Sewer Networks. ISBN 9781439881774 Ashley, R.M., Bertrand-Krajewski, J.L., Hvitved-Jacobsen, T., Verbanck, M. Solids in Sewers. Characteristics, Effects and Control of Sewer Solids and Associated Pollutants. ISBN13: 9781900222914 Lecture notes (via Brightspace) Papers and chapters from relevant literature (via Brightspace) Assignment manual with requirements and data (via Brightspace)
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (R&G): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Calculator
Collegerama	Yes

ENVM1502	River Basin Hydrology and Water Management	9
Responsible Instructor	Dr. M. Hrachowitz	
Instructor	Prof.dr.ir. N.C. van de Giesen	
Instructor	Dr.ir. R.W. Hut	
Instructor	Dr.ir. R.J. van der Ent	
Instructor	Dr.ir. G.H.W. Schoups	
Instructor	Dr. S.C. Steele-Dunne	
Instructor	Dr. M. Hrachowitz	
Instructor	Prof.dr.ir. R. Uijlenhoet	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Floods and droughts are among the most concerning water system threats and may be aggravated by land use and climate change. Water resources engineers use process understanding, mathematical models and observation technology including remote sensing to study floods, droughts and the effects of interventions. In this module students will learn about hydrological processes that occur at different scales in river basins and how they are affected by land use change, climate change, and/or human interventions. Focus lies on floods, droughts and the role of vegetation. Students will learn about the physics of hydrological processes and how to observe and model them and to evaluate the effects of management strategies on the river basins taking ethical considerations into account. Processes in river basin hydrology and water management Link to learning objectives of the module: A3: LO1. In this first topic students will focus on understanding the physical processes at different scales through lectures and discussing recent journal papers. The specific topics include: - Macroscale Laws, structure and organization in hydrological processes. - Role of Vegetation & land management in the Water Cycle - Flood processes & management - Cold region processes - Drought processes & management Remote sensing data and data assimilation in river basin hydrology and water management Link to learning objectives of the module: A3: LO3. In this second topic students will learn, via tutorials and assignments, about observation techniques, data handling and how to use them in modelling. - Remote sensing data types, sources and properties - Accessing, handling and processing data - Automatizing data access - Learning from data: basic concepts of data assimilation Models in river basin hydrology and water management Link to learning objectives of the module: A3: LO2. The third topic aims to conceptualize the hydrological processes such that the effects of climate and land use change can be quantitatively evaluated. Typical hydrological and river basins model concepts will be explained and how to calibrate and test them. - Model types and applications in Hydrology - Flow paths - Model parameter calibration and testing - Sources of uncertainty and the role of inverse, ill-posed problems - Model evaluation and selection techniques	
Study Goals	(a) Describe and conceptualize the role of different components of the terrestrial water cycle for the distribution and movement of water through the landscape with particular emphasis on the role of vegetation and the critical zone in hydrological systems towards quantitative descriptions (Unit 1). (b) Select and apply remote sensing technologies for observation and monitoring of hydrological processes (Unit 2). (c) Develop mathematical models to explain water processes in order to predict floods and droughts and the effectiveness of interventions (Unit 3).	
Education Method	Teaching and learning that encourages students to take responsibility for their learning process: Processes in river basin hydrology and water management The teaching method is a combination of interactive lectures and exercises. Prior to lectures corresponding book chapters/handouts/papers should be studied. To engage students, short mini exercises are given during the lectures. Remote sensing data and data assimilation in river basin hydrology and water management After a short introduction, students follow tutorials to get hand-on experience with working with spatial data. Additionally, they work on weekly assignments. These assignments are part of the students portfolio. Models in river basin hydrology and water management The teaching method is a combination of interactive lectures and exercises. Prior to lectures corresponding book chapters/handouts/papers should be studied. To engage students, they work on assignments where they develop their own model concept including testing different calibration methods and assessing its performance.	
Assessment	Written Exam (40 %), in which being able to explain, critique and apply knowledge of all theory covered in the course will be assessed. In the case of low student numbers the responsible instructor can change the written exam into oral exams. This decision will be communicated through brightspace in the first week of the course Written Report and Poster Presentation (60 %) on an individual project where the student demonstrates that they can apply the learning goals to a context (region) different from the one taught in the lectures and practicals. The opening lectures of the course	

	will present all the details on this project. Depending on student numbers the responsible instructor can decide to have the project be conducted in groups.
Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.
Academic Skills	Knowledge of fundamental hydrological concepts and methods
Literature & Study Materials	Analytical thinking; Application of theory to new situations using the relevant equations; Making inferences of physical situations by analysing observational data; Reading and discussing journal papers. Processes in river basin hydrology and water management Material for the course will include (modified versions of): https://ocw.tudelft.nl/courses/hydrological-measurements/ (accessed 7-4-2021) https://ocw.tudelft.nl/courses/hydrology-of-catchments-rivers-and-deltas/ (accessed 7-4-2021) Selected papers Remote sensing data and data assimilation in river basin hydrology and water management Material for the course will include (modified versions of): https://ocw.tudelft.nl/courses/spatial-tools-in-water-resources-management/ (accessed 2-4-2021) Models in river basin hydrology and water management Material will be taken from publications on hydrological modelling including: Kirchner, J. W. (2006). Getting the right answers for the right reasons: Linking measurements, analyses, and models to advance the science of hydrology. <i>Water Resources Research</i>, 42(3). Beven, K. (2006). Searching for the Holy Grail of Scientific Hydrology: $Q \neq H$ (S? R?) A as closure. <i>Hydrology and Earth System Sciences Discussions</i>, 3 (3), 769-792. Hrachowitz, M., Savenije, H. H. G., Blöschl, G., McDonnell, J. J., Sivapalan, M., Pomeroy, J. W., ... & Cudennec, C. (2013). A decade of Predictions in Ungauged Basins (PUB) a review. <i>Hydrological sciences journal</i>, 58(6), 1198-1255. Hrachowitz, M., & Clark, M. P. (2017). HESS Opinions: The complementary merits of competing modelling philosophies in hydrology. <i>Hydrology and Earth System Sciences</i>, 21(8), 3953-3973. Blöschl, G., Bierkens, M. F., Chambel, A., Cudennec, C., Destouni, G., Fiori, A., ... & Renner, M. (2019). Twenty-three unsolved problems in hydrology (UPH) a community perspective. <i>Hydrological sciences journal</i>, 64(10), 1141-1158. This list will be updated each year with new and relevant publications.
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Hand calculator, English dictionary
Collegerama	Yes

ENVM1600	Water Treatment Technologies	15
Responsible Instructor	K.M. Lompe	
Instructor	Dr.ir. J.G. Langeveld	
Instructor	Prof. L.C. Rietveld	
Instructor	Prof.dr.ir. M.K. de Kreuk	
Instructor	Dr.ir. S.G.J. Heijman	
Instructor	Prof.dr.ir. D. van Halem	
Instructor	Prof.dr.ir. J.B. van Lier	
Instructor	Prof.dr.ir. J.P. van der Hoek	
Instructor	M. Laureni	
Instructor	K.M. Lompe	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Water treatment is needed to provide humans with water of sufficient quality to use it safely for purposes ranging from drinking to industry and to avoid pollution of the living environment. In module Water Treatment Technologies one will learn how to develop, research and design water treatment plants for the treatment of groundwater, surface water, municipal sewage and industrial water. Specific attention will be given to health impact, resource recovery and closing the urban water cycle. The module will cover separation, biological and chemical conversion technologies, including, membrane filtration, advanced oxidation, aerobic granular sludge and anaerobic treatment. In addition to project work and lectures, the module offers field visits to water treatment plants, experimental work in the laboratory and modelling exercises (Biowin, WAVE). The technologies covered in this module, include: Anaerobic digestion: slurry digestion and high-rate anaerobic treatment Aerobic sludge: flocculent and attached (granules) growth Biofiltration Pressure-driven membranes (reverse osmosis, nanofiltration, ultrafiltration) (Bio) sand filtration Ozone, chlorine, H2O2 and UV Electro(photo)chemical oxidation Activated carbon, GAC/ PAC/PACAS Ion exchange Resource recovery from sludge-based systems The course is organised with: Project (one can choose the "designer" or "researcher" track), assessed by group report, presentations and individual oral exam Lectures on water treatment technologies, assessed with written exam Laboratory, model, excursion and in-class assignments (formative assessments)	
Study Goals	Identify process conditions for drinking, industrial and sewage water treatment technologies based on the mechanistic understanding of coupled biological, chemical and physical processes. Assess the applicability of a water technology based on their effectiveness for target compounds, operational criteria, sustainability, and local conditions/ regulations. Design a water treatment plant including dimensions, hydraulic line, process scheme and control, considering budget, health impact, resource recovery and closing the urban water cycle. Research a water treatment technology, including hypothesis testing, experimental design, modeling and evaluating scientific literature. Evaluate project and research outcomes with the project team, clients and the scientific audience using communication and management tools.	
Education Method	Teaching and learning that encourages students to take responsibility for their learning process: Project: Water treatment research and design Lectures (individual) Case studies (group work), Simulations if chosen by student (group work), Project and problem-based teaching, Excursions The students are expected to read background material in the form of research articles from the literature and chapters from several handbooks available in digital form in the TU Delft library. The assignments need to be carried out during the lecture period. Online lectures and on campus workshops Treatment Technologies Lectures (individual) Laboratory experiment and report (group work) Self-study of lecture and practice material provided on Brightspace Preparation of laboratory roadmap for practical Laboratory experiment on campus, Lectures online,	

Assessment	Students complete the module successfully when passing the 3 summative assessments: Written Exam: Water Treatment Technologies (6 ECTS) Group Presentation Project (1 ECTS) Oral Exam on Final Project Report (8 ECTS) A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment. Moments of formative assessments include: 2 or 3 Laboratory experiment reports Biowin and WAVE assignment Treatment train assignment Opening day challenge
Academic Skills	Critical thinking; Analytical thinking; Problem solving; Collaborate with peers, Communicate results in presentations and project/scientific reports; Organizational and interpersonal skills; Ethical reflection/consideration.
Literature & Study Materials	The study material will be offered via Brightspace and consists of knowledge clips (with written hand-outs), scientific papers, conceptual process design lecture notes and Biowin handbook. Lectures will be available via opencourseware (OCW).
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations

ENVM1601	Operation, Control, Management and Adoption of Urban Water Infrastructure Systems	15
Responsible Instructor	Dr.ir. J.G. Langeveld	
Instructor	Dr.ir. J.G. Langeveld	
Instructor	Dr.ir. M.M. Rutten	
Instructor	E.J.M. Blokker	
Instructor	Prof.dr. G.J. Medema	
Instructor	Prof.dr.ir. J.P. van der Hoek	
Instructor	Dr. E. Abraham	
Instructor	Prof.dr. Z. Kapelan	
Instructor	Dr. R. Taormina	
Instructor	M.K. Mehta	
Contact Hours / Week x/x/x/x	24/x/x/x	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Urban water infrastructures must sustain urban living by supplying reliable drinking water, minimising contact risk with pathogens, maintaining urban flood risk at an acceptable level, and protecting the aquatic environment. On top of this, the infrastructures must contribute to circularity. This module deals with the operation, control, management and adaptation of urban water infrastructure systems, comprising both networks and off-grid systems. All the B-modules in the Water Resources Engineering track will have a workshop on ethical responsibility of engineers. Operations and control of urban water systems Link to learning objectives module B2: LO1, LO2 and LO7. Lectures about this topic will focus on operation of drainage systems, control of urban water infrastructures and data processing: Conventional operation of both drinking water and urban drainage systems Automated operation via real time control of urban water infrastructures. Cybersecurity of urban water infrastructure Data collection, storage, validation and mining for operation and maintenance of urban water infrastructures. Data sources are (internal) monitoring networks, call data, risks databases, insurance data and social media (especially for urban flooding). Sensors and sensor network design. Data validation. Asset management of urban water infrastructure systems Link to learning objectives module B2: LO3, LO4 and LO7. Lectures about this topic will focus on asset management of urban water infrastructure systems. Topics that will be covered are: Overview, inspection, deterioration, interventions and other related topics. Introducing AM concepts, highlight process steps, AM management methods, i.e., performance appraisal, financial accounting, portfolio modelling and management, and (technical) requirements and data needs per process step. Inspecting techniques for urban water infrastructures, related to the field of application/related failure mechanisms, both conventional and more advanced Deterioration modelling concepts ranging from deterministic models (impact of uneven soil subsidence on failure of rigid pipes) to cohort survival statistical models. Different methods for prediction of asset deterioration via pipe condition grades, burst rate, blockage and collapse rate will be shown with examples. Different types of interventions (from regular maintenance to rehabilitation and replacement). Deterioration modelling. Adaptation of urban water infrastructures Link to learning objectives module B2: LO5, LO6 and LO7. Lectures about this topic will focus on adaption to climate change, water demand management and urban pollution. Interaction with other infrastructure systems and related governance issues will also be addressed. Adaptation to climate change / uncertain future including urban flood risk management: Dealing with excess of water; basic definitions and concepts, flood risk assessment, different interventions and management options, urban pollution management, thermal energy recovery from piped systems, nature-based solutions Water demand management Emerging water quality issues and urban pollution management. Water quality impacts of emissions from urban water infrastructures: CSO, SSO, WWTP effluent. Entire range of impacts from acute toxic to long term build up. Including new topics: Emerging contaminants (include Coronavirus), microplastics Public health aspects related to urban water infrastructures, focussing on water supply (drinking water + alternatives) and wastewater and stormwater management (reducing contact risk with pathogens). Urban pollution management. Interactions with other infrastructure systems and related governance issues Physical interactions between water and wastewater, gas, road, telecom etc infrastructure; Material interactions: water-energy-food nexus	
Study Goals	B2-LO1: Design, evaluate and optimise control strategies for urban water infrastructures. (ILO: c) B2-LO2: Develop data driven management of urban water infrastructure systems using data mining and modelling technique. (ILO: e, f) B2-LO3: Apply asset management to urban water infrastructures. (ILO: c) B2-LO4: Study deterioration processes in urban water infrastructures using models and monitoring data and identify knowledge gaps in scientific literature on deterioration processes. (ILO: d, e, f, g, h) B2-LO5: Incorporate climate change adaptation and circularity in the design of urban water infrastructures. (ILO: c)	

Education Method	B2-LO6: Assess the impact of urban development/urban transition processes on urban water infrastructure. (ILO: b, h)
	B2-LO7: Conduct a design and a research project in a team and communicate results with relevant audience. (ILO: i, j, k)
	Teaching and learning that encourages students to take responsibility for their learning process:
	Operations and control of urban water systems Lectures 1 day workshop on cybersecurity Assignments excursion
	Asset management of urban water infrastructure systems Lectures Assignments
	Adaptation of urban water infrastructures Lectures Group research project Excursion
Assessment	Students complete the module successfully when passing the 4 summative assessments:
	Oral exam, single exam covering material taught in the three subthemes (50%).
	Operations and control of urban water systems (16,6%): Group design exercise
	Asset management of urban water infrastructure systems (16,6%): Group research report
	Adaptation of urban water infrastructures (16,6%): Group research report
	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.
Tags	Water management
Expected prior Knowledge	A2 module
Academic Skills	Information collection and synthesis; Application of theory to new situations using the relevant equations; Making inferences of physical situations by analysing observations/data; Ethical reflection/consideration.
Literature & Study Materials	Operations and control of urban water systems Lecture notes and papers/chapters from relevant literature will be provided via BrightSpace. The list of papers will annually be updated and will contain papers like: Langeveld, J. G., L. Benedetti, J. J. M. de Klein, I. Nopens, Y. Amerlinck, A. van Nieuwenhuijzen, T. Flameling, O. van Zanten, and S. Weijers. 2013. "Impact-Based Integrated Real-Time Control for Improvement of the Dommel River Water Quality." Urban Water Journal 10 (5): 312-329 van Daal P., Gruber G., Langeveld J., Muschalla D., Clemens F. (2017). Performance evaluation of real time control in urban wastewater systems in practice: Review and perspective. García, L., Barreiro-Gomez, J., Escobar, E., Tellez, D., Quijano, N., Ocampo-Martinez, C., 2015. Modeling and real-time control of urban drainage systems: a review. Adv. Water Resour. 85, 120e132. http://dx.doi.org/10.1016/j.advwatres.2015.08.007 . Asset management of urban water infrastructure systems Background material, papers and chapters from relevant literature will be provided via BrightSpace. The list of papers will annually be updated and will contain papers like: Tscheikner-Gratl, F., Caradot, N., Cherqui, F., Leitão, J.P., Ahmadi, M., Langeveld, J.G., Le Gat, Y., Scholten, L., Roghani, B., Rodríguez, J.P., Lepot, M., Stegeman, B., Heinrichsen, A., Kropp, I., Kerres, K., do Céu Almeida, M., Bach, P.M., Moy de Vitry, M., Sá Marques, A., Simões, N.E., Rouault, P., Hernandez, N., Torres, A., Wery, C., Rulleau B., & Clemens, F. (2020). Sewer asset management state of the art and research needs. Urban Water Journal. Dirksen, J., Clemens, F. H. L. R., Korving, H., Cherqui, F., Le Gauffre, P., Ertl, T., Snaterse, C. T. M. (2013). The consistency of visual sewer inspection data. Structure and Infrastructure Engineering, 9(3), 214-228. van Riel, W., Langeveld, J.G., Herder, P.M., Clemens, F.H.L.R.. (2014). Intuition and information in decision-making for sewer asset management. Urban Water Journal, 11 (6), pp. 506-518.
Judgement	Adaptation of urban water infrastructures Background material, papers, chapters from relevant literature and an excursion report will be provided via BrightSpace. The list of papers will annually be updated and will contain papers like: Marlow, D. R., Moglia, M., Cook, S., Beale, D. J., Land, C., & Road, G. (2013). Towards sustainable urban water management : A critical reassessment. Water Research, 47(20),71507161. https://doi.org/10.1016/j.watres.2013.07.046 Nieuwenhuis, E., Cuppen, E., Langeveld, J., & De Bruijn, H. (2020). Towards the integrated management of urban water systems. Journal of Cleaner Production Wong, T.H.F. , Rogers B.C. Brown, R.R. 2020. Transforming Cities through Water-Sensitive Principles and Practices. One Earth. 3 (4), pp 436-447
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Collegerama	none
	No

ENVM1602	Regional Hydrology	15
Responsible Instructor	Dr. T.A. Bogaard	
Instructor	Prof.dr.ir. M. Bakker	
Instructor	Dr. T.A. Bogaard	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Our regional water systems are far from natural anymore, with human-made or modified rivers, canals, and subsurface water systems. This infrastructure cannot be seen separately from its surrounding environment, while keeping the relation with the desired functionalities. In this module students will learn about open water systems, with their canals (human-made or modified) and structures, as well as the subsurface water system. Focus will be on the hydraulics of the water infrastructure (flows), but issues of water quality, ecology and morphology are included as well. Project: Water systems Link to the learning objectives of the module: B3: LO1, LO2, LO5 and LO8. Students learn how to connect the individual behaviors of infrastructural elements in a water system, how to model water systems, using available software, and how to define goals of selected (types of) systems in terms of desired water flows (like drainage-irrigation, flood-drought, urban-rural) by focusing on the following topics: Existing water systems are analyzed, using several sessions specifying quantification of behavior - allowing linking to the other sub themes. Exchanges between teams (peer reviews) and students working on different modelling approaches allow students to develop insight into model use and system behavior. Specific problems in selected water systems are addressed, and students propose measures relating explicitly to the other sub themes. All the B-modules in the Water Resources Engineering track will have a workshop on ethical responsibility of engineers. Groundwater modelling Link to learning objectives of the module: B3: LO2 and LO4. In this unit, students learn how to model groundwater systems, by quantifying groundwater flow and heads. Different model approaches are applied in exercises to assess the environmental impact of man-made changes in the geohydrological system. Specific topics include: The analytic element method and the finite difference method (including MODFLOW). Comparison of grid-based methods and analytical methods. Design of well fields including capture zones and wellhead protection Pumping test analysis Travel times Time series analysis of groundwater time series. Prediction of groundwater heads in wet and dry periods. Environmental (eco)hydrology Link to learning objectives of the module: B3: LO2 and LO3. This sub theme focuses on quantifying and measuring the regional water system in its dynamic surrounding environment. Hands-on measuring techniques will be applied in a field setting, where after students critically analyse and reflect on the collected data. Next to the focus on quantifying the fluxes, special attention will be paid on water quality, ecology and geomorphology. Specific topics include: Geomorphological processes Measuring e.g., precipitation, evaporation, interception, soil composition, infiltration, hydraulic conductivity, piezometry Applying river discharge measurements by different methods Observation of water quality, ecosystem characteristics and geomorphology Understand the influence of discharge characteristics on water quality, ecosystem characteristics and geomorphology Tracer hydrology	
Study Goals	B3-LO1: Define how a water systems shape relates to the function(s) of the water system, in terms of (surface and subsurface) water fluxes. (ILO: b, h, l) B3-LO2: Quantify the water fluxes in selected systems, including non-steady states and interactions between fluxes. (ILO: a, f) B3-LO3: Use of measurement methodologies for hydrology in its dynamic biotic and abiotic environment and critically interpret the obtained data. (ILO: a, e, g) B3-LO4: Develop mathematical models to explain groundwater flow and heads to predict environmental impact. (ILO: a, f, g) B3-LO5: Propose measures to realize desired functionalities in water systems, considering short- and long-term criteria. (ILO: c, d, h, l) B3-LO6: Conduct a project in a team and communicate results with relevant audience. (ILO: i, j, k)	
Education Method	Teaching and learning that encourages students to take responsibility for their learning process: Project: Water systems The sub-theme builds on a series of workshops, with focus areas per week, that relate to the project a team works on. Project teams meet at least once every week with their supervisor to discuss progress.	

Assessment	Groundwater modelling After the lectures, students will work in Jupyter notebooks to practice with the theory. In each notebook students will work on a model assignment. The notebooks are part of one's portfolio of the lecture series. Environmental (eco)hydrology The teaching method consists of a fieldwork, where students will get hands-on experience with carrying out hydrological fieldwork. The collected field data has to be screened and analysed. At the end of the day, one will present the results. Additionally, lectures on the theory of morphodynamics and ecology will be provided. During these lectures students will study scientific papers on the interaction of (a)biotic and hydrological systems, that will be discussed in class. Students complete the module successfully when passing the 3 summative assessments: Project: Water systems (40%): Assessment 1: Final exam testing quantification of water systems, with at least 30% of the questions related to assignments prepared during the course. Assessment 2: Team report, including details on individual contributions, plus reflection per team member on personal skills: teamwork, reporting, communication. Groundwater modelling (40%): The weekly assignments that are part of the student's portfolio are graded. Environmental (eco)hydrology (20%): Group Assignment.
Academic Skills	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.
Literature & Study Materials	Analytical thinking; application of theory to new situations using the relevant equations; making inferences of physical situations by analysing observational data; reading and discussing journal papers; Ethical reflection/consideration. Project: Water systems Building on the study material from the Stepping Stone I and II Modules (which cover water design formula for example), additional study material is provided in the shape of guided assignments, with a problem description plus two or three papers relevant for the topic at hand. Per project, a starting kit will be collected, including a project description, key data, three relevant papers, and a list of stakeholders. During the project work, the teams collect additional material, in close contact with the supervisor.
Judgement	Groundwater modelling Weekly Jupyter Notebooks, with a model assignment, will become available. Environmental (eco)hydrology Part of field course syllabus CIE5471 Hydrological and Ecological Fieldwork in River Systems; List of scientific articles, book chapters, hand-outs and powerpoint presentations (see CIE5580 Ecology and Morphodynamics in Catchments) For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations

ENVM1603	Water Resources Engineering and Management	15
Responsible Instructor	Dr.ir. M.W. Ertsen	
Instructor	Dr.ir. M.W. Ertsen	
Instructor	Dr. E. Mostert	
Instructor	Dr. R.R.P. van Nooijen	
Instructor	Dr.ir. O.A.C. Hoes	
Instructor	Dr. S. Pande	
Instructor	Dr. E. Abraham	
Contact Hours / Week x/x/x/x	28/x/x/x	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This module focuses on water systems infrastructures of different types. Regional, rivers and subsurface water systems are included. The module combines analysing water systems with finding options for their (re)design and operation. Project: Water systems Link to the learning objectives of the module: B4: LO1, LO2 and LO7. Students learn how to connect the individual behaviour(s) of infrastructural elements in a water system, how to model water systems, using available software, and how to define goals of selected (types of) systems in terms of desired water flows (like drainage-irrigation, flood-drought, urban-rural) by focusing on the following topics: Existing water systems are analysed, using several sessions specifying quantification of behaviour - allowing linking to the other sub themes. Exchanges between teams (peer review) and students working on different modelling approaches allow students to develop insight into model use and system behaviour. Specific problems in selected water systems are addressed, and students propose measures relating explicitly to the other sub themes. Integrated Assessment of Water Systems Link to the learning objectives of the module: B4: LO3, LO4 and LO7. The changing climate and socioeconomic conditions will likely challenge the functioning of current water systems. Assessing such uncertainties requires understanding how the functioning of systems is valued in the short and long run. Such understanding allows assessing diverse water engineering options that are available at present. Based on the observation and predictions of effects of engineering solutions, this subtheme develops creative interventions for sustainable water futures (e.g., in context of SDGs) through an interdisciplinary approach for designing and implementing such solutions by addressing the following topics: Design and simulation of relevant climate and socioeconomic scenarios Economic and non-market valuation of system functioning in the short and long run based on the scenarios. Analysis and proposal of engineering solutions with functionalities in water systems that are climate resilient and economically viable. Control of Water Systems Link to the learning objectives of the module: B4: LO5, LO6 and LO7. In a rapidly changing world, where the demands of society approach or exceed the limits imposed by natural resources, engineering solutions need to make efficient use of resources and quickly adapt to change. This sub theme teaches how to derive indicators and design a control algorithm for a water system by focussing on the following topics: Rapid adaptation requiring indicators of the desired functioning of the system that can be calculated frequently and automatically. Semi-automated set of rules (algorithms) to steer indicators to their predefined reference values.	
Study Goals	All the B-modules in the Water Resources Engineering track will have a workshop on ethical responsibility of engineers. B4-LO1: Define how a water systems shape relates to the function(s) of the water system, in terms of (surface and subsurface) water fluxes. (ILO: a, b, c, e, f) B4-LO2: Quantify the water fluxes in selected systems, including non-steady states and interactions between fluxes. (ILO: a, b, c, d, e, f, g) B4-LO3: Assess engineering options for water systems in terms of societal benefits, environmental impact, livelihoods, and adaptive capacity. (ILO: a, c, f, h, l) B4-LO4: Assess an economic analysis of these engineering options in terms of current and future investments, costs and benefits. (ILO: a, c, f, h, l) B4-LO5: Propose measures to realize desired functionalities in water systems, considering short- and long-term criteria. (ILO: a, b, c, d, e, f, g, h, l) B4-LO6: Determine how a proposed set of measures to realize desired functionalities may be implemented. (ILO: b, d, e, g, h, l) B4-LO7: Design and implementation of a proposed set of measures to realize the desired system performance. (ILO: a, b, c, e, f)	
Education Method	Project: Water systems The sub-theme builds on a series of workshops, with focus areas per week, that relate to the project a team works on. Project teams meet at least once every week with their supervisor to discuss progress. Integrated Assessment of Water Systems Problem based learning, select group/individual assignments (multiple, linked to specific case studies/problems (e.g., investing in geothermal energy storage, water for food and environment quality nexus at basin scale), with assignments per session. The analysis builds on relevant MUDE elements and the EE programme module. Control of Water Systems Problem based learning, select group/individual assignments, supported by some specific lectures. Building on elements that have been covered in the MUDE, the sub theme introduces control issues and options, plus the relevant underlying mathematics	

Assessment	Students complete the module successfully when passing the 4 summative assessments, related to the sub-theme's as indicated below. Project: Water systems (40% of module grade): Assessment 1: Final exam testing quantification of water systems, with at least 30% of the questions related to assignments prepared during the course (20% of module grade). Assessment 2: Team report, including details on individual contributions (20% of module grade). Integrated Assessment of Water Systems (30% of module grade): Assessment 3: Written exam, with at least 30% of the grade related to assignments prepared during the course. Control of Water Systems (30% of module grade): Assessment 4: Part a) An open book supervised assignment with laptop in week 4 or 5 (depending on scheduling) (0.4 of grade), Part b) Written exam (0.6 of grade).
Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Assessment. Steady flow in surface water flows in canals and structures; basic aspects of dynamics in water systems; basic aspects of hydrology; basic aspects of societal aspects of water systems; mathematics
Connection to other modules	This module can be selected from each A module in the Water track of the MSc Environmental Engineering. This module can be selected from each A module in the tracks Hydraulic Engineering and Hydraulic&Offshore Structures of the MSc Civil Engineering.
Academic Skills	Problem analysis; Data application; Quantification; Integration; Writing; Communication; Teamwork; Ethical reflection/consideration.
Literature & Study Materials	Project: Water systems Building on the study material from the Stepping Stone I and II Modules (which cover water design formula for example), additional study material is provided in the shape of guided assignments, with a problem description plus two or three papers relevant for the topic at hand. Per project, a starting kit will be collected, including a project description, key data, three relevant papers, and a list of stakeholders. During the project work, the teams collect additional material, in close contact with the supervisor. Integrated Assessment of Water Systems Study material is provided in the shape of guided assignments, with a problem description plus two or three papers relevant for the topic at hand. As hydro-economic modelling is a key element of this module content, Roobavannan et al., 2020. Sustainability of agricultural basin development under uncertain future climate and economic conditions: socio-hydrological analysis. Ecological Economics, 174, 106665. (in press) is foreseen as study document. Further specific content depends on what is covered in the MUDE and EE programme module. Control of Water Systems Study material is provided in the shape of guided assignments, with a problem description plus two or three papers relevant for the topic at hand. One overarching study document is envisioned, but the decision to use an existing document (book) or compose a (more limited) sub-theme specific one will be made later, as the content depends on MUDE elements.
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	TBD
Collegerama	No

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Environmental Engineering

ENVE Track Resource & Waste Engineering (R&WE)

ENVM1100	Fundamentals of RW and E	15
Responsible Instructor	Dr. H.M. Jonkers	
Instructor	Dr.ir. M.C.M. Bakker	
Instructor	Dr. H.M. Jonkers	
Instructor	Prof.dr.ir. T.J. Heimovaara	
Instructor	Dr.ir. M. Ottele	
Instructor	Dr. M.W.N. Buxton	
Instructor	Prof.dr. P.J. Vardon	
Instructor	Dr. D.F.J. Schraven	
Instructor	Dr. B.M. van Breukelen	
Instructor	F.S. Desta	
Instructor	Dr. J. Gebert	
Instructor	Dr.ir. A.A.M. Dieudonné	
Instructor	Q.A.M. Reijtenbagh	
Instructor	S. Hiller Bernal	
Practical Coordinator	Dr. F. Di Maio	
Contact Hours / Week x/x/x/x	x/6/6/x	
Education Period	2 3	
Start Education	2	
Exam Period	2 3 4	
Course Language	English	
Course Contents	The stepping stone Fundamentals of Resource & Waste Engineering is compulsory for all students following the Resources and Waste Engineering (R&WE) track. This module provides an introduction to three scientific fields, which will be specifically addressed in the three respective themes: (1) Circularity concepts in resource and waste engineering, aiming for reduction of use of primary resources and emission of harmful components to the soil-, water-, and atmosphere compartments; (2) Environmental impacts of resource production and use and associated production of waste, with focus on primary and secondary resources typically produced for and used in the geo- and civil engineering building sector; (3) Biogeochemical processes occurring in the soil-, water-, and atmosphere compartments, with specific focus on how biogeochemical processes transform specific geo- and building sector related contaminants in these compartments and what the consequences are for the environmental quality of these compartments. Goal: The objective of the Resource & Waste Engineering stepping stone is to gain conceptual knowledge and understanding of environmental problems arising from geo- and civil engineering practices and how circular building and waste management strategies and (stimulation of) specific biogeochemical processes can mitigate these negative effects. The module content is organized in the following three themes: (1) Circularity in resource and waste engineering Link to learning objectives of the module: SS: LO1, LO2, LO3, LO5, LO6 and LO7. Central topic of this theme is the assessment of the theoretical background of resource efficiency principles and circular resource and waste concepts including value creation from waste and development of circular business models. The obtained knowledge will provide insight on how circular resource and waste management practices can be used to support the economy and achieve political goals for reducing use of primary resources and emissions of harmful components to the environment. This theme particularly focuses on the following topics: Concept, history, scope and reach of the circular (building) economy; General principles of circularity and resource efficiency; Implications and implementation of circularity and circular economy in the context of supply chains and supply; networks; Value creation from secondary resources; Sustainable and circular business models; Legislative framework addressing circularity and resource efficiency; Quantification of circularity performance through LCA methods. This theme builds further on prior knowledge obtained in the ENV programme module and provides theoretical background knowledge as a basis for further deepening and elaboration on specific circular resource and waste management techniques, cases and applications in subsequent R&WE track A and B modules. (2) Environmental impacts of resource utilisation Link to learning objectives of the module: SS: LO1, LO5 and LO7. Main aim of this theme is gaining understanding of the solid waste flows associated to mining- and civil engineering building practices and how these are linked to the production, use and end-of-life of primary and secondary resources and the environmental effects they cause in the soil-, water- and atmosphere compartments. This theme particularly focuses on the following subjects: Introduction to types, properties and quantities of the main construction- and mining-related solid waste sources and streams; Environmental effects (impacts) and hazard risks of construction- and mining-related solid waste sources and streams; Legislative frameworks and ethical aspects of construction- and mining-related waste; Solid waste recycling- and disposal options in relation to material and energy flows Introduction to technical routes for converting- and valorising waste; Evaluating environmental effects and hotspots occurring in the exploration-, extraction- and production stage of mineral resources; The theory of physical and chemical transport phenomena Quantitative assessment of source emission and its impact on objects; Environmental impacts associated to geothermal energy production; Introduction to (overview of) possible remediation measures of environmental impacts related to solid waste production and processing; Ethical aspects of resource utilization will also be discussed in an interactive lecture organized together with the Faculty of Technology, Policy and Management. This theme builds further on prior knowledge obtained in the ENV programme module and provides theoretical background	
Course Contents Continuation		

	knowledge as a basis for further deepening and elaboration on remediation measures of environmental impacts of geo- and civil engineering related waste production and processing in subsequent R&WE track A and B modules. (3) Effect of biogeochemical processes on environmental quality Link to learning objectives of the module: SS: LO4 and LO7. This theme builds further on the introduction to environmental quality provided in the Environmental Engineering core programme module. The main aim of this theme is gaining understanding of how biogeochemical processes occurring in soil, water and atmosphere are involved in the conversion of specific contaminants and what consequences are for the environmental quality of these three compartments. Theories are explained and calculations are done and applied to various cases. This theme particularly focuses on the following subjects: Composition and structure of soil Hydraulic and mechanical behaviour of soil Retention and reactivity of soils sorption of organics, ions, metals Transfer of gasses between soil, water and air Alkalinity, pH-buffering, and Acid / Base equilibria Dissolution and precipitation of minerals, calculation of activity coefficients Basics of chemical modelling (PHREEQC): speciation and water-mineral interaction Biogeochemical reactions (Gibbs free energy, Standard Potential, van t Hoff equation) Microbial conversion of organic and inorganic contaminants Basics of redox reactions including calculation of oxidation states and chemical oxygen demand plus introduction on the occurrence of the main redox processes in groundwater and lakes This theme provides theoretical background knowledge as a basis for further deepening and elaboration on contaminant remediation measures of geo- and civil engineering related waste production and processing in subsequent R&WE track A and B modules.
Study Goals	On the successful completion of this module, the students are able to: SS-LO1: Identify the possibilities and limitations of resource circularity and solid waste management options and have knowledge on (critical) resource and waste flows. (ILO: g) SS-LO2: Create value from waste, and specifically implement concepts that enable circular principles to be applied to resource and waste streams relevant for the geo- and civil engineering building sector. (ILO: l) SS-LO3: Recognise environmental challenges and economic opportunities in solid waste management and translate these into scientific research questions and/or engineering opportunities. (ILO: d) SS-LO4: Develop numerical models and explain and verify, using scientific methods, the underlying biogeochemical processes that affect the environmental quality of soil, water and air. (ILO: a, f) SS-LO5: Take into account the national and EU goals and regulations when developing engineering solutions for waste management options, including collection, separation, treatment, transport, re-use, recovery, recycling and final disposal. (ILO: h) SS-LO6: Quantify circularity and environmental performance of materials (resources and wastes) and building processes relevant for the geo- and civil engineering sector using Life Cycle Assessment (LCA) tools. (ILO: g) SS-LO7: Communicate the essential principles of resource and waste engineering both orally and in writing. (ILO: i)
Education Method	Form of instruction per theme is the same: Class lectures and workshops of in total 6 times 45 minutes per week for a total of up to 12 weeks (Q2 and Q3) will be the primary mode of instruction to deliver the theoretical concepts of this theme. Background information in form of research articles and chapters from textbooks will be provided to the students in digital form via the TU Delft library. A group case study assignment will be given to students comprising a literature study on a specific geo- or civil engineering related waste stream and its associated environmental effects, reporting of findings in a written report, and presentation of results to fellow students. The assignment needs to be carried out and finalized within the lecture period.
Assessment	Obtained theoretical knowledge will be assessed through a (written) digital (Ans) individual exam (50% of final grade) and literature review-, reporting- and presentation skills through an individual project portfolio (30% of final grade) and one group case study assignment (20% of final grade). An intermediate self-reflection and fellow student peer feedback on performance of individual students to the group process is part of the group case study assignment.
Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment. Knowledge obtained in preceding and parallel running Core Program Module ENVM1000 Principles of Environmental Engineering
Academic Skills	Analytical assessment, reporting- and communication skills are the core competencies that will be gained in this stepping stone module. In addition to individual assignments students will also gain experience in collaborating in groups and learn how to effectively communicate obtained results. Project planning, organizational management-, reporting- and presentation skills will also be gained at individual level.
Literature & Study Materials	Lecture slides, papers, and chapters from relevant literature will be provided via Brightspace. Some examples per theme include: (1) Circularity in resource and waste engineering Platform CB'23 - Measuring circularity (2019) Core method for measuring circularity in the construction sector MAJ Huijbregts et al (2017) ReCiPe 2016: a harmonized life cycle assessment impact method at midpoint and endpoint level. Int J Life Cycle Assess 22: 138-147 (2) Environmental impacts of resource utilisation Zhengfu Bian et al. (2012) The Challenges of Reusing Mining and Mineral-Processing Wastes. Science 337: 702-703 DM Franks et al. (2011) Sustainable development principles for the disposal of mining and mineral processing wastes. Resources Policy 36 (2): 114-122 G Lopez et al (2017) Thermochemical routes for the valorization of waste polyolefinic plastics to produce fuels and chemicals. A review. Renewable and Sustainable Energy Reviews 73: 346-368 (3) Effect of bio-geo-chemical processes on environmental quality Bertron (2014) Understanding interactions between cementitious materials and microorganisms: a key to sustainable and safe concrete structures in various contexts. Materials and Structures 47: 1787-1806 DA Nimick et al (2011) Diel biogeochemical processes and their effect on the aqueous chemistry of streams: A review. Chemical Geology 283: 317 K Soetaert et al (2007) The effect of biogeochemical processes on pH. Marine Chemistry 105: 30-51
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (R&GE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Dictionary
Collegegrama	Yes

ENVM1200	Resource Engineering	9
Responsible Instructor	A. Varveri	
Instructor	Dr.ir. M.C.M. Bakker	
Instructor	Dr. F. Di Maio	
Instructor	Prof.dr.ir. J.W.G. van de Kuilen	
Instructor	Dr. H.M. Jonkers	
Instructor	Dr.ir. M. Ottele	
Instructor	A. Varveri	
Instructor	Prof. M. Veljkovic	
Instructor	Dr. J. Gebert	
Instructor	A.T. Gebremariam	
Instructor	Dr. Y. Mosleh	
Instructor	A. Jagadeesh	
Instructor	S. Hiller Bernal	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The R&WE track will start with only one module A. This module A on Resource Engineering is therefore compulsory for all students following the Resources and Waste Engineering (R&WE) track. The module builds further on the topics addressed in the R&WE stepping stone and gives students an introduction to the flows of resources and wastes, as well as to the types and properties of waste that result from human activity. Students will learn about using methodologies to quantify changes in resource demand and evaluate these in terms of their environmental implications and resource availability (critical materials). The module will further explore the issues and variables that play a role in using secondary resources more efficiently, and in closed loops. In this context, the limitations for circularity and the role of bio-based solutions (renewable and bio-based materials) in achieving circularity will also be addressed. Module A offers an engineering perspective to address key aspects related to circular production and construction processes and introduces a range of engineering tools to assess environmental performance. Goal: The objective of the Resource Engineering module is to gain in-depth knowledge of managing material flows and waste streams in closed loops and learn how circular and bio-based economy approaches can be implemented in production and construction processes, thus contributing to increasing the use of renewable and recyclable resources while minimizing the consumption of energy and raw materials. The module content is organized in the following two themes: (1) Flows of materials, resources and residual wastes (Link to learning objectives of the module: A-LO1, A-LO2, A-LO3, A-LO7 and A-LO8) Central topic of this theme is the analysis and management of material flows and waste based on circularity and bio-economy knowledge, with the aim of reducing environmental impacts. (2) Circular construction processes and sustainable materials (Link to learning objectives of the module: A-LO3, A-LO4, A-LO5, A-LO6 and A-LO7) Central topic of the theme is the application of circularity and bio-economy concepts on construction processes and materials and their evaluation based on tools for the quantification of the environmental performance at system and material levels.	
Study Goals	Upon successful completion of this module, the students are able to: A-LO1: Explain principles, methodologies and concepts related to waste management, circularity and construction materials. (ILO: a) A-LO2: Evaluate waste management options, considering the possibilities and limitations of resource circularity. (ILO: a, c, d, l) A-LO3: Assess technologies for separation, sorting of and resource recovery from solid waste by implementing concepts that enable material circularity. (ILO: a) A-LO4: Evaluate the state-of-the-art circular and bio-based economy approaches and methodologies for production and construction processes. (ILO: c, i, l) A-LO5: Evaluate the environmental impacts and performance of production and construction activities by applying both quantitative and qualitative methodologies. (ILO: b, d, g) A-LO6: Develop engineering solutions for implementing circularity and bio-based concepts in production and construction processes based on environmental, economic, societal and legislative grounds. (ILO: c, d, h, l) A-LO7: Demonstrate the capacity to effectively communicate (both written and oral) and share results and opinions with peers, engineers, and academics. (ILO: i) A-LO8: Provide constructive feedback, managing conflicts in diverse teams, and tackle together challenges in implementing circularity and bio-based economy. (ILO: j, k).	
Education Method	(1) Flows of materials, resources and residual wastes - Interactive lectures (on-campus or online) will be the primary mode of instruction to convey the theoretical concepts of the module. - Research articles from the literature and chapters from several handbooks available in digital form in the TU Delft library will be provided to the students as background material. - An assignment will be given to students in groups (of 2 persons) to formulate solutions for management of waste (household, industrial, agricultural, etc.) and allow students to learn from each other. The assignments need to be carried out during the lecture period. (2) Circular construction processes and sustainable materials - Interactive lectures (on-campus or online) will be given to teach the theory part and provide specific knowledge and discuss the tools and methodologies for determining and improving circularity and sustainability in production and construction processes. - A real-life case study (project) will be presented to the students (in groups of 2-3 persons) early in the theme. The aim of the group project is to quantify and evaluate the environmental performance of a product/process/structure and propose engineering	

Assessment	solutions to improve it. To successfully complete the module, students must pass the three (3) summative assessments as described below: (1) Written exam The knowledge of theory and problem-solving skills for A1 theme are tested by a written exam (20%). (2) Group assignment - Theme A1 The analysis skills are tested through a group assignment (20%). (3) Project portfolio - Theme A2 The design, writing and presentation skills are assessed through a group project (60%). In this module, the group processes are also assessed. Students are asked to individually provide a reflection about their contribution and the individual contributions of their group members (group dynamics). A successful completion of the module requires a minimum grade of 5.8 per exam/assessment.
Tags	Design Group work Project Small groups Sustainability
Connection to other modules	Module A consolidates and builds on prior knowledge obtained in the ENV programme module and R&WE stepping stone themes, and prepares for more advanced topics in the R&WE B-modules.
Academic Skills	Collection and synthesis of information, critical thinking, analytical thinking and problem solving are the main academic skills which will be trained. In addition, students will need to collaborate with their colleagues, produce a group report and present their work, so organizational, interpersonal, writing and presentation skills will be developed as well.
Literature & Study Materials	Lecture slides, papers, and chapters from relevant literature will be provided via Brightspace. Some examples include: (1) Flows of materials, resources and residual wastes M. Agnoletti, S. N. Serner (editors), Ch7: History of Waste Management and the Social and Cultural Representations of Waste, The Basic Environmental History, 2014 G. Tchobanoglous, H. Theisen, S. Vigil, Integrated Solid Waste Management, McGraw-Hill. (2) Circular construction processes and sustainable materials Sustainable steel Buildings, ed. B.Hauke, M. Kuhnhenne, M. Lawson, M.Veljkovic, 2016, p.360, Willey. Mantalovas, K.; Di Mino, G.; Jimenez Del Barco Carrion, A.; Keijzer, E.; Kalman, B.; Parry, T.; Lo Presti, D. European National Road Authorities and Circular Economy: An Insight into Their Approaches. Sustainability 2020, 12, 7160. F. Di Maio et al, Measuring resource efficiency and circular economy: A market value approach, Res, Cons. & Rec, 2017 (122), pp. 163171. R., U., Ayres, Economic growth: politically necessary but not environmentally friendly, Ecological Economics 15 (1995) 97-99 United Nations Environment Programme, 2014. Sand, rarer than one thinks. Environ. Dev. 11, 208218. Gebremariam et al., Comprehensive study on the most sustainable concrete design made of recycled concrete, glass, and mineral wool from C&D wastes; Construction and Building Materials.
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations

ENVM1300	Waste Processing Technologies	15
Responsible Instructor	Dr. F. Di Maio	
Instructor	Dr.ir. M.C.M. Bakker	
Instructor	Dr. P.C. Rem	
Instructor	Dr. Y. Yang	
Instructor	Dr. H.M. Jonkers	
Instructor	A. Varveri	
Instructor	Dr. M.W.N. Buxton	
Instructor	Dr. D.F.J. Schraven	
Instructor	Dr. M. Soleymani Shishvan	
Instructor	Dr. J. Gebert	
Instructor	A.T. Gebremariam	
Contact Hours / Week x/x/x/x	x/x/x/5-10 (exact number of contact hours is not clear, but ranges between 5 and 10 hours)	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Module B1 focuses on routes and technologies that produce high-quality raw materials from different solid waste sources with high economic and societal impacts and a degree of urgency. Examples are household waste, WEEE, plastics, metals, industrial wastes, incineration residues, and tailings from mines. Routes are designed to uphold and further develop the circular economy's generic principles and concepts as taught in the stepping stone and Module A: i.e., effective solutions for recovery of materials from End-of-Life constructions, design out harmful residues and emissions to water, air, and soil, and prevent the depletion of non-renewable resources. This module will further explore the issues and variables that play a role in preparing and using secondary resources more efficiently, safeguarding the environment, and in closed loops. In this context, the limitations of circularity will also be addressed. The student will be challenged to apply the knowledge and skills provided by the expert teaching teams in a final project of design and analysis (B1-3), in compliance with regulations and discussing ethical aspects. This module also addresses the development of soft skills such as presentation techniques, professional reporting, team working, and efficient communication with internal and external experts. Goal: Gain knowledge and skills to understand materials and waste flows and to develop technologies for recycling, resource recovery, waste separation, characterisation, and disposal in the context of the circular economy. The module content is organised in the following three themes: (1) Resources recovery from End-of-Life constructions Link to learning objectives of the module: B1: LO1, LO2, LO3, LO5, LO8, and LO10. With a focus on the environment and the economy, this theme offers an engineering perspective to address critical aspects related to circular production and construction processes, advanced technologies for recycling, and recovery of resources from construction and demolition waste in a circular economy. The following topics will be addressed: - Regulations - Processes and procedures - Material labels - Waste flows - Dismantling strategy - Demolition techniques - Collection strategies - Primary and secondary raw materials - Innovative technological solutions that foster circularity - Routes to market - Environmental aspects (2) Waste collection and separation techniques Link to learning objectives of the module: B1: LO3, LO4, LO5, LO6, LO7 and LO8. Knowledge of solid waste separation principles and techniques is key to efficient recycling and improving the sustainability of material resources. This theme provides detailed knowledge about the flows of resources and wastes resulting from human activities. This theme will further explore the issues and variables that play a role in using secondary resources more efficiently and in closed loops. In this context, the practical limitations of circularity will also be addressed. The theme concentrates on the different steps and processes comprised in the solid waste management chain, from waste collection to treatment, recycling, and disposal. Taking into account the issues of sustainability and circularity, the theme provides knowledge on the processes costs, the quality, and the value of the secondary materials resulting from the different processes/treatments discussed. The following topics will be addressed: - Collection methods - Physical waste properties - Particle segregation and agglomeration - Sampling theory - Pre-treatments - Pre-processing methods - Performance-based modelling - Separation principles - Separation technologies - Ore - Tailings - Household waste - Waste incineration and Rankin cycle - Optimizing separation lines for value or recovery - Sensor feeding techniques - Sensor principles - Sensor applications - Detection-limited sensor data	
Course Contents Continuation		

(3) Recovery and quality of raw materials from waste
 Link to learning objectives of the module:
 B1: LO2, LO9, LO10 and LO12.

A challenging design and analysis exercise in student groups, spanning the route from waste source to extraction and further purification and/or refinement to high-quality raw materials.
 Specific knowledge about mining operations, minerals processing, metallurgical refinement techniques, and final solutions for harmful residues (e.g., landfilling techniques) can be provided in a series of starting lectures or workshops, depending on the specified design topics. The following topics will be addressed:

Waste sources
 Volumes and capacity
 Conditions and limitations
 Design of separation and treatment strategy and technologies
 Quality of raw materials
 Purification
 Refinement
 Treatments
 Residues
 Emissions
 Energy consumption
 Ethics
 Ecological and societal impacts.

Quantification and discussion of environmental impacts in connection to regulations on residues and emissions are part of the evaluation of the technical design.

Moreover, students are challenged to identify and discuss possible ethical implications of their design in a dedicated workshop organised in collaboration with the Faculty of Technology, Policy, and Management.

Study Goals

On completion of this module, the students are able to:

B1-LO1: Recognise logistical procedures and key technologies that safeguard the recovery and quality of recyclable materials and reusable objects. (ILO: d, i)

B1-LO2: To take account of the main Dutch legislation and EU directives aimed at the circularity of materials from waste. (ILO: h, l)

B1-LO3: Make a considered choice for an efficient waste collection method. (ILO: c, d, l).

B1-LO4: Characterise solid waste heaps and streams (e.g. demolition waste/ore). (ILO: a, b, e)

B1-LO5: Match a pre-treatment to meet the requirements for optimum separation and quality of the raw materials. (ILO: a, d)

B1-LO6: Recognise appropriate design steps leading from the physical principle to an operational separation technology. (ILO: a, f, g)

B1-LO7: Make a considered choice for a separation technique or quality inspection technique. (ILO: a, g)

B1-LO8: Design and assess a waste/ore processing line in terms of recovery and quality or added value. (ILO: a, g)

B1-LO9: Implement efficient treatment steps that minimise solid residues and harmful emissions. (ILO: h, l)

B1-LO10: Quantify the non-circular impacts of a treatment option or process. (ILO: g, l)

B1-LO11: Take into account the main legislation concerning special wastes, residues, and emissions. (ILO: h)

B1-LO12: Design efficient treatment and processing steps that safeguard the recovery and quality of raw materials. (ILO: i, j, k)

Education Method

(1) Resources recovery from End-of-Life constructions

Physical lectures

Workshops followed by work in groups

Self-study

Students will be asked to prepare for the lectures/workshops, studying reading materials and/or scientific articles prepared for the lectures.

Following a thematic workshop, students will work in groups to produce a short report where, as an example, the pros and cons of a circular strategy are discussed, compared, and assessed quantitatively.

Background reading material is provided, which is sourced from research articles and book chapters from the scientific literature.

(2) Waste collection and separation techniques

Physical lectures

Group assignments

Lab practical work in groups

(3) Recovery and quality of raw materials from waste

Physical lectures on special topics

Regular workshops and feedback sessions.

Assessment

(1) Resources recovery from End-of-Life constructions

Group assignment (15%): Group assignment: final report (10%) and oral presentation (5%). During the module, a peer review is conducted where each student submits feedback about his/her personal experiences on the level of collaboration and communication in the group.

(2) Waste collection and separation techniques

Group assignment: This assignment will involve experimental work in the laboratory (15%)

Written exam (20%)

(3) Recovery and quality of raw materials from waste

Group project (50%): final report (40%) and oral presentation (10%). During the module, a peer review is conducted where each student submits feedback about his/her personal experiences on the level of collaboration and communication in the group.

A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment.

Expected prior Knowledge

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Academic Skills

Being an interdisciplinary module, flexible and yet critical thinking, reflective functioning, and maintaining a collaborative and yet problem-solving attitude are the main targeted academic skills.

In addition, students will need to communicate and collaborate with teachers and technical assistants, which stimulates their organizational and interpersonal skills.

Literature & Study Materials	Hand-outs, articles/book chapters (via Brightspace), and open-source online material. (1) Resources recovery from End-of-Life constructions The European Green Deal, COM(2019) 640 final. A., Akhtar, A., K., Sarmah, Construction and demolition waste generation and properties of recycled aggregate concrete: A global perspective, Journal of Cleaner Production 186 (2018) 262-281 AT Gebremariam et al., Innovative technologies for recycling End-of-Life concrete waste in the built environment, Cons and Rec 2020, (163), 104911 Gálvez-Martos et al., Construction and demolition waste best management practice in Europe, Res, Cons and Rec 2018, (136), pp 166-178 (2) Waste collection and separation techniques Hand-outs, articles or book chapters via Brightspace and open-source material through the Internet. M.C.M. Bakker, Challenges in Advanced Solid Waste Separation, 2019, http://resolver.tudelft.nl/uuid:64a13fd4-edfe-4610-b20f-491278908281 EU Commission, Assessment of separate collection schemes in the 28 capitals of the EU, 070201/ENV/2014/691401/SFRA/A2, Final report 2015. G. Tchobanoglous, H. Theisen, S. Vigil, Integrated Solid Waste Management, McGraw-Hill. (3) Recovery and quality of raw materials from waste Hand-outs, scientific articles/books chapters (via Brightspace) and open-source online material; which will be tailored to the specific group topics.
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	The B1-2 exam will be a limited open-book exam. No laptop will be permitted.
Collegerama	No

ENVM1301	Reactive Resources and Wastes	15
Responsible Instructor	Dr. J. Gebert	
Instructor	Prof.dr.ir. M.K. de Kreuk	
Instructor	Prof.dr.ir. T.J. Heimovaara	
Instructor	Prof.dr.ir. J.B. van Lier	
Instructor	Dr. M.W.N. Buxton	
Instructor	Dr.ir. R.E.F. Lindeboom	
Instructor	Dr. J. Gebert	
Instructor	E. Maio	
Instructor	A. Soleymani	
Instructor	Dr. S.C. Buisma-Yi	
Contact Hours / Week x/x/x/x	x/x/x/10-16 (exact number differs over the weeks but is between 10 and 16)	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This module addresses the processes in different types of reactive resources/wastes and their possible impact on the environment. These vary in relation to the different ways of material treatment, use or storage and disposal and will be evaluated for the different possible impact pathways (water path, gas path). On this basis, the potential of wastes to serve as secondary resources will be developed. Furthermore, this module will teach engineering applications to avoid and minimise environmental impacts during waste and resource storage, treatment, use and disposal, and to remediate and stabilise affected sites. The pertinent legislation will be taught as a continuous topic spanning all themes. The module content is organized in the following themes: (1) Environmental impact of reactive resources and wastes Link to learning objectives of the module: B2: LO1, LO2 and LO5. The following topics will be addressed: Water path: Leachate quality, transport of water through wastes and modelling thereof, leachate and groundwater monitoring, wastewater treatment technologies. Gas path: Gas production and modelling thereof; transport of gas through wastes and emissions into the atmosphere, gas concentration and emissions monitoring and treatment technologies. Greenhouse gas accounting and clean development mechanisms (CDM). Source-Pathway-Receptor-concept. Sorption, dispersion, transport. Point-of-compliance-point-of-emission concept. Acceptable emissions. Regulations on protection of soil, groundwater, surface water and atmosphere. (2) Waste as a resource Link to learning objectives of the module: B2: LO3 and LO5. The following topics will be addressed: Valorisation of wastes, recovery of raw materials from wastes. Concepts of waste-to-resource and waste-to-energy. End-of-waste criteria and procedures. Resource potential of specific waste streams: A- Biowastes: Anaerobic digestion B- Contaminated soils: Soil separation, treatment and remediation technologies (biological, chemical, physical). Re-use of soil. C- Sediments as a resource, sediments as a waste: Sediment-to-Soil ripening processes. Beneficial use of sediments. Sediment disposal. D- Mining residues: Properties, hazards, use potential. 'Hidden flows'. E- Construction and demolition wastes, bottom and other ashes. Regulations on re-use and recovery of wastes. (3) Final sinks Link to learning objectives of the module: B2: LO4 and LO5. The following topics will be addressed: Waste separation and treatment technologies Principles of sanitary landfill engineering including sealing, leachate collection and gas extraction systems Landfill processes and monitoring of emissions and deformation behaviour Landfill aftercare, end of aftercare and in situ waste stabilisation technologies Waste incineration Legacy sites Regulations concerning operation, monitoring and aftercare of sanitary and liability for legacy landfills.	
Study Goals	On completion of this module, the students are able to: B2-LO1: Explain the fundamental principles that guide contaminant mobilisation and emission from different types of waste. B2-LO2: Quantify contaminant emissions from waste via the water and gas/atmospheric pathway, analyse their impact on human health and the environment and conceptually design treatment options. B2-LO3: Apply end-of-waste criteria on potential secondary resources and analyse opportunities for and limitations to recovery, reuse and recycling of contaminated reactive materials. B2-LO4: Assess options and conceptually design develop stabilization and monitoring strategies for long-term safe use of legacy sites, sanitary landfills and mine tailing sites. B2-LO5: Understand current European waste-related legislation and apply the management options for reactive wastes as regulated therein.	
Education Method	Interactive lectures on theme-specific content Theme-specific self-study using scientific articles, reports or book chapters including presentation thereof to the class	

	Group project intersecting all three themes: case study As part of the case study, a workshop will be organized in collaboration with the Faculty of Technology, Policy and Management, in which students will discuss the ethical implications of the engineering solutions for the management of reactive resources and wastes. Field group work intersecting all three themes, including field measurements Excursion intersecting all three themes, e.g. to a waste sorting or treatment plant, wastewater treatment plant, landfill, incinerator, soil remediation project, dredged material treatment site.
Assessment	For the successful completion of the module, students have to pass the following assessments: Exam at module level (50%) Group project (30%) intersecting themes 1 to 3: technical report (25%) and final oral presentation (5%) Field group work, intersecting themes 1 to 3 (20%): report in the format of a scientific publication An intermediate self-reflection and fellow student peer feedback on performance of individual students to the group process is part of the group case study assignment.
Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment. Contents taught in the preceding modules. High school level knowledge of chemistry and biological conversion processes. Basics of advective and diffusive flow.
Academic Skills	Reading of scientific literature, self-responsible and inquiry-based information collection and evaluation, planning and execution of a project, time management and anticipation, scientific writing and presenting, critical thinking and teamwork.
Literature & Study Materials	Textbook chapters, scientific papers, technical reports and lecture slides will be provided via Brightspace. Examples include: Christensen, T. (ed) (2010): Solid Waste Technology & Management, 1 & 2. Wiley, DOI:10.1002/9780470666883 Abdel-Mohsen & Paleologos (2017): Fundamentals of Geoenvironmental Engineering. Elsevier, ISBN: 9780128048306 Blume et al. (2016): Scheffer/Schachtschabel Soil Science. DOI:10.1007/978-3-642-30942 Hill, M. (2020): Understanding Environmental Pollution (4th ed.). Cambridge: Cambridge University Press. DOI:10.1017/9781108395021 Fitts, C. (2012): Groundwater Science. Academic Press, ISBN: 9780123847058 Appelo & Postma (2005): Geochemistry, Groundwater and Pollution, Second Edition. Taylor & Francis. ISBN: 978-0-415-36421-8 Hou, D. (2019): Sustainable Remediation of Contaminated Soil and Groundwater. Elsevier, ISBN: 9780128179826.
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Open book, all materials except electronic devices which are connected to the internet.
Collegerama	No

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Environmental Engineering

ENVE Year 2 2023

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Environmental Engineering

ENVE Joint Interdisciplinary Project

TUD4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Project Coordinator	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Summary	JIP consists of three sets of activities: 1. The project which takes place at a company. Students are responsible for the project work (collaboration, integration of different perspectives, the content quality of their work). They plan their activities via a scrum method, are challenged to realise field trips, consult with experts and realise the necessary research work. The team keeps a Scrum log, run a blog for interaction with the outside world and every team member keeps a personal development log. 2. Meeting Professionals lectures and workshops about specialised topics, by academic staff or senior professionals from the companies involved. 3. Plenary meetings and design reviews for all the teams, company coaches and academic staff. At the meetings the students present their intermediate resp. final outcomes and plans, and are provided with feedback.	
Course Contents	The aim of the Joint Interdisciplinary Project is to prepare students to contribute to solving impactful technological challenges. The projects not only demand good engineering working knowledge but also experience with interdisciplinary and systems theory, and both knowledge and mindsets of innovation and entrepreneurial behavior. The project brief is provided by renowned companies like Airbus, Arcadis, etc. Teams of interdisciplinary student teams guided by a company coach and offered academic and industry expertise, are invited to realise an innovative problem solution to a complex problem and contributing to the sustainable development goals.	
Study Goals	1. Cognitive abilities attributable to interdisciplinary learning Demonstrate the ability to engage in perspective-taking; Develop structural knowledge pertaining to the problem; Integrate knowledge and modes of thinking drawn from two or more disciplines; Produce an interdisciplinary understanding of complex problem or intellectual question. 2. Scientific and intellectual development Capable to analyze scientific and societal consequences (economic, social, cultural, environmental) of the innovation; 3. Research and design capabilities Demonstrate engineering skills: technical skills, interpreting results, creativity, usability for company/institute; Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; 4. Collaboration and communication in an interdisciplinary team Demonstrate behavioral competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to respect different disciplines and adapt to different cultures); Show ability to write a technical report: structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (30 pages); Present work performed in a structured way through an oral presentation to their peers and customer. 5. Self-adjustment and reflection capabilities Plan and control the project efficiently considering resources and methodology; Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career. Cognitive abilities attributable to interdisciplinary learning: The ability to integrate (scientific and practical technological) knowledge from different disciplines to solve complex problems Scientific and intellectual development The capacity to evaluate the ethical, scientific and societal consequences of the proposed innovation Research and design capabilities The ability to create reasonable and relevant research or design , according to the academic standards of the involved disciplines Collaboration and communication in an interdisciplinary team Demonstrate behavioural competences and skills relevant for teamwork and effective communication with different stakeholders Self-adjustment and reflection capabilities To carry out regular reflections on professional and personal development and being able to improve upon those reflections Understand contemporary and societal issues in their work.	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	The assessment criteria pertain to the process, product and presentation and are assessed at an individual and team level, during 3 review sessions at three levels of accomplishment (each review assesses at a different level): Interdisciplinary work Scientific reasoning and ethical mindset Innovation process Presentation and communication Academic staff, supported by the input from company coaches, grade the students during the Problem Statement, the Midterm and the Final Review. The final group mark for JIP is differentiated per person and is based on: The team presentations - 30% The Problem definition, progress report and final report 30% The individual contribution to the team 20% The final individual reflection report 5% The Blog 5%	
Remarks	Project Outcomes (product, model, system) are presented at: 1. Problem Statement Review 2. Midterm meeting 3. Final Review 4. Symposium presentation	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Environmental Engineering

ENVE Multidisciplinary Project

CEGM3000	Multidisciplinary Project (MDP)	15
Responsible Instructor	Dr.ir. M.M. Rutten	
Module Manager	Dr.ir. M.M. Rutten	
Instructor	Y. de las Heras	
Contact Hours / Week x/x/x/x	N/A	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	English	
Course Contents	In the Multidisciplinary Project (MDP) students investigate a problem for a client in a team that consists of students from multiple tracks within the Faculty of Civil Engineering and Geosciences and develop a solution strategy or other advice. The client can be external or faculty staff. Students from other faculties can be involved, yet permission from the module coordinator is required for this. The faculty encourages students to perform a MDP abroad and to acquire a project themselves by reaching out to external organisations. Project topics are also offered via the online learning platform Brightspace. After acquisition, the MDP consists of three phases: Phase 1: Preliminary investigation. Students start with a problem analysis resulting in a problem statement and an objective. Coupled to that, a project strategy will be formulated. This strategy includes which methods will be used, which contribution can different disciplines provide to the project, which steps have been passed through successively, which information is still necessary, where that information can be found etc. In addition, the organization of the group will be set up in this phase. This phase ends with a presentation of the project proposal to the client and supervisors. Phase 2: Design and/or research. During this phase students will execute the strategy developed under phase 1 will and adjust it if needed. The work may exist of research and/or of developing design alternatives or solution alternatives. Alternatives may be worked out for the complete problem or for sub-problems. Half-way this phase the progress is presented in a midterm presentation. Phase 3: Round-off. In the round-off, students deliver the project results to the client and supervisors. First, a draft report is discussed with the supervisor and client, whereupon the definite version is made. The participants evaluate the project, both regarding the content and regarding the project process. Also a public final presentation is given.	
Study Goals	After completion of this module the students are able: MDP-LO1: Communicate project plans, intermediate and final results with a client and other relevant stakeholders in written and oral form. (ILO: i) MDP-LO2: Initiate, devise and plan project, in order to meet requirements, set by a client. (ILO: j) MDP-LO3: Monitor and manage a project in a team considering the client and other stakeholders involved. (ILO: j) MDP-LO4: Analyze a complex problem in the field of civil engineering, environmental engineering and/or geosciences. (ILO: j) MDP-LO5: Select and apply appropriate research and design methods to address this complex problem. (ILO: j) MDP-LO6: Understand the effect of diversity in disciplines, talents, skills, characters and cultures on a group process and address this effectively in a group process. (ILO: k) MDP-LO7: Apply working methods that stimulate combining and integrating knowledge and methods from different disciplines. (ILO: k)	
Education Method	MDP starts with a kick-off in which the steps of the project and methods for complex problem analysis and multidisciplinary working will be introduced. During the project students have regularly discussions with their supervisors and client and peer review and intervision sessions with other groups. All these activities will be organized online to allow participation from abroad.	
Prerequisites	As of academic year 2023-24 Q2: students from must have obtained at least 45 ECTS. Mail the application form together with a study progress overview from each group member (all combined in one mail or one pdf) to studentoffice-esa-ceg@tudelft.nl. In the study progress overview EC's of parts of the modules can be used. This needs to be made clear by the student.	
Assessment	Formative assessment (feedback): Feedback from supervisors and client during presentations of project proposal and midterm. The draft report will be discussed with the group. Peer review on project proposals and draft reports will be organized. Summative assessment: Students complete the module successfully when passing the assessment that consists of: 1. Quality written report and oral presentation (10%) 2. Quality research / final design as presented in written and oral report (50%) 3. Group process and multi disciplinarity (40%).	
Special Information	As of academic year 2023-24: all students from cohort 22-23 and later performing a MDP will be rewarded with 15 ECTS. For cohort 21-22 and earlier it will be divided into 10 EC + 5 EC extracurricular. See TER/Annex of your cohort.	
Expected prior Knowledge	A large part of the first year of your program. See prerequisites.	
Academic Skills	During the project, students work on their skills regarding inquiry, collaboration, communication, integration of different perspectives, and reflection on personal functioning.	
Literature & Study Materials	Students will be offered handouts and web resources on problem analysis, working in diverse teams, project management, reporting a presenting. For problem analysis following material (or comparable resources) will be used together will dedicated material on Brightspace: https://ocw.tudelft.nl/courses/ngi-2/subjects/8-bonus-module-problem-demarcation-actor-analysis-skills-2/, (accessed 2-4-2021) For teamwork and project process following material (or comparable resources) will be used together will dedicated material on	

Brightspace:
 Schwartz, S. H. (2014). National culture as value orientations: Consequences of value differences and cultural distance. In Handbook of the Economics of Art and Culture (Vol. 2, pp. 547-586). Elsevier.
<https://www.belbin.com/about/belbin-team-roles>
<https://www.mbtionline.com/>
<https://scrumguides.org/>
<https://online-learning.tudelft.nl/project-management-skills/>

Students can use the resources below for report writing and presenting:
<http://eloquentscience.com/>
<https://faculty.washington.edu/heagerty/Courses/b572/public/StrunkWhite.pdf>
 Cambridge Academic English - An integrated skills course for EAP (Student's Book; C1 Advanced). Authors: Martin Hewings and Craig Thaine
 Publisher: Cambridge University Press, 2012
 ISBN: 978-0-521-16521-1

Judgement	N/A
Permitted Materials during Exam	N/A
Collegerama	No

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Environmental Engineering

ENVE Internship

CIEM0120	Research Internship	10
Responsible Instructor	Dr.ing. M.Z. Voorendt	
Module Manager	Dr.ing. M.Z. Voorendt	
Contact Hours / Week x/x/x/x	N/A	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	English	
Course Contents	A Research Internship (RI) is a civil engineering research or design project conducted in a professional environment. The aim of an RI is twofold. On the one hand, the RI is about formulating and conducting a research or design project directly related to some topical quest, raised by either an external party (such as private company, (non-)governmental body, scientific institute, pressure group, either in the Netherlands or abroad) or an internal party (research or advisory projects of any kind, for instance related to externally funded projects, private-public co-operation and so on). This basically means that the topic of an RI must be of evident societal, entrepreneurial or scientific relevance and that it needs to be conducted in the professional environment that raised the quest. On the other hand, the professional environment the concerned project is conducted in, must be reflected on. This means that students conducting an RI are required to evaluate the influence they experience from their working environment on their performance and deliverable. This has a broad notion. From an organisational perspective, this comprises at least supervision, guidance, availability of relevant information, technical and administrative facilities. Besides, the student is required to reflect on her/his role in and contribution to the organisation where the RI is conducted, as well as her/his personal development. It is important to note that conditions may apply, for instance regarding confidentiality, intellectual property and so on. It is the responsibility of the student to ensure that such conditions are agreed upon by all parties involved prior to the start of the project and in compliance with TU regulations (see TER).	
Study Goals	On completion of this course, the student will be able to Formulate a research or design question, aligned with the clients demands, as well as produce and conduct a plan of work evidently converging to an answer to that question; [ILO 7, 8, 9, 10] Integrate and apply academic knowledge together with engineering and design skills acquired in previously taken modules of the master programme and relevant previous education; [ILO 8, 10] Deliberate, communicate and report both progress and results with the client and other stakeholders; [ILO 9] Reflect on obtained results in regard to conclusiveness, relevance for the clients demands, ethical aspects; [ILO 7, 9, 10] Reflect on personal performance in the context of the clients organisation and in regard to personal development during the project. [ILO 7, 9, 10] Reflect on the formal and informal performance of the organization where the project has been conducted. [ILO 6, 9] The Research Internship, as it is about a research or design project conducted in a professional environment, links to practically all ILOs, but from a generic perspective it links primarily to ILOs 7-10.	
Education Method	Internship at an external or internal party	
Assessment	Formative assessment: Feedback is given by the university supervisor (1) on the project plan (including motivation) and (2) at least twice during the project, either in progress meetings or based on progress reports and (3) on the draft final report. Feedback is given also by the involved representative of the institution where the RI is conducted (the company supervisor). Summative assessment: - Engineering skills (25%) Application of theory, motivation of methods/techniques Experimental work, programming, calculations Ambition level, amount of work Usability of results for company - Academic approach (25%) Formulation of a goal and approach Argumentation Extension of knowledge and/or methods Achievement of the goal, reflection, judgement - Reflection component (20%) Reflection on personal objectives and future career Indication personal strengths, weaknesses and future personal development Reflection on the performance of the company Reflection on ethical and societal aspects of main project - Reporting (15%) Structure and consistency Language proficiency, style Use of literature, sources Lay-out, use of figures, tables, equations, etc. - Behavioural competences & skills (15%) Initiative, own contribution	

	Responsibility Communicative skills Independency For each of the above five criteria, a partial grade is set by the university supervisor (70%) and the company supervisor (30%). In the case of a substantial difference (more than 1 point on a 10 point scale), the responsible instructor is consulted. The final grade is the weighted average of the five partial grades. To pass, the rounded final grade must be at least 6.0 and all grades for the main five assessment categories are at least 5.0.
Expected prior Knowledge	MUDE, Programme Base and from the track the concerned student has selected the Track Base and Module A.
Academic Skills	The student has completed at least 45 EC's from the ISP, including partial results.
Literature & Study Materials	Analytical skills, report writing, project planning, communication
Judgement	Recommended: Elling et al (2011): Report Writing for Readers with Little Time. 1st edition, Noordhof Uitgevers.
	See Assessment
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	N/A
Collegerama	No

ENVM0121	Extension Research Internship	5
Responsible Instructor	Dr.ir. M.M. Rutten	
Module Manager	Dr.ir. M.M. Rutten	
Contact Hours / Week x/x/x/x	N/A	
Education Period	1 2 3	
Start Education	1 2 3 4 5	
Exam Period	1 2 3 4 5	
Course Language	English	
Course Contents	The CIEM0120 Research Internship (RI) is an engineering research or design project conducted in a professional environment. The topic of an RI must be of evident societal, entrepreneurial or scientific relevance and that it needs to be conducted in the professional environment that raised the quest. The professional environment the concerned project is conducted in, is reflected on. The elective course ENVM0121 acts as an extension of the base-module CIEM0120 Research Internship, for those students that would like to spend a whole quarter on the internship. As this is an extension of an existing course, with existing learning goals, the elective as such does not include additional learning goals.	
Study Goals	On completion of this elective, the student is able to: 1. Formulate a research or design question, aligned with the clients demands, as well as produce and conduct a plan of work evidently converging to an answer to that question; 2. Integrate and apply academic knowledge together with engineering and design skills acquired in previously taken modules of the master programme and relevant previous education.	
Education Method	Research activities within a professional environment. See CIEM0120	
Prerequisites	The student has completed at least 45 EC's from the ISP, including partial results.	
Assessment	See CIEM0120. The elective does not require additional, separate reporting on content. Assessment items need to include evidence/explanation of the implementation of the additional 5 credits. These additional credits are expected to be spent on additional data collection and/or analysis.	
Expected prior Knowledge	MUDE, Programme Base and from the track the concerned student has selected the Track Base and Module A.	
Academic Skills	Analytical skills, report writing, project planning, communication	
Literature & Study Materials	TBD	
Judgement	NA	
Permitted Materials during Exam	NA	
Collegerama	No	

ENVM3000	Internship Environmental Engineering	15
Responsible Instructor	Dr.ir. M.M. Rutten	
Module Manager	Dr.ir. M.M. Rutten	
Contact Hours / Week x/x/x/x	N/A	
Education Period	1 2 3 4	
Start Education	1 2 3 4 5	
Exam Period	1 2 3 4 5	
Course Language	English	
Course Contents	During the internship, the student learns how a professional organisation operates with its clients and other organisations and demonstrates the ability to work in a professional organisation at an academic level. It is essential that the internship is performed within a professional working environment. This can for example be a consultant, a contractor, a governmental board or department, a ministry or similar. An internship may consist of an assignment or project completely carried out by the student, a contribution to a large-scale project or an analysis of for instance a construction process. The goals of the company are leading in the project, be it that the learning objectives of the internship must be met and the activities are of sufficient academic level. The work of the student resides within the field of environmental engineering and the student is mainly supervised by the company.	
Study Goals	Please note the difference with the Research Internship (CIEM0120 and its extension ENVM0121). Upon completion of the Internship, the student is able to: 1. identify at least three personal learning objectives specific to the internship before the start of the internship, based on a critical reflection to what extent personal learning objectives are reached. (ILO 3, related to Portfolio ENVM) 2. explain the setting in which the organisation operates, adds value (in economic and societal terms), including the main function of the organisation in relation to the domain of environmental engineering, the type of clients which form the primary source of income, and the workflow within the organisation. (ILOs 8, 10, 12) 3. explain the personal position concerning the internship activities in relation to the skills, expertise and responsibilities of academic engineers in the organisation (ILOs 3, 4, 12) 4. explain the role in the professional team the internship was conducted within (e.g. personal responsibility and initiative, independency, ability to work together). (ILO 11) 5. reflect on the ethical and societal aspects and consequences of the main project(s) worked on during the internship, including the own role and actions to these aspects and consequences. (ILO 12) 6. apply the academic knowledge and engineering skills he has developed over the course of his BSc and MSc education in a professional setting in the field of environmental engineering. (ILOs 3, 4) 7. Present internship findings in concise written and oral formats (ILO 9)	
Education Method	See course content	
Practical Guide	Enrolment/application:	
Prerequisites	The registration of the course Internship Environmental Engineering is processed through the digital information system OnStage. All information regarding the registration of the internship can be found in the OnStage manual on Brightspace, as well as the general internship manual containing all further internship information. Due to faculty regulations, internship applications must be completed and approved by the internship office BEFORE the formal start of your internship. Otherwise, your internship will not be accepted and graded as part of your master education. The student has completed at least 45 EC's from the ISP, including partial results.	
Assessment	The internship is arranged with: 1. Kick of meeting with company supervisor and academic supervisor 2. Academic supervisor joins in an activity that the student performs at the company (e.g. presentation at team meeting, workshop etc) 3. Performance interview at the end of the internship based on the documents provided with company supervisor and academic supervisor The internship is assessed on the basis of a report consisting of three distinct sections: 1) a company reflection (30% of the final grade) 2) a personal reflection (40% of the final grade) and 3) a technical section (30% of the final grade). Ad 1. Company reflection (min. 4000 words (about five pages excl. figures). This part contains a description of the main function of the company and its position on the environmental engineering market. The role or concerns of other key stakeholders is addressed. The student describes the formal organisation of the company and provides an overview of the processes from acquisition of a project until the final product (project management). The student analyses strengths and weaknesses of the organization, in relation to the work done. The internal organisation of the department where the intern is working is analysed regarding cultural aspects or group roles (e.g., using the methods provided by Meyer, Hofstede and Belbin). The student is able to reflect on the ethical and societal aspects and consequences of the main project on during the internship, and relates the own role and actions to these aspects and consequences with focus on the position of the academic engineers and own position as an intern with a university background. Ad 2. Personal reflection (min. 4000 words (about five pages excl. figures). This part contains the students reflection on the extent to which the personal learning objectives have been met during the internship and the professional experience gained (knowledge and skills). The student is able to identify and reflect upon at least three personal learning objectives specific to the internship before the start of the internship. The student is able to critically reflect to what extent these personal learning objectives are reached and define new learning goals. The student is able to reflect on own aspirations to have a similar role in a similar company after graduation. Ad 3. Technical part (min. 20 pages, excl. appendices). This section contains a description of the project or assignment performed within the context of the internship (as defined in the internship proposal) and the technical results achieved during the internship period. The student shows the ability to apply the relevant academic knowledge and engineering skills from the ENVM program in a professional setting in the field of environmental engineering. In the case of major discrepancies between the findings of the internship supervisors, the responsible lecturer decides on the grading of the internship at hand.	

Expected prior Knowledge	First year modules in the ENVM program
Connection to other modules	During an internship knowledge gained in other courses/modules is applied.
Academic Skills	Academic thinking (originally, systematically, analytically, critically, reflectively) Academic skills (reading, writing, listening, speaking, planning)
Literature & Study Materials	On Brightspace: - Internship Manual - OnStage Manual - downloadable template for summary - internship offers
Judgement	N/A
Permitted Materials during Exam	N/A
Collegerama	No

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Environmental Engineering

ENVE Study abroad

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Environmental Engineering

ENVE C-modules

ENVM2100	Industry Water	5
Responsible Instructor	H.L.F.M. Spanjers	
Instructor	Dr. B.M. van Breukelen	
Contact Hours / Week x/x/x/x	Total: 28 contact hours/lectures + 32 hours excursion	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction into industrial classification, industrial symbiosis, water footprint, BAT reference documents; Water intensive industries, including mining, chemical, textile, steel, paper, food and drinks; Water intensive processes, including steam production and use, cleaning in place, cooling; Water and energy, including water-energy nexus, heat recovery; Water treatment technologies, including electrochemical, membrane assisted separation; Corrosion, scaling and fouling: interactions, prevention and mitigation; Safety, ethics and risk assessment; Modelling of corrosion and scaling; Water use and reuse in selected companies: analyzing real cases and proposing solutions. In this module a large variety of water qualities will be considered, ranging from ultra-pure water (e.g. makeup water for steam boilers) to concentrated residual water (e.g. brines from membrane filtration). Week number Week Activities Week 1 Introduction; Lecture assignment 1 Week 2 PHREEQC assignment introduction and setup; Lecture assignment 2 Week 3 Water intensive industries; Lecture assignment 3 Week 4 Water intensive processes; Lecture assignment 4 Week 5 Water and energy; Lecture assignment 5 Week 6 Water treatment technologies; Lecture assignment 6; Visit/assignment Company 1 Week 7 Corrosion, scaling and fouling; Lecture assignment 7; Visit/assignment Company 2 Week 8 Safety, ethics and risks; Visit/assignment Company 3 Week 9 PHREEQC group discussion; Visit/assignment Company 4 Week 10 Questions and answers	
Study Goals	Students will develop understanding of the importance of water use in the industry. They will gain knowledge of water intensive production processes and their related water qualities. The course will enable them to identify opportunities for water saving, reuse and resource recovery, and apply best available technologies. The student will be equipped with skills to take a unique position as a water professional in the industry, and will be able to integrate disciplines of industrial process engineering and water process engineering. Ultimately the student will care about industrial water technology for a sustainable society. After the course, the students will be able to: - describe water intensive production processes and their related water qualities; - identify opportunities for water saving, reuse and resource recovery, and to apply best available technologies (ILO1, ILO3, ILO4, IOL9); - translate interfering physical, chemical and biological processes into models, and use the models to analyze the interferences and to propose approaches to alleviate them (ILO1, ILO6);	
Education Method	Lecture preparation, interactive (guest) lectures, demonstrations, lecture assignments, visit to companies with assignment and interaction with company people, PHREEQC modelling assignment, group and individual reporting. Lecture assignments concern theory taught during the modules, company assignments are based on real case problems proposed by the companies, and PHREEQC assignments concern modelling studies of corrosion, scaling and fouling problems relevant in industrial water operations.	
Assessment	Assessment (Formative): Seven lecture assignments. Assessment (summative): PHREEQC report (group), 25%; Written exam (individual), 75%	
Remarks	Elective Environmental Engineering, Water Resources Engineering Resource and Waste Engineering, Chemical Engineering (prior consultation with responsible instructor required)	
Expected prior Knowledge	ENVM1600 Water treatment technologies	
Connection to other modules	ENVM1600 Water treatment technologies	
Academic Skills	Critical and analytical thinking, interpretation, writing reports, problem solving and cooperation. Excursions to industrial sites will encourage active reflection on own core skills and help develop academic professionalism, and create awareness of and reflection on and responsibility towards the social context and consequences of technology and scientific actions.	
Literature & Study Materials	Lecture slides; Collegerama; Quizzes and exercises; Syllabus; Company material; Papers, videos and other supporting materials.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Calculator	
Collegerama	Yes	

ENVM2102	Water and Health	5
Responsible Instructor	Prof.dr. G.J. Medema	
Instructor	Prof.dr.ir. D. van Halem	
Contact Hours / Week x/x/x/x	4 hrs per week, 28 hrs in total	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The contribution of civil and environmental engineers to the present excellent state of health in our society was and is enormous. Civil and environmental water engineers are also health engineers. Good design, operation and maintenance of water systems is key in health protection and central to the Sustainable Development Goals. The global burden of waterborne diseases is still very high and largely preventable! The disease burden is particularly high in low income countries, but also high income countries have water and health challenges. Look at the Flint drinking-water crisis (lead and Legionella), the increasing number of outbreaks of Legionella via cooling towers and hot water systems, increasing reuse, antibiotic resistance in the environment, water in the city etc. This course aims to empower you with knowledge to incorporate health-thinking in your engineering skills. Knowledge on human diseases related to water; pathogenic microbes, antimicrobial resistance and toxic chemicals and the diseases they cause, how these contaminants behave in the water environment and how they can be transmitted through water systems and cause disease. Many different health hazards, and you will learn to design a water system that is safe or evaluate the safety of an existing water system using a uniform health impact assessment approach. The WHO frameworks and guidelines on risk assessment of water and health are central to this course.	
Study Goals	Protection of public health is an important element of civil and environmental engineers. In this course, students get knowledge about how disease agents are transmitted via water systems and how this can be prevented. Students will be able to evaluate whether a water system is safe, design water systems that are safe, and After the course the students will: - be able to critically evaluate the health risks associated with engineered water systems (ILO 4); - understand agents that cause disease and how they can be transmitted via (engineered) water systems (ILO 2&8); - identify the importance of barriers in water systems to control disease agents (ILO 3); - assess how to design/operate water systems to protect public health and prevent health risks to occur (ILO 12); - describe the potential health risks associated with water systems in a systematic, quantitative manner using Health Impact Assessment and Quantitative Microbial Risk Assessment (ILO 7); - work in a team to recommend solutions for safe water systems (ILO 11); - report to stakeholder on risk management options for a safe water system on the basis of a scientifically justified quantitative assessment (ILO 9).	
Education Method	Lectures, discussion, exercises, practical work, group assignment, report, presentation	
Assessment	Assessment (Formative): Assessment is based on a group assignment to perform a quantitative Health Impact Assessment (Quantitative Microbial Risk Assessment or otherwise) of a selected exposure to pathogenic micro-organisms (or toxic chemicals) via a (engineered or non-engineered) water system. You obtain the knowledge on the methodology to conduct this assignment partly from the lectures, and must collect and critically evaluate information from the scientific literature or databases. The assignment projects must be executed in groups of 2 persons. At the end of the course, you and your fellow student have to present this assignment and submit your group report. In addition, individual group members have to submit a written recommendation section about their groups assignment. The report should contain 10.000 words (with illustrations, tables, literature references). Assessment (summative): Assignment, judged on three components: presentation (25%), report (50%) and individual recommendations (25%).	
Remarks	Elective, WRE, EE	
Expected prior Knowledge	BSc Civil or environmental engineering or equivalent	
Connection to other modules	ENVM1501 (safe infrastructure design); ENVM1601 (health issues and data from water infrastructure systems)	
Academic Skills	Concept development, model design, critical review of literature data, data analysis, uncertainty analysis, scientific report writing, scientific presentation	
Literature & Study Materials	Scientific literature and guidance documents provided	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

ENVM2104	Design Nature Based Solutions in Water Management Using Aquatic Ecology and Morphodynamic Principles	5
Responsible Instructor	Dr. T.A. Bogaard	
Contact Hours / Week x/x/x/x	1 day per week (8 hours)	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	In Water management nature Based Solution are promoted as a sustainable and cost-effective solution for a variety of water related problems in society such as prevention of droughts and floods, reducing heat stress and disaster risk reduction. This course will address the underlying ecological and morphological principles. The course will introduce and discuss key principles and apply those in various NBS in water management. More specifically: The course introduces ecological concepts related to water management. It will detail on aquatic ecosystems functioning for stagnant water, running water and wetlands. Points to cover will be: biological, physical and chemical aspects of aquatic ecosystems, trophic levels, characteristic organisms, population dynamics, natural and induced shifts in ecosystems and the aquatic ecological effects of climate change. The course will introduce hydro-geomorphological concepts and detail on especially water erosion, sedimentation and landslide mechanisms and the role of vegetation and animal activity on it in different environments (mountains, rivers, estuaries) and different climates (arid, tropical and humid). Feedback mechanisms between ecology and hydro-geomorphological processes will be discussed. Furthermore, the course will discuss and evaluate case-studies of Nature Based Solutions (NBS) in Water Management. Week number Week Activities Week 1 4 hrs interactive lectures, assignments and reading; 4 hrs workshops Week 2 4 hrs interactive lectures, assignments and reading; 4 hrs workshops Week 3 4 hrs interactive lectures, assignments and reading; 4 hrs workshops Week 4 4 hrs interactive lectures, assignments and reading; 4 hrs workshops Week 5 4 hrs interactive lectures, assignments and reading; 4 hrs workshops Week 6 4 hrs interactive lectures, assignments and reading; 4 hrs workshops Week 7 4 hrs interactive lectures, assignments and reading; 4 hrs workshops Week 8 4 hrs interactive lectures, assignments and reading; 4 hrs workshops Week 9 Poster presentations workshop Week 10 Exam	
Study Goals	After successfully finishing the course, the student is able to LO1 co-design nature based solutions in water management, including evaluating impacts of civil engineering interventions on geo- and ecosystems. (ILO 3) Hereto, the student : LO2 can apply knowledge about ecological concepts and ecological points of view to analyze ecosystems functioning. ILO 1-2) LO3 can apply knowledge about hydro-morphodynamical processes of the earth surface (erosion, landslides, sedimentation) to evaluate the interrelationships between the abiotic and biotic environment. (ILO 1-2) LO4 knows the jargon of the disciplines of ecology and geomorphology (ILO 9)	
Education Method	Interactive class room lectures, discussions, poster presentation and workshops	
Assessment	Assessment (Formative): Feedback during case study evaluations Feedback on assignments (poster presentation) Assessment (summative): Written exam on aquatic ecology and morphodynamic principles used in NBS in Water Management	
Remarks	Elective, all with proven background in hydrology, water resources, river engineering and similar	
Expected prior Knowledge	Sound understanding of water systems, hydrology, rivers	
Academic Skills	N/A	
Literature & Study Materials	Study material comprises of a list of scientific articles, book chapters, hand-outs including case studies	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

ENVM2105	Water Law & Organisation	5
Responsible Instructor	Dr. E. Mostert	
Contact Hours / Week x/x/x/x	+ - 6 per week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	How safe should a flood defences be? How clean should surface and groundwater be? Who is entitled to use scarce water? Who is responsible for flood risk management, water supply and water quality? Who should pay for water management? And what can someone do if they think the rules are not followed? These are just some of the questions that will be addressed in the course water law and organisation. In this course you will write a legal advice concerning a specific water management issue (e.g. diffuse pollution, major river works or drought management in a specific country or region). The central question to be addressed is whether the relevant law can and should be changed to manage this issue better. In the process, you will learn about the most important water laws, how they are made (and changed), how they are interpreted and applied, how they can affect the water management practice and how engineers could cope with this. Depending on the topic of the advice, it may be written either in English or in Dutch.	
Study Goals	After following this course, the student should be able to: 1. analyse the law that is relevant for a specific water management issue (ILO 2, 8) 2. identify the practical implications of the law for resolving or management this issue (ILO 8) 3. identify the need and possibilities for legal change (ILO 8) 4. write a proper, well-balanced and well referenced legal advice (ILO 3, 8, 9) To be able to this, the student should: 5. obtain basic information about the functioning of (water) law, including its sources, major regulations, how it is developed, interpreted and applied (or not), and the role of politics, the courts and stakeholders (ILO 8)	
Education Method	Lecturing In-class presentations, discussion and (peer) feedback Self study/ paper writing	
Assessment	Assessment (Formative):Feedback on proposal and draft advice Assessment (Summative):Paper (including short discussion about the paper)	
Remarks	Elective, all programmes.	
Expected prior Knowledge	None	
Connection to other modules	ENVM1000, ENVM1603 (no prerequisites)	
Academic Skills	Independent literature study, analysis, academic writing	
Literature & Study Materials	For Dutch language students: E. Mostert, Nederlands waterrecht van oud naar nieuw, TU Delft Open Textbooks, Delft 2022 For English language students: Fontaine, P.: Europe in 12 lessons, Publications Office of the European Union, and Mostert, E., Law and Politics in River Basin Management: The Implementation of the Water Framework Directive in The Netherlands, Water 2020, 12(12), 3367; https://doi.org/10.3390/w12123367	
Judgement	See assessment. A rubric for the paper will be published on Brightspace. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

ENVM2106	Engineering and Development	5
Responsible Instructor	Dr.ir. M.W. Ertsen	
Contact Hours / Week x/x/x/x	8 hours/week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Focusing on engineering in the Global South, this elective relates engineering applications (in global engineering) to a societal debate (what is (desirable) development). These themes span domains, programs and tracks within the CEG Faculty. Students define their expertise in relation to the themes and other students expertise. They learn about similarities and differences of techniques and methods from different domains. Building on the idea that engineering solutions useful and needed as they are are always positioned within a societal debate, students discover their own position in this arena. They are challenged to define and discover how their own expertise can be usefully mobilized for societal development. Given that engineers are looked at to propose solutions, students learn how to engage their own expertise within debates with other experts and other stakeholders. Students are engaged in a discussion-based format to learn and exchange on issues of development, political aspects of engineering, options/examples of interventions, comparisons between contexts, etcetera. Examples on differences and similarities in debates and positions on technological development influencing engineering solutions are explored. The exact setup of the sessions is decided on for each study year, as themes and topics are adapted to debates as they are encountered in relevant circles. These sessions are related to this list of key topics: Development theory (societal development, power relations, economic issues, globalization); Role of technology (engineering decisions, innovation ideas, history of technology); Co-creation and design interventions (participatory design, tinkering, values in design); Sustainable development goals (Differences and connections, development cooperation, role of states and UN institutes). Week activities: 2 sessions of 4 hours each, with associated reading, group discussions and individual reflection. As the actual themes are different each year. In general, the first weeks will be used to introduce a series of themes/topics, with the later weeks more focused on case studies and/or more complex debates.	
Study Goals	After completion of this module the students are able: 1. Explain their definition of and position towards societal development 2. Recognize existing approaches on the role of technology in society 3. Recognize how different stakeholders link engineering solutions to (ideas on) societal development 4. Relate their own expertise to other fields of expertise when developing technologies	
Education Method	In weekly sessions, with associated reading, group discussions and individual reflection, the first weeks will be used to introduce a series of themes/topics, with the later weeks more focused on case studies and/or more complex debates. Through workshops, with discussions, guest lectures, and design sessions, students are introduced to themes and applications in the Global Engineering domain. Workshops are followed by group discussions, after which individual students prepare further documentation on their own position(s).	
Assessment	Assessment (Formative): - Individual reports: reflection of the learning goals overall in relation to the session - Active participation in sessions (demonstrated in minutes and during sessions) Assessment (summative): - Individual essay on a theme that relates to the sessions and study material (consistency of the argument, references).	
Remarks	Elective Students from all MSc programmes can take this elective.	
Expected prior Knowledge	N/A	
Connection to other modules	This elective is also a part of the CrossOver Engineering for Global Development (CEGM2002)	
Academic Skills	Analysis of societal positions; reading texts; discussion skills; written and oral communication.	
Literature & Study Materials	The study material is collected each year, related to actual debates. Students read papers (about 5 to 7) and discuss these.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Environmental Engineering

ENVE D-Modules (Cross-overs)

CEGM2000	Suspensions, Sludges and Soils	10
Responsible Instructor	Dr. C. Chassagne	
Module Manager	Dr. C. Chassagne	
Instructor	Dr.ir. E.C. Slob	
Instructor	Dr.ir. S.G.J. Heijman	
Instructor	O. Kirichek	
Instructor	Prof.dr.ir. J.B. van Lier	
Instructor	Dr. J. Gebert	
Contact Hours / Week x/x/x/x	2x4 uur = 8 uur in lecture room 2 x 8 uur = 16 uur practicum	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Sediments and sludges are mixtures of mineral clay, organic material and water. Sediment and slurries are dredged for port maintenance purposes. Sludges found in (waste)water treatment plants are stabilized and dewatered for resource recovery and/or incineration. Some of these slurries/sludges are used in engineered constructions like dikes. In this module, the biochemical and physical properties of sediments and slurries will be reviewed. The way to assess their properties will be studied from the aqueous phase to the solid phase. The module consists in lectures, group assignments and practical work (lab/in-situ). Learning Unit 1: From Suspensions to Slurries EC: 4 Weeks: 1-3 The microscopic/mesosopic properties of clays and organic matter will be reviewed from the point of view of colloid science. The flocculation of particles in combination with different environmental conditions will be studied and modelled. The properties of slurries in terms of settling thickening, rheology and consolidation will be measured and modelled, as function of composition and environmental conditions. List of key topics (Elements of Learning Unit): - Nature of clays (mineralogy) and clay properties - Nature of organic matter and sludge (origin, composition, stability) - Interfacial properties of colloidal particles (surface charge, Derjaguin Landau Verwey Overbeek (DLVO) theory, polymers), - Living organic matter interactions with clay and sludge - Flocculation, coagulation - Rheology of slurries - Settling, thickening, dewatering and consolidation Laboratory experiments and measurements to determine the following properties: - particle size distribution - surface charge - settling rate - coagulation/flocculation - rheological properties - conductimetry - capillary suction time - compressibility - biodegradation of sediment organic matter (SOM) - methane potential of sludges Learning Unit 2: From Slurries to Soils EC: 3 Weeks: 4 - 6 This unit addresses the changes that sediment properties undergo during post dredging treatment and ripening on land. Students will understand and be enabled to engineer the coupled processes of physical, biological and chemical ripening which transform sediments from saturated, soft, anaerobic material to an unsaturated, well-structured and aerated soil fit for use as earthen construction material. Students will understand the associated environmental impacts, for example, related to sediment contamination or degradation of their organic components and the benefits related to sparing excavation of natural soil resources for construction purposes. List of key topics (Elements of Learning Unit): - Mechanical/physical, chemical and biological ripening processes - Methods and criteria to assess physical/mechanical, biogeochemical and biological stability of sediments and soils - Carbon cycling in the sediment-to-soil process - Environmental behaviour of conditioned slurries - Beneficial use of conditioned slurries and sediment Laboratory experiments - geochemical processes during ripening - biodegradation of organic matter - redox potentials - basic soil physical/mechanical and geochemical properties Learning Unit 3: In-situ Characterization of Sediments EC: 3 Weeks : 6-8 The student will learn to use the methods and equipment to monitor the important properties of mud in a natural environment. These properties are assessed for mud in the water phase as well as on land. These methods will be connected to the knowledge acquired in Unit 1 and 2. List of key topics (Elements of Learning Unit): - Monitoring of suspended sediment (submersible laser-diffraction, camera, optical back-scattering equipment, Acoustic Doppler current profiler) - Acoustic methods used for navigation in ports and waterways - Determining in-situ density and flow properties of slurries - Electromagnetic and seismic methods for monitoring of soils and fluid waste repositories	
Study Goals	1. Understand the principal processes that determine behaviour of fine-grained sediments (mud) and sludge in the aqueous phase	

	and in the solid phase. 2. Formulate hypotheses regarding the effect of specific environmental conditions on the behaviour of sediments, sludges and soils. 3. Apply analysis and monitoring methods to study sediments, sludges and soils properties in different environmental conditions (in the laboratory and in-situ) 4. Use the corresponding model(s) to extract the relevant parameters from the raw data obtained from the specific characterization method (point 3) and be able to analyse the results. 5. Assess the properties of mud in relation to a specific engineering problem such as the improvement of water clarity, stabilization of dikes, impact of dredging plumes, navigability of fairways in mud-rich areas.
Education Method	For all units: Interactive lectures (videos, Q&A, discussion about recently published or seminal papers) Targeted self-study using scientific articles, reports or book chapters Laboratory (see description in Units) and field work. Field work will be 1 or 2 days to put theory in practice and will be organized with our partners. Field work is complementary to lab work. At module level: Group project: Case study (cross-over all units) Material and equipment: MUDNET laboratory / sanitary eng. laboratory / Field equipment (partnership) / Geo lab
Assessment	Formative assessment (feedback) Oral feed-back to students (contact hours) will be planned during the whole course, as teachers will be present for demonstrating the use of the equipment, the way to analyse results and the way to report these results. (see in particular LO1, LO2, LO5) Contact hours will also be used to clarify fundamental points that are unclear (in plenary lectures, books, exercises, articles, etc. that students will have to study). The course will be given in the spirit of flipped classroom . Students will be asked to read and comment one or two articles every week. These articles will then be discussed during plenary sessions. The experiments realized during the course of a unit will be formalized in the form of a small report. The design of this report will be discussed and assessed with the teacher, so that students will be prepared for the case study, whereby a full report should be delivered (and that students should design themselves, see summative assessment). During lively discussions, the students will be challenged by the teachers and fellow students to explain their choices and comments their results. The students will acquire the scientific critical mind and their out-of-the-box thinking that is expected from good engineers. Summative assessment Exam each student (60%) At this point, oral examinations are foreseen, which are possible due to the expected amount of students attended the CO module. If this number rises, it will be discussed to have a written exam. During this exam, each student will be asked about the theoretical aspects of the course. Group project report (40%) A case study will be performed and each group of students will deliver a common report on the work that is performed. The case studies will bridge the knowledge acquired in the course of all units. The groups will present the work to an audience and will have to reply to questions. Each student within the group will be challenged, and the final mark for each student will be a weighted coefficient between group report, presentation and individual attitude during the case study and individual response to questions. All subgrades must be > 5,8.
Expected prior Knowledge	N/A
Academic Skills	Analysis of complex (multi-cause) problems, information collection and evaluation, multi-disciplinary and integral problem solving, project management, written and oral communication, teamwork.
Literature & Study Materials	All units : State-of-the-art articles (pdfs available on BS) Video recordings (to be recorded available on BS) Slides (pptx) corresponding to the given lectures Unit 1 New Lecture Notes have been made (available on BS), partially based on the book Introduction to Colloid Science application to sediment characterization https://books.open.tudelft.nl/home/catalog/book/16 and several leading publications exercises are being prepared (pdfs available on BS) Chassagne, Claire, Zeinab Safar, Zhirui Deng, Qing He, and Andy Manning. "Flocculation in estuaries: Modeling, laboratory and in-situ studies." In Sediment Transport-Recent Advances. IntechOpen, 2021. handouts for the use of equipment will be prepared and made available Unit 2 Blume et al. (2016): Scheffer/Schachtschabel Soil Science. https://link.springer.com/book/10.1007%2F978-3-642-30942-7 Weil & Brady (2016): The nature and properties of soils. Pons, L. J., & Zonneveld, I. S. (1965). Soil ripening and soil classification : initial soil formation of alluvial deposits with a classification of the resulting soils = Pedogenese initiale et classification des sols = Initiale Bodenbildung und Bodenklassifikation = Pedogenesis initial y clasificacion de los suelos. (Publication / International Institute for Land Reclamation and Improvement; No. 13). Veenman. https://edepot.wur.nl/60013 Journal papers on sediment ripening and beneficial use, exemplary journal: Journal of Soils and Sediments, Springer. https://www.springer.com/journal/11368 Lawrence et al. (2008). Biosolids Engineering and Management, Handbook of Environmental Engineering. ISBN 978-1-59745-174-1 Spinoza and Vesilind (2001). Sludge into biosolids. IWA publishing, London, UK. ISBN-10: 1900222086; ISBN-13: 978-1900222082 Published case studies on beneficial use of sediments: https://www.dredging.org/resources/ceda-publications-online/beneficial-use-of-sediments-case-studies Unit 3 https://library.seg.org/doi/book/10.1190/1.9781560801719 Shakeel, Ahmad, Alex Kirichek, and Claire Chassagne. "Rheology of Mud: An Overview for Ports and Waterways Applications." Sediment Transport-Recent Advances (2021). Draganov, D., Ma, X., Buisman, M., Kiers, T., Heller, K., & Kirichek, A. (2021). Non-intrusive characterization and monitoring of fluid mud: laboratory experiments with seismic techniques, distributed acoustic sensing (DAS), and distributed temperature sensing (DTS). In Sediment Transport-Recent Advances. IntechOpen.
Judgement	N/A

Permitted Materials during Exam	N/A
Collegerama	No

CEGM2001	Sustainable Cities	10
Responsible Instructor	Dr. H.M. Jonkers	
Module Manager	Dr. H.M. Jonkers	
Instructor	Dr.ir. M. Ottele	
Instructor	Dr. D.F.J. Schraven	
Contact Hours / Week x/x/x/x	approximately 12 hours/week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Sustainable cities addresses the need to transform current cities not only to climate mitigating- (substantially lower CO2 and environmental footprint), climate resilient-(adaptive to future climate changes) and circular- (minimal requirement of primary resources and minimal damage to environment) but simultaneously to desirable ones, meaning that they support overall wellbeing of its inhabitants. Some and specifically larger Dutch cities are currently characterized by the presence of aged and desolate city blocks, quarters and infrastructures with often co-occurring environmental and social problems which negatively affect wellbeing of inhabitants. Challenges to overcome are clearly manifold and solutions should therefore be integral in order to be both economically and functionally effective. The indicators that cover the four targeted sustainable (wellbeing) built environment performance criteria (climate mitigation, climate resilience, circularity and wellbeing) and addressed in this module are the following seven: 1. Quality (chemical composition) of outdoors air, water and/or soil (including presence of litter), 2. State of maintenance and quality of present buildings and infrastructure (service life-, circularity-, energy- and environmental performance), 3. Risk of heat stress occurrence (specifically in summertime), 4. Risk of flooding (specifically during intense rain showers), 5. Noise occurrence, 6. Recreational- and social interaction opportunities (quantity and quality of present public spaces) and 7. Quality and quantity of present nature (natural capital) comprising biodiversity and beneficial goods and services that nature can provide to city inhabitants. These seven living quality indicators are specifically addressed in this module as these are considered to be representative for liveability and desirability of sustainable cities. In this Cross Over module specific Dutch city quarters will be analysed both for the qualitative and quantitative state of these seven wellbeing indicators, followed by reviewing available and potential technical solutions for improvement of shortcomings related to individual indicators, and finalized by defining and designing integral solutions for often co-occurring problems. Specific focus of this module will be on the potential of nature-based solutions as these can typically provide integral solutions to multiple climate mitigating-, climate resilience- and wellbeing-related problems. E.g. the Dutch PBL (Netherlands Environmental Assessment Agency) states in their every two-years updated report that more public green (vegetation) in cities is required to increase wellbeing of its inhabitants, and that this can be achieved in balance and in concert with other urgent and space-requiring activities. Latter relate to the societal challenges for the need for more houses and residences, renewable energies, and climate resilience of- and circular building performance in cities. The module Sustainable Cities is practice oriented and case based. Each year specific city quarters will be designated for analysis. Although for practical reasons Dutch cities located close to Delft will be investigated, the problem of insufficient climate mitigation, climate resilience, circularity performance, and lack of wellbeing of inhabitants is relevant for metropolitan cities worldwide. Integral solutions found for Dutch cities are therefore also expected to be applicable to cities worldwide.	
Study Goals	After completion of this module the students are able to: CO-SC-LO1. List and analyse socio-environmental problems occurring in the built environment with specific focus on the air-, water- and soil compartments in city and infrastructural environments CO-SC-LO2. Formulate research and design questions related to socio-environmental problems in the built environment and conduct a multi-disciplinary project in a small group to find answers to the defined question, dealing with deadlines and requirements set in advance CO-SC-LO3. Use written and oral communication skills to effectively communicate theory, facts and results with respect to urban-specific health and socio-environmental problems with researchers and stakeholders in a multi-disciplinary context in order to achieve integral design solutions and strategies CO-SC-LO4. Work effectively in international, multicultural, inter-disciplinary teams CO-SC-LO5. Acquire new knowledge and skills to achieve multi-disciplinary and integral design solutions and strategies for multiple climate-, environmental- and wellbeing related problems occurring in the urban environment CO-SC-LO6. Quantify the environmental impact of linear and circular building practices using LCA (Life Cycle Assessment) techniques CO-SC-LO7. Define the socio-environmental-economic benefits of circular- and nature based building practices using SCBA (Societal Cost Benefit Assessment) related techniques CO-SC-LO8. Develop integral nature-based design solutions for multiple challenges in cities related to the need for climate mitigation and resilience, circularity performance of built structures, and wellbeing of inhabitants	
Education Method	Teaching and learning that encourages students to take responsibility for their learning process: - Lectures. Book chapters and handouts specifically relevant to each lecture should be studied beforehand. In addition to theory explanation, students will be engaged through exercises and quizzes. Small scale experimental demonstrations are given during the lectures in which students participate. - Workshop, hands-on, on campus. Students will learn how to build water-, air-, soil- or noise quality sensors and measuring systems. These will be subsequently used in a practical measuring campaign and obtained results will be analysed and reported. - Visits to designated city quarters is part of the assessment theme. Students will make an inventory of quality of living conditions based on the 7 sustainability performance indicators listed above, and report obtained results. - Case study portfolio. Students will prepare reports of tasks (individual and group tasks) and gather these in an individual case study portfolio. This portfolio will ultimately comprise reports from all assignments covering the module themes assess, review and design. Delivery of assignments will be biweekly formatively assessed. - Case study assignments. Students will form multi-disciplinary groups related to their background expertise during the design theme. Each group will work on a specific set of socio-environmental problems occurring in the designated city quarter and develop multidisciplinary integral solutions based on the knowledge and expertise obtained in the preceding assess and review themes. Groups will report findings to other groups in form of oral presentations. These outputs will be formatively assessed. Final integral solutions proposed by groups will be communicated written (in form of report) and orally (presentation). The final group assignment will be formatively assessed and added to the students individual case study portfolio.	
Assessment	Formative assessment (feedback): During the module, composed of the three themes Assess, Review and Design, tasks have to be delivered. Tasks comprise written reports and oral presentations, in part individual and in part in form of group assignments. The reports will be formatively assessed on completeness and quality by involved lecturers and in one specific case by students in form of a peer-review assignment and feedback will be provided to the students. All tasks will be assembled in an individual case study portfolio which will be summatively assessed for completeness and quality in the last week of the module. Summative assessment:	

	1. One Mobius (computer) individual examination in the last week of the module, covering the theory treated in the given lectures in the three themes (counting for 30% of final grade) 2. The individual case study portfolio will be summatively assessed for completeness and quality in the last week of the module (see Formative assessment, case study portfolio tasks) (counting for 70% of final grade) Minimal grade for both subgrades must be 5.8 to pass the module
Special Information	The Sustainable Cities Cross Over module is divided in the three consecutive and connected themes Assess, Review and Design. A short description of the three themes and its contents is as follows: Module theme 1 Assess: Quantitative assessment of city quarter quality covering 7 performance indicators. In the first theme assess the designated city blocks will be visited and quantitatively analysed for current state of quality based on the 7 indicators listed above: Topics: - Quality of built structures in relation to environmental footprint, circularity- and service life performance- Environmental quality in relation to air-, water- and soil quality and presence of litter and occurrence of noise - Build your own air-, water-, soil- or noise quality sensor (practical; including analysis and interpretation of obtained data and data uncertainty assessment) - Wellbeing quality in relation to risk of heat stress, risk of flooding, quantity and quality of public spaces (recreational- and social interaction opportunities) and quantity and quality of present nature - Methods and techniques for quantification of indicator values- Sustainability assessment methods: LCA (Life Cycle Assessment) and SCBA(Societal Cost Benefit Assessment) Module theme 2 Review: Methods for quality improvement of the built environment In the second theme review specific techniques and methods potentially suitable for mitigation or solving of problems related to the 7 sustainability indicators addressed in module unit 1 will be reviewed. Specific focus will be on green solutions, i.e. through application and implementation of vegetation in the built environment as a way to integrally tackle multiple problems simultaneously. Topics: - Methods and techniques for improvement of air-, water-, soil and noise quality- Circular and other sustainable building strategies for improvement of environmental performance of and wellbeing in city quarters - Eco-engineering solutions for improvement of liveability, climate resilience and climate adaptation of cities. Module theme 3 Design: Integral solutions for improved wellbeing in cities In the third theme design, integral solutions that can significantly mitigate problems, as identified in theme assess, through implementation of technologies reviewed in theme review, will be proposed and quantitatively evaluated. Specifically in theme design students from different MSc program and track backgrounds (EE, AES, CE, Architecture) will work in multidisciplinary teams to apply own background and new obtained knowledge in the assess and review themes to formulate and develop effective nature-based integral solutions to the multiple problems occurring in the analysed city quarters. Specific goal of theme review is to convert current non sustainable city quarters to climate resilient and desirable ones through application of integral sustainable solutions. Topics: - Integral building- and ecoengineering design solutions - Multidisciplinary creation and presentation of design solutions - Critical (qualitative and quantitative) review of design solutions including LCA and SCBA analysis
Expected prior Knowledge	N/A
Academic Skills	Analysis of complex (multi-cause) problems, information collection and evaluation, multi-disciplinary and integral problem solving, project management, written and oral communication, teamwork.
Literature & Study Materials	Examples of books: Urban Climates (Oke, Mills and Christen 2017); The city of the future by BNA (Branchevereniging Nederlandse Architectenbureaus), publisher Blauwdruk (2019); Stad x Klimaat Het gebouw als watermachine by BNA and TUDelft (2020) (in Dutch, but English handouts will be provided to non-Dutch speaking students); TEEB Manual for Cities: Ecosystem Services in Urban Management (2011) (by The Economics of Ecosystems and Biodiversity (TEEB): Geneva; teebweb.org). These books comprise theory and review a number of practical cases concerning urban heat stress and integral ecoengineering solutions for additional socio-environmental problems occurring in the built environment. Additional specific literature in form of scientific papers and book chapters on analytical techniques will be provided via BrightSpace. Experimental material for the sensor workshop will be provided during the practical.
Judgement	N/A
Permitted Materials during Exam	N/A
Collegerama	No

CEGM2002	Engineering for Global Development	10
Responsible Instructor	Dr.ir. M.W. Ertsen	
Module Manager	Dr.ir. M.W. Ertsen	
Instructor	Prof.dr.ir. D. van Halem	
Instructor	Dr. H.M. Jonkers	
Instructor	Dr. E. Abraham	
Co-responsible for assignments	M.D.M. Palmeros Parada	
Contact Hours / Week x/x/x/x	12 hours/week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Focusing on engineering in the Global South, this module relates engineering applications (in global engineering) to a societal debate (what is (desirable) development), a theme spanning domains, programs and tracks within the CEG Faculty. Students learn about similarities and differences of techniques and methods from different domains. Building on the idea that engineering solutions useful and needed as they are are always positioned within a societal debate, students discover their own position. Students learn how to engage their own expertise within debates with other experts and other stakeholders on issues of development, political aspects of engineering, options for interventions, contexts, etcetera. Learning Unit 1: Engineering (and) Development 1 (Credits: 5) In a series of workshops, with discussions, guest lectures, design sessions, etcetera, students are introduced to themes and applications in the Global Engineering domain. After group discussions, individual students prepare documentation on their own position(s). Differences and similarities in debates and positions on technological development influencing engineering solutions are explored. The exact setup of the sessions is decided on for each study year, as themes and topics are adapted to debates as they are encountered in relevant circles. These sessions are related to this list of key topics: Development theory (societal development, power relations, economic issues, globalization); Role of technology (engineering decisions, innovation ideas, history of technology); Co-creation and design interventions (participatory design, tinkering, values in design); Sustainable development goals (Differences and connections, development cooperation, role of states and UN institutes). Learning Unit 2: Engineering (and) Development 2 (Credits: 5) Teams of 5 to 7 students work on a real-life issue, relevant to at least a number of different stakeholders (design, planning, methods, etcetera), in the Global Engineering domain. The teams mobilize content from unit 1 in their project, in applying concepts and bringing their team work into unit 1. The projects are made available through the Delft Global network and will vary each year. Examples of possible projects include water provision in urban or rural settings, plastic pollution, affordable housing, access to energy. Team members will discuss the project with members of other teams, to facilitate reflecting on team results. List of key topics: Design teams working on a specific challenge; Team working sessions under guidance of a moderator; Exchange meetings between teams (discussions and workshops); Reporting on team results; Presenting team results in different formats. Please note that the exact setup of the sessions is decided on for each study year, as themes and topics are adapted to debates as they are encountered in relevant circles. The projects are made available through the Delft Global network and will vary each year.	
Study Goals	After completion of this module the students are able: CO-EGD-1. Explain their definition of and position towards societal development CO-EGD-2. Recognize existing approaches on the role of technology in society CO-EGD-3. Recognize how different stakeholders link engineering solutions to (ideas on) societal development CO-EGD-4. Relate their own expertise to other fields of expertise when developing technologies CO-EGD-5. Activate their own expertise to design engineering solutions for given development issues CO-EGD-6. Specify their own role in processes of co-creation in engineering	
Education Method	Teaching methods involved in Unit 1: Interactive setups, different per session (discussions, role plays, etcetera); Exercises (from closed to more open); (Guest) lectures (a few) Teaching methods involved in Unit 2: Team training (discussion, designing, team roles, team task division); Reporting and presentation skills workshops; Exchange workshops	
Assessment	Formative assessment (feedback): Unit 1: Individual reports: reflection of the learning goals overall in relation to the session; Active participation in sessions (demonstrated in minutes and during sessions) Unit 2: Team: report (quality of content, societal relevance, team work process); Individual: presentations during exchange (quality of content, presentation skills); Individual: debates during exchange between teams Summative assessment: - Unit 1: Individual essay on a theme that relates to the sessions and study material (consistency of the argument, references) - Unit 2: Individual feedback essay on the team project (position in the team, reflection, writing skills) - The final grade is the average of the two grades. Each grade required to be at least 5.8.	
Special Information	Week 1 - week 8: 3 sessions of 4 hours each, with associated reading, group discussions and individual reflection. The exact setup of the sessions is decided on for each study year, as themes and topics are adapted to debates as they are encountered in relevant circles. The projects are made available through the Delft Global network and will vary each year. Week 9 - week 10: Self study	
Expected prior Knowledge	This course is open to all students from MSc programs.	
Academic Skills	Analysis of societal positions; reading texts; discussion skills; written and oral communication; team work	
Literature & Study Materials	The study material is collected each year, related to actual debates. Students read papers (about 5 to 7) and discuss these.	
Judgement	N/A	
Permitted Materials during Exam	N/A	
Collegerama	No	

CEGM2003	Data Science and Artificial Intelligence for Engineers	10
Responsible Instructor	Dr. R. Taormina	
Responsible Instructor	Dr. H. Wang	
Responsible Instructor	Dr.ir. I. Barcelos Carneiro M Da R	
Module Manager	Dr. R. Taormina	
Instructor	F.P. van der Meer	
Instructor	Dr. K. Anupam	
Instructor	Dr. O. Cats	
Instructor	P.K. Krishnakumari	
Instructor	Dr. D.V. Voskov	
Instructor	Dr. M.A. Schleiss	
Instructor	J.O. Colomes Gene	
Instructor	Dr. G. Rongier	
Contact Hours / Week x/x/x/x	5-7 blocks a week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	Elective for MSc programmes AES, CE, ENVM	
Course Contents	Module Introduction Data Science (DS) and Artificial intelligence (AI) are poised to revolutionize all aspects of Civil Engineering, Environmental Engineering and Applied Earth Sciences. Our interdisciplinary module in Data Science and Artificial Intelligence for Engineers (DSAIE) will offer 2nd year Masters students the opportunity to learn these powerful tools and apply them from the very beginning. We will focus primarily on Probabilistic Machine Learning and Deep Learning methods, effective data-driven approaches to deal with the uncertainty and the complexity characterizing the challenges of the interconnected natural, living and built environments. During this module, the students will broaden their engineering understanding through collaboration with peers from diverse backgrounds, following instruction from experts across various programs, and applying what they learn on real-world case studies. The course will also provide plenty of opportunities to further develop their Python coding skills through hands-on exercises and workshops in areas such as collaborative coding, high performance computing, building automation and testing, and advanced data analytics. By the end of the module, the students will have acquired a practical and holistic understanding of DS/AI and its applications in their fields of choice, which will be extremely valuable for the students future careers. Module Units The proposed cross-over module is divided into three units: Unit 1. Data Science & Probabilistic Machine Learning (4 EC) Unit 2. Deep Learning (3 EC) Unit 3. Project (3 EC)	
Study Goals	After completion of this module, the students can: LO1. Explain core concepts in data science, machine learning and artificial intelligence, including the fundamentals of probabilistic and deterministic data-driven techniques. LO2. Translate a real-world problem related to the natural, the living and the built environment into a DS/AI application by following a multidisciplinary approach. LO3. Adapt suitable probabilistic and deterministic DS/AI techniques to solve real-world problems related to the natural, the living and the built environment. LO4. Implement DS/AI pipelines in Python using best coding practices for execution robustness, result reproducibility and validation. LO5. Present the results of a DS/AI analysis in oral and written formats, being able to i) motivate their choices, ii) defend their opinions, iii) cater to diverse audiences, and iv) critically evaluate feedback. LO6. Build workable DS/AI solutions for a multidisciplinary set of realistic problems in cooperation with peers of various backgrounds.	
Education Method	Unit 1 & Unit 2 In a typical lesson, the instructor will illustrate theoretical concepts to the class in a plenary session. When suitable, the instructor will alternate theory with exercises (individual or group) and coding walkthroughs/demos in Python. These will be based on selected datasets chosen by the instructor. While most of the lectures will be on campus, lectures for advanced topics will be delivered via recordings. Unit 3 Theory. Instructors can provide limited content related to the applications via recordings. Advanced topics of UNIT2 can be chosen in combination with the specific project selected. Applications. There will be 3 applications on very diverse datasets for students to apply the concepts of Unit 1 and Unit 2 while preparing for the final project. Students will work in groups of 5; instructors will be available with TAs for support in class. Each application will last for approximately 4 hours, as done in the MUDE. Projects. In week 4, project instructors will introduce case studies/datasets that students can exploit for developing their final projects. Contrary to applications,	

the proposed projects may not have clearly defined research questions. Students will form groups of around 5 to work together on their final projects. Students can also come up with their own dataset and case study, provided they find the support of an instructor that can act as a coach/mentor (open inquiry, see Figure 3). Groups will be formed in Week 5. Students will preferably link the self-study advanced topics of Unit 2 (Week 6) to the project they choose.

Coding Workshops.

The coding workshops will be usually delivered as a short lecture followed by hands on exercises (e.g., on notebooks, HPC terminals, ...) to master specific coding knowledge related to AI and Data Science. Instructors and TAs will be available in class for the entire duration of the workshops.

Practical Guide

Unit 1. Data Science & Probabilistic Machine Learning (4 EC) [LOS 1, 3, 4]
Week 1 to Week 3

In this Unit we will first introduce the module and reactivate prerequisite knowledge from the MUDE with respect to Supervised Regression. We will then explore Supervised Classification, the other major family of Supervised Machine Learning methods. We will then study the Bayesian perspective on model selection, before exploring tree-based methods and kernel based methods for regression and classification. After refreshing MUDE knowledge on clustering and PCA, we will explore more advanced topics in unsupervised machine learning and generative machine learning. Students will work on two different application sessions to reinforce their knowledge.

Unit 2. Deep Learning (3 EC) [LOS 1, 3, 4]
Week 4 to Week 6

This unit will mainly deal with Deep Learning methods, representing the state-of-the-art of AI-based applications in many fields, including CEG-related ones. We will focus on key topics such as representation learning, and the use of inductive biases and symmetries to overcome the curse of dimensionality. These will be introduced with respect to convolutional and recurrent neural networks, but we will explore other powerful and more recent approaches such as physics-based deep learning and transformers. Students will work on one application session to reinforce their knowledge.

In Week 6, students will select two advanced topics to study independently (i.e., videos, materials and notebook). These will be preferably linked to the projects they select in Week 4 (see Unit 3).

Unit 3. Project (3 EC) [LOS 2, 3, 4, 5, 6]
Week 4 to Week 10

Projects are an essential part of the module and constitute a dedicated unit. Students will work in groups on a selected CEG-related project to apply and reinforce the skills acquired in Unit 1 and Unit 2 while collaborating on real case studies. All available projects will be showcased by the responsible instructors in Week 4, and students have time until Week 5 to choose their project.

Potential case studies

Here follows a tentative list of potential case studies from which students can select their projects. Students can choose their projects regardless of their background. Students can also come up with their own project from scratch, provided they find a mentor by Week 5.

Environmental Engineering

Rainfall-runoff modelling using Deep Learning methods
Prediction of drinking water demands
AI-based modelling of water quantity and quality processes in urban water networks
Plastic detection using AI-based computer vision
Defect detection in sewer systems with AI-based computer vision

Civil Engineering

Surrogate modelling for flow through porous structures
Accelerate finite elements methods for structural mechanics with AI.
Defect detection of railway infrastructures using deep convolutional neural networks and on-board camera images.
Flood Risk Mapping with Deep Learning methods.
Unsupervised learning for rail health condition classification.
Unravelling the Dynamics of Urban Demographics
Incorporating safety and beauty perceptions into travel journey planners using street view images

Applied Earth Sciences

Rainfall nowcasting using Deep Learning methods.
Machine Learning for applications in Energy Transitions
One-shot learning for generative modeling of geological structures
Exploring the effect of different formulations of Gaussian processes on spatial predictions

Assessment

Formative assessment (feedback)

Theory

Computer-based quizzes with multiple answers for each topic, including coding exercises. These will be made available every week. A mock-up theory exam will be made available in Week 5.

Project

Formative feedback provided in class during applications 1-3 which serve preparatory activities for the project. Feedback from instructors and peers on project progress at the end of Week 7.

Summative assessment on module level

There will be two summative assessments: 1) an ANS written exam for the theoretical part, and 2) a group presentation with interview for project evaluation.

Students understanding of theoretical content/coding skills will also be assessed as part of the project learning line, specifically their ability to apply the theory and reflect on the meaning and limitations of produced results. This justifies the equal value of the two summative assessments (i.e., 50% each).

The project deliverables include the final presentation plus a code repository. The grade will reflect the originality and impact of the final solution, the quality of presentation, and the quality of code repository and documentation. Dedicated assessment tools for group work will be employed to facilitate marking of individual contributions.

All subgrades must be >5.8 to pass the module.

Special Information

Weekly Syllabus

10 ECs = 280 hours. We consider 28 hours per week divided in 7 blocks of 4 hours each. There will be around 50 hours of lectures, mainly focused on the first half of the module. The second half focuses on the project. Sufficient time is reserved for

self-study during the entire module. The students are free to organize their self-study as they want, but instructors and/or teaching assistants will be available in a classroom for explanations after lectures. When the project runs, there is a block reserved each week to interact with project instructors and TAs.

Week 1

Introduction to the course with overview.

Translating real-world problems to build data science and artificial intelligence solutions.

MUDE Recap: supervised regression, deterministic perspective on model selection: bias-variance trade-off, regularization, cross-validation, neural networks.

Coding Workshop #1: PyTorch

Data types: time series & sequences, tabular data, images & raster data, unstructured vs structured data

Probabilistic perspective on model selection: informative/uninformative priors, hyperpriors, Maximum a Posteriori (MAP) estimates, Bayesian model comparison, Empirical Bayes, Variational Bayes

Week 2

Supervised classification: classification with decision regions, decision theory for classification, classification with basis functions, logistic regression, classification with neural networks

Feature engineering, input selection and dealing with unbalanced data.

Coding Workshop #2: collaborative coding for DS/AI projects.

Frequentist and Bayesian kernel-based methods, instance-based learning

Application #1: Supervised Classification

Week 3

MUDE Recap: unsupervised machine learning, k-means and PCA

Inference and learning with discrete and continuous probabilistic latent variables, introduction to Generative Machine Learning, Probabilistic Principal Component Analysis (PPCA)

Application #2: Unsupervised & Generative Machine Learning

Week 4

From Probabilistic Machine Learning to Deep Learning: curse of dimensionality, representation learning, inductive biases & symmetries for data efficiency

Projects showcase

Convolutional Neural Networks: 1D/2D/3D convolutional layers, pooling, composition, feature extraction, translational invariance and equivariance, data augmentation, image regression and classification.

Regularization: early-stopping and dropout layers.

Pre-trained models and transfer learning.

Coding Workshop #3: AI on DelftBLUE

Application #3: Deep Learning

Week 5

Recurrent Neural Networks: sequential inductive bias, simple RNNs, gated recurrent neural networks (LSTMs and GRUs), time series data.

Intro to Physics-based deep learning: basics of neural ODE/PDE solvers, neural operators, physics informed neural networks, physics guided machine learning.

Other type of inductive biases: quick introduction on transformers, graph neural networks, bayesian deep learning.

Presentation of advanced topics that students can select (see below).

Project selection

Initial work on projects

Week 6

Advanced Topics

Project work

Week 7

Project work

Project preliminary presentations + feedback

Week 8

Project work

Week 9

Project work

Written exam

Week 10

Project work

Project submission and presentation

Expected prior Knowledge

CEGM1000 - Modelling, Uncertainty and Data for Engineers

Academic Skills

Analysis of complex (multi-cause) problems related to the built/natural/living environment; information collection and evaluation; multi-disciplinary and integral problem solving; project management; oral communication; teamwork; basic coding skills.

Literature & Study Materials

Unit 1 & Unit 2

Study material: lecturers slides, recorded videos/Collegerama, Jupyter notebooks, GitHub code repositories + documentation, datasets.

Literature:

Bishop, Christopher M. "Pattern Recognition and Machine Learning", 2006.

Murphy, Kevin P. "Probabilistic Machine Learning: An introduction.", 2021.(open source <https://probml.github.io/pml-book/book1.html>)

Murphy, Kevin P. "Probabilistic Machine Learning: Advanced Topics.", 2022. (open source <https://probml.github.io/pml-book/book2.html>)

Goodfellow, Ian, et al. Deep learning, 2016. (open source <https://www.deeplearningbook.org/>)

Molnar, Christoph. "Interpretable machine learning", 2020. (open source <https://christophm.github.io/interpretable-ml-book/>)

Wilke, Claus O. Fundamentals of data visualization: a primer on making informative and compelling figures, 2019. (open source <https://clauswilke.com/dataviz/index.html>)

Selected seminal papers in Data Science, Machine Learning, Deep Learning and Interpretable Machine Learning.

Select webpages and blogs.

Unit 3

Study material: datasets with description, code repositories, recorded videos (optional), all material of Unit 1 and Unit 2.

Literature: each instructor will provide a selected list of scientific papers and other publications (e.g., MSc Theses of previous

students) for the applications/projects they propose.

Coding Workshops & Computational Resources

The coding workshops will be partially based on high-quality, worldwide known, open-source Data Science and Computing workshops provided by The Carpentries (<https://carpentries.org/>), CodeRefinery (<https://coderefinery.org/>), and Aalto Scientific Computing (<http://scicomp.aalto.fi/training/>). The instructors will select, summarize, modify, and expand the material available from these resources to fit the scope of this Module. Students will be able to access the DelftBLUE supercomputer for the entire module (<https://www.tudelft.nl/en/dhpc>).

Judgement

See assessment

Permitted Materials during Exam

N/A

Collegerama

No

CEGM2004	Noise and Vibration: Generation, Propagation and Effect on Humans and Environment	10
Responsible Instructor	A. Tsouvalas	
Module Manager	A. Tsouvalas	
Instructor	Prof.dr. A. Metrikine	
Instructor	Dr. A.A. Nunez Vicencio	
Instructor	A. Varveri	
Instructor	Dr. K. Anupam	
Instructor	Dr. Z. Yang	
Instructor	Dr. A. Cicirello	
Contact Hours / Week x/x/x/x	approximately 4*2HC Theory and 1*2HC project with instructors per week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	General description: Noise is sound considered unpleasant, loud or disruptive to hearing, which has become an increasing concern throughout the world. While urban noise (from transportation, construction, etc.) brings a negative influence on human health, underwater noise (from pile driving, seismic surveys, etc.) threatens the leaving space of marine mammals and fish. Similar to noise, vibration is undesirable in many cases because it causes damage to engineering structures and discomfort to people. From the viewpoint of physics, noise is vibrations transmitted through a medium, such as air or water. Therefore, effective control of noise and vibration can be possibly be achieved at the source, path, or receiver. In this module, students learn the fundamentals of noise and vibration, including structural vibrations, sound generation and transmission, perception of noise by inhabitants, conduct in-class/lab tests to obtain and analyse vibro-acoustic data, and use numerical tools to mitigate the adverse effects on the urban and marine environment. Overall structure of the module: The module structure consists of two interrelated themes; Theory and Project. The Theory theme starts first and students are exposed to fundamentals of noise and vibrations, important sources of noise in air and underwater and dynamics of structures. The coupling between the structure and the surrounding media, in which waves are transmitted, is central in the Theory theme. Vibro-acoustic modelling techniques are discussed and software (that the students use for the Project theme) is introduced. Besides modelling, students are also exposed to measurement setups and equipment regularly used to measure and quantify noise and vibration levels. The theme on Theory is followed by the Project theme in which students apply the theoretical concepts to realistic environmental noise and vibration problems. Case studies, in which levels of vibrations and/or noise are unacceptable, are selected by the instructors together with the relevant industry. Students form groups to tackle the environmental challenging case. Results of this investigation are reported by the group and presented at the end of the quarter. In the theme theory the mathematical background of acoustics and vibration is taught with an eye on real-life applications. Emphasis is placed on an in-depth analysis of structure-borne noise emission and its effective cancellation either at the source or at the propagation path. This unit is developed such that it includes illustrative applications of the different noise sources in the environment including the effects on people and animals. This unit is fundamental and interlinked to the theme project in which a practical implementation of the theoretical concepts is accomplished. In the theme Project, students apply the theoretical concepts to realistic environmental noise and vibration problems. The students are divided into groups. Groups are formed by 2-3 students following different MSc programmes in order to enrich the expertise within the groups and to be able to reach scientific depth in the final result of the project.	
Study Goals	After completion of this module, the students are able to: LO1 Formulate the governing equations for problems in which structural vibrations and sound generation and propagation are involved. LO2 Construct analytical formulae and numerical models to simulate structure-borne acoustic emission. LO3 Analyse the environmental noise and translate those measurements to (internationally) accepted noise metrics. LO4 Evaluate the impact of noise and vibrations on the environment and humans using metrics for noise and vibration perception. LO5 Design noise mitigation strategies to tackle urban and underwater noise pollution. LO6 Compose scientific reports on vibration- and noise-related topics using professional terminology.	
Education Method	Teaching and learning that encourages students to take responsibility for their learning process. - Lectures. Pre-study material (book chapters and handouts) will be provided to students before the lectures. The course content will be described with real case studies, examples and interactive demonstrations with the active participation of the students. Students will be asked to refresh their knowledge at parts to avoid repetition of material covered earlier in the curriculum. - Hands-on assignments. While the students learn about the theory and mathematical descriptions related to the physics of noise and vibration, they should apply the outcomes to the case studies. The students should complete bi-weekly tasks in small assignments which will help them to execute the project. Delivery of bi-weekly assignments will be optional and formative feedback will be provided by the instructors. - Workshop. The students will learn the basic noise and vibration measurement equipment, use them, and analyse the collected data. The measurements sessions will be performed either in-class, in the lab facilities or, ideally, in the campus during the workshop. - Self-study: During weeks 1-5, CE students read selected topics related to acoustics and EnvEng, AES students topics that relate to structural dynamics. This way students get exposed systematically to theory from another discipline. The Project theme is organized by means of feedback/plenary sessions. The students form multi-disciplinary groups related to their backgrounds. Each group can choose one of the provided urban or underwater noise problems as a project topic under the supervision of the instructors so that all subjects are covered equally. Each group should develop innovative solutions based on the obtained knowledge in the preceding theory theme. Groups will present their process and results to other groups during bi-weekly meetings with the aim to initiate a live discussion and provide a formative assessment of the work-in-progress. Each group will prepare a detailed final report related to their project. These outputs (presentation and final report) will be summatively assessed.	
Computer Use	The software to be used by the students for the execution of the project will consist of: - available licensed software from TUD (e.g. Comsol Multiphysics) and - in-house software prepared by the teaching team.	
Practical Guide	Weeks 1 and 2: Introduction to noise and vibrations - Fundamentals of acoustics: acoustic metrics, sound wave propagation. - Fundamentals of structural vibrations: vibrations of beams and plates, coupling to the acoustic domain and sound radiation. - In-class examples with analytical solutions, homework.	

	Week 3: Noise and vibration sources and impact on environment (distribution of projects and group forming at the end of week 3) - Noise and vibration sources in the urban and marine environment (traffic, rail, underwater and air noise) - Underwater noise and vibration sources, defining the acoustic problems. - Audibility range of species, levels of injury, etc. - WHO reports, MSFD, biological research, noise metrics, international standards. - Levels of vibrations acceptable in buildings. - Impact of traffic, rail and air-induced noise and vibrations on human life - Underwater noise impact on marine life: Noise from shipping, pile driving, seismic surveys, explosions and wind. - In-class examples with analytical solutions, homework. Weeks 4 and 5: Propagation of waves and modelling - Brief information related to ray theory, mode theory, finite elements, and boundary elements (air/water/soil layered waveguides). - Why do we need simulations? How to choose among the different simulation techniques? - Model validation and interpretation of the modelling results. - Generation of noise maps. - In-class numerical examples (homework related to simulations with FE or inhouse software; the software to be used will be determined by the instructors depending on the type of problem analyzed). Week 6: Sound and vibration measurements - Noise measurement equipment: sound level meters, hydrophones. - Lab-scale (or in-class) demonstration of working principles and data analysis. - Analysis of noise and vibration data, converting pressure to relevant metrics for the noise impact (homework related to the analysis of measurement data). - Tire-pavement measurement methods. - Rail-wheel measurement methods. Weeks 7 and 8: Noise mitigation and vibration suppression - Noise mitigation and vibration suppression strategies, smart mitigation strategies (acoustic black holes, metamaterials), examples for the noise mitigation related to pile-driving. - Measurements in lab/field. - In-class discussion of case studies: noise mitigation and vibration suppression. Project - Conducting a project regarding either urban or underwater noise: Each group will conduct a project (distributed in the end of the 3rd week of the module) with the focus being placed in one of the subjects listed below: 1. Structural-borne noise and vibrations in urban environments: traffic noise including modelling of tire-pavement interaction, railway noise including modelling of rail-wheel interaction, noise from civil construction in urban environments. 2. Structural-borne noise and vibrations in offshore environments: seismic surveys offshore, noise and vibrations from installation of foundations offshore, shipping noise, drilling. To accomplish the project, students are required to conduct simulation, analyse the results of noise and vibration data (data sets will be provided by the instructors), design a strategy of noise mitigation and/or vibration suppression, and compose a final report.
Assessment	Formative assessment (feedback): Students will discuss their results and formative feedback will be provided by the instructors and based on plenary sessions during the quarter on a weekly basis. Formative assessment of weekly assignments regarding theory unit. Summative assessment: The module is assessed by means of the project work with a report and an in-class presentation.
Expected prior Knowledge	N/A
Connection to other modules	The CO module brings together engineers of various backgrounds and challenges them with a problem in which both the source of noise and vibrations, the emission of waves and the mitigation of noise and vibrations are tackled in a holistic manner. This requires teams consisting of students of various expertise which is mobilized during the execution of the project. Given the focus of the module, there is some overlap with the following parts of the individual programmes: - Overlap with the AEE track in the EnvEng programme on the fundamentals of acoustics and measurement of noise; - Overlap with the SE track in the CE programme on fundamentals of structural vibrations and measurements; - Overlap with AES programme on elements of wave propagation. Despite the overlap though, the students here will need to go beyond their comfort zone, tackle a multi-disciplinary problem, and work together to find a suitable remedy to reduce the disturbance to inhabitants. They will also deepen their knowledge further by tackling not only the individual elements dealt with earlier in their programmes but also their complex interaction.
Academic Skills	Analysis of complex (multi-cause) problems, information collection and evaluation, multi-disciplinary and integral problem solving, project management, written and oral communication, teamwork.
Literature & Study Materials	Theory: Mathematical description and physics related to noise and vibration, analysis of practical problems of structure-borne noise and vibrations. Selected chapters of the following books: [1] Allan. D Piece, Acoustics: An Introduction to Its Physical Principles and Applications, 2019, Springer. [2] Steven L. Garrett, Understanding Acoustics: An Experimentalists View of Sound and Vibration, 2020, Springer [3] M. Möser, G. Müller, Handbook of Engineering Acoustics, 2013, Springer [4] D. Thompson, Railway Noise and Vibration, Mechanism, Modelling and Means of Control, 2010, Elsevier https://www.sciencedirect.com/book/9780080451473/railwaynoise-and-vibration. [5] F. Fahy, P. Gardonio, Sound and Structural Vibration: Radiation, Transmission and Response, 2008, Academic Press. [6] Robert Bernard and Roger L. Wayson, An Introduction to Tire-Pavement Noise of Asphalt Pavements, 2007. [7] TRBs National Cooperative Highway Research Program (NCHRP) Report 630: Measuring Tire-Pavement Noise at the Source examines a Suggested Procedure for Measuring Tire-Pavement Noise at the Source Using the On-board Sound Intensity (OBSI) Method. Additional specific literature (if any) in form of scientific papers, other book chapters, or public reports will be provided in class. Project: Conducting a project regarding either urban or underwater noise Books: same as theme Theory above. Additional materials including the manuals of the measurement equipment and the papers reviewing the existing studies of the specific topics on urban or underwater acoustics.
Judgement	N/A
Permitted Materials during Exam	N/A
Collegerama	Yes

CEGM2005	Probabilistic Modelling of real-world phenomena through ObseRvations and Elicitation (MORE)	10
Responsible Instructor	E. Ragno	
Module Manager	E. Ragno	
Instructor	Ir. F.C. Vossepoe	
Instructor	G.F. Nane	
Instructor	Dr.ir. O. Morales Napoles	
Instructor	Dr.ir. R.C. Lanzafame	
Contact Hours / Week x/x/x/x	approximately 4 hours lectures, 4 hours workshop per week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Probabilistic Modelling of real-world phenomena through ObseRvations and Elicitation (MORE) is a methodological type module oriented and structured around advanced topics in probability and statistics applied to engineering problems. Rainfall, water flowing into rivers, cars and trucks crossing bridges, ocean waves, and earthquakes are all examples of phenomena of which we wish to quantify the magnitudes, frequencies, uncertainties, and risks to make our daily life safer and more manageable. Such phenomena are characterized by some degree of randomness, meaning that their future outcomes are unknown. Hence, how do we design sewer systems to deal with precipitation in urban areas, estimate the loads that will compromise the structural integrity of a bridge or protect coastlines from flooding? In this module, students will learn methods of probability and statistics to infer characteristics of random and uncertain phenomena around us when observations (data) and physics-based numerical models are available, by using (non)stationary extreme-value analysis, dependence modelling in higher dimensions and data-assimilation techniques. When data pose challenges or are simply missing, students will learn about structured expert judgment methods to quantify uncertainty by using expert opinion. The module will provide students with a theoretical understanding of different probabilistic techniques so that they can discern which method(s) is(are) most suitable for analysing the problem at hand. At the same time, students will learn how to translate their theoretical knowledge into computer codes. As part of this course, students will apply the methods learned in an interdisciplinary group project around specific themes, that will vary on a yearly basis. Example of projects themes are (i) shoreline protection in the face of sea level rise; (ii) subsidence as a results of ground water extraction; (iii) water management assessment under urbanization and climate change; (iv) reliability of aging infrastructure under climate change. Students will work in groups, about 4 members each, and based will come up with specific research questions for their theme that they will answer using methods learned. The total 10 EC will be divided as follows: 6 EC in Unit 1 - Theory and Example applications and 4 EC in Unit2 - Group Project.	
Study Goals	After completion of this module the students will be able to: LO1 Discuss the concepts of non-stationary extreme value analysis, multivariate probability distributions, multivariate uncertainty quantification, data assimilation, and expert judgment. LO2 - Apply the concepts of non-stationary extreme value analysis, multivariate probability distributions, multivariate uncertainty quantification, data assimilation, and expert judgment to problems in civil engineering, applied earth sciences and environmental engineering. LO3 - Analyse and discuss the engineering applications (for example climate change, extreme discharge, induced seismicity, maximum bending moments, traffic loads, infrastructure design, pollution health effects) that accompany the techniques in probability and statistics taught in this module with an emphasis on the added value of uncertainty quantification. LO4 Write computer codes using open-source, e.g., Python, or commercial, e.g., Matlab, programming languages to translate key methods and techniques presented in the module and to carry out analysis related to specific track-related problems. LO5 Make use of existing open-source code and software to carry out analysis using more advanced methods which require higher-level coding skills. LO6 - Design and carry out an interdisciplinary research project of interest in science or engineering that incorporates methods taught in this module within a guided inquiry-based learning framework. LO7 - Work effectively in international, multicultural, inter-disciplinary team	
Education Method	Theory and example applications Lectures: the module will offer a mix of classical style lectures from an instructor, collaborative group sessions and educational videos. Dedicated classroom time will be provided each week for the lectures and collaborative sessions with students and instructors. Hands-on learning via computer practicum and think-pair-share activities will be implemented in class. The students will benefit from the interaction with their peers and from the need to re-interpreting theoretical concepts in consecutive executable actions. At the same time, teacher-center lectures are necessary to show the students the theory and rationale underlying the statistical methods covered in class. An example application of these activities can be related to extreme value analysis. Students will be provided with a time series for a general variable (for example maximum yearly wave height in the North Sea for CE, maximum PM2.5 concentrations in Amsterdam for EE, or expected probability of earthquake occurrence in AES) and they are asked to extrapolate a sub-sample of the observations containing only extremes. First, they will do this individually and then they will share ideas with their partner. After gathering students ideas, the instructor will explain how extremes can be sampled based on current methods in extreme value theory. Students will then apply such methods into executable coding scripts to sample extremes from a time series of their interest. Applications: Students will work on weekly assignments related to practical applications of the theory learned in class. Such assignments will derive from the three programs so that the students can appreciate the interdisciplinary character of statistical methods. The applications proposed will be related to a specific track in such a way that also students that do not come from that track can understand them, as exemplified in the Module Content section. Students are encouraged to follow the think-pair-share approach. In this way, they will first try to solve the problem individually and then compare their work with their peers for feedback. Dedicated classroom time will be made available for peer feedback and collaboration.	

Group project

A guided inquiry-based learning approach will be followed, similar to that implemented in the MUDE module (Q1 and Q2 of the first year MSc programme). This module builds on the project-related skills developed in MUDE by allowing students to identify their own topic of interest to further analyse it using at least two methods covered in class. Students progress will be followed by the instructors starting from week #4, in which an introduction on the remaining topics covered in the module will be given to facilitate the project definition. Guidance documents for students (e.g., effective group work strategies, formulation of research questions, report writing) will be provided. Students will have all the material from the theory section. However, students will be encouraged to carry out a literature review to get familiar with how the scientific community has approached similar cases. Teaching methods in Unit 2 are largely focused on guidance with individual groups during the Q&A sessions in weeks 7 and 8. For example, some students may need help solving coding issues, whereas others may need advice extending the methods covered in Unit 1.

Practical Guide

Theory and example applications:

In this unit, students will learn theoretical concepts related to advanced techniques in probability and statistics. The unit content is spread over a period of 6 weeks. In each week, one topic will be treated comprehensively, from theory to practical applications.

The following topics will be covered:

(w1) Extreme value theory: Power law distributions, rank ordering statistics (including tail index estimation - e.g., Hills method), Goodness of Fit diagnosis (Kolmogorov-Smirnov, Anderson-Darling etc.), Heavy tailed distributions; Review of extreme value theory peak over threshold, block maxima; Non-stationary extreme value theory. Example assignment: data analyses of (extreme) observations for inferring, e.g., design values.

(w2) Data Assimilation: maximum a posteriori probability methods, ensemble methods, variational methods and particle methods. Example assignment: quantification of subsidence due to fluid extractions.

(w3) Expert Judgment: Cooks method for uncertainty quantification, Elicitation of probabilistic dependence (multivariate uncertainty). Example assignment: Elicitation of future hydro-meteorological variables due to climate change, or, Assessment of uncertainty regarding mortality and morbidity due to PM10 and PM2.5 through Cookes method.

(w4) Bivariate distribution using copulas: Introduction to copulas compared to traditional bivariate distribution; Covariance, product moment correlation coefficient, rank correlation coefficients (Spearman's rho, Kendall's tau), tail dependence coefficient, estimation (maximum likelihood, moments); One parameter copulas: Gaussian copula, Gumbel copula, Frank copula, rotations of copulas. Examples of 2 parameter copulas (t-copula), and 3 parameter copulas goodness of fit diagnostic tools: semi correlations. Example assignment: modelling correlation between PM2.5 (air pollutant) and health effects (e.g., daily mortality) in the context of regression analysis for long and short term effects.

(w5) Multivariate probability via Bayesian Networks: Advanced topics on discrete Bayesian networks; Non-Parametric Bayesian Networks; Dynamic Bayesian Networks. Example assignment: risk and reliability assessment, e.g., Calculation of flooding probabilities, stochastic modelling of vehicle loads in bridges, calculation of maximum bending moments in bridges, health risks, others.

(w6) Multivariate probability using vine-copulas: multivariate distribution via pair-copula construction. Example assignment: multivariate probabilistic design of coastal structures subjected to multiple forces, e.g., wind speed, wave height, wave periods.

Group project:

Students work on a research topic within the proposed themes that they choose themselves where at least two techniques covered during the module will be applied. During week #4, the topics covered in weeks #5 and #6 will be introduced briefly so that students can better identify potential projects. The students will propose a project topic beginning in week #5, which is reviewed by the instructors and finalized by the end of week #6, including a work plan (analysis methods and anticipated results) and definition of the project data set. Weeks #7-8 include a number of Q&A sessions by the instructors for checking students progress and discuss possible common questions. While there will be some review of material, these weeks are dedicated to finalizing the group project. A project is performed in groups of 2 or 3 students (depending on the size of the class), preferably from different programs.

Assessment

Formative assessment (feedback):

Formative assessments on common errors identified in weekly assignments and during projects progress meetings will be provided to the class. Group and individual formative assessments will be provided when necessary.

Peer review based on weekly assignments will be implemented. This in particular for the programming components of each week.

At the end of week #8 and before the final deadline for project report submission, the students will have the chance to present their partial results to their peers and receive peer feedback. The students will receive a feedback form to fill in during the group presentation. Feedback will be on the structure of the presentation, relevance of the project topic, and clarity of the methods implemented.

Summative assessment:

Theory and applications

Summative assessment will consist of an individual oral exam covering both theoretical aspects and applications of the statistical methods taught in class. The oral exam will build upon the analyses carried out in the report and expand to the rest of the topics (theory) discussed in the module. The progress (depth and extent that each student can reach) of the oral exam will depend on the ability of each student to answer to questions. The depth and extent that each student can reach during the oral exam will determine the individual grade. The oral examination is 50% of the total grade. As stated earlier, this grade will be individual and hence may differ from the group project grade per student. LOs 1,2,3 will be tested.

Group project

Summative assessment consists of a written report summarizing project setup and results. The purpose of the group project is to LOs 2,4,5,6,7 will be tested. The group project is 50% of the final grade.

All subgrades must be ≥ 5.8

Special Information

The methods are first discussed theoretically and then exemplified via applications. Specific applications will be drawn from applications related to the three programs in the faculty. The applications proposed will be related to a specific track in such a way that also students that do not come from that track can understand them. For example, a nonstationary statistical model of sea level height will be developed to include the signal of sea level rise. This example is relevant to coastal engineers but also to earth system scientists. At the same time, environmental engineers can understand the relevance of it without too much in-depth pre-knowledge of coastal dynamics and climate change.

Students will write their own code to solve simple problems using the fundamentals of each method. In-house open-source Matlab- and python-based software will be used for more advanced problems. Example applications will be provided by members from the faculty, some of which will also be linked to the MUDE example applications and project proposals to ensure a coherent learning line in probability and statistics in the new master curriculum.

During week #1 a general overview of all topics in the module will be presented, including project-related examples, to allow students time to choose the methods to implement in their final project. Starting from week #4, topics covered in weeks #5 and #6 will be introduced briefly so that students can better identify potential projects.

Expected prior Knowledge

MUDE CEGM1000

Connection to other modules	Overlap with other parts of the programmes. In the MUDE, the students will get familiar with basic concepts of probability and statistics (e.g., axioms, discrete and continuous random variables, central limit theory, bivariate distributions, Bayes theorem). These basic concepts represent a necessary pre-knowledge for understanding the topics covered in this module. We expect some overlap in extreme value analysis, and bivariate probability distributions, but this would be propaedeutic to the related more advanced topics as nonstationary extreme value analysis, and Bayesian Networks and Vine-copulas respectively. Similarly, students will be introduced to data assimilation techniques that link the data, uncertainty and modeling learning lines in the MUDE, which will also be extended in the MORE. We foresee a small overlap in HOS regarding reliability-based design approach that includes extreme value analysis. Some overlap with CO modules Data Science and Artificial Intelligence for Engineers and Monitoring of Structural Health and Geohazards mostly with respect to Bayes rule (Bayesian updating) and other data assimilation techniques is also expected.
Academic Skills	Analysis of complex problems, communicate scientific knowledge and findings in inter-disciplinary context, critical thinking, project management, team work, written and oral communication
Literature & Study Materials	Theory and applications Group project Students will receive lecture slides, references to relevant scientific literatures, text book, MOOC, recordings of past lectures on related courses, and open source software developed over the year by the instructors. All the material will be accessible via Brightspace. Example literature: Coles, Stuart, et al. An introduction to statistical modeling of extreme values. Vol. 208. London: Springer, 2001. Genest, C. and Favre, A. (2007). Everything You Always Wanted to Know about Copula Modeling but Were Afraid to Ask. J. Hydrol. Eng. 12, SPECIAL ISSUE: Copulas in Hydrology, 347368. Morales Nápoles, O., Worm, D., van den Haak, P., Hanea, A., Courage, W., Miraglia, S.: Reader for course: Introduction to Bayesian Networks, TNO-060-DTM-2013-01115, TNO, Delft, The Netherlands, 2013. Sections: 1, 2. Anca Hanea, Oswaldo Morales Napoles, Dan Ababei, Non-parametric Bayesian networks: Improving theory and reviewing applications, Reliability Engineering & System Safety, Volume 144, December 2015, Pages 265-284 MOOC material for Expert Judgment: https://online-learning.tudelft.nl/courses/decision-making-under-uncertainty-introduction-to-structured-expert-judgment/ Software: Treatment of Uncertainty: OpenTURNS, https://openturns.github.io/. Open source. Extreme value ProNEVA matlab based. Available for distribution. Discrete BNs. Netica & Hugin (commercial but a limited free version). Powerful graphical user interface Matlab & Python (open source) options less friendly graphically but easy to implement coding. Continuous BNs UniNet (commercial but a limited free version). Powerful graphical user interface BANSHEE. Matlab version available in SoftwareX (own development). Python (open source) version under development available by the end of 2021. Vine-copulas R-vine. Open source Expert Judgment ANDURIL matlab based. Available in SoftwareX (own development) ANDURYL python based. Available in SoftwareX (own development) Data Assimilation Python-based Jupyter scripts
Judgement	N/A
Permitted Materials during Exam	N/A
Collegerama	No

CEGM2006	Subsurface Storage for Energy, Water and Climate Applications	10
Responsible Instructor	Dr. H. Hajibeygi	
Module Manager	Dr. H. Hajibeygi	
Module Manager	Prof.dr.ir. M. Bakker	
Instructor	Dr. K.H.A.A. Wolf	
Instructor	Dr.ir. J.M. Bloemendal	
Instructor	Dr. B.M. van Breukelen	
Instructor	Dr.ir. A.A.M. Dieudonné	
Instructor	Prof.dr. S. Geiger	
Contact Hours / Week x/x/x/x	Diverse: lectures, computer practicals and lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The subsurface is increasingly used to store fluids associated with energy and for climate change mitigation. There are many potential technologies, and each has its place. In this module, the key technologies which are available and under-development are detailed, including key aspects of flow of fluids in the subsurface. These concepts are utilised to evaluate the technologies, such that we can efficiently use the subsurface. Students will move from broad concepts and how to assess the applicability (Learning Unit 1), through theoretical aspects of how the systems work (Learning Unit 2) to evaluation and design (Learning Unit 3).	
Study Goals	After completion of this module the students are able: CO-SS-LO1. Explain how earth underground resources can be used and sustained in an environmentally responsible manner for cyclic storage of fluids and energy (ILO: 4, 10). CO-SS-LO2. Design a conceptual framework for a given storage project, considering the geology and geomechanics of the host rock and physics of fluids under dynamics (ILO: 1, 3, 4, 8, 11). CO-SS-LO3. Evaluate the performance of storage systems, including the influence of their design parameters and nearby systems (ILO: 4, 5, 6, 7, 9). CO-SS-LO4. Evaluate geochemical processes in order to evaluate the environmental performance of systems (ILO: 4, 5, 6, 7, 9). CO-SS-LO5. Weigh the various advantages and disadvantages of subsurface storage options against each other and propose an optimal solution (ILO: 3, 6, 9, 10). CO-SS-LO6. Communicate their findings clearly in an appropriate comprehensive manner for various technical and non-technical stakeholders (ILO: 2).	
Education Method	Teaching and learning that encourages students to take responsibility for their learning process: Learning Unit 1: Subsurface storage technology - Lectures. Book chapters and handouts specifically relevant to each lecture should be studied beforehand. In addition to theory explanation, students will be engaged through exercises and quizzes. Learning Unit 2: Heat, solute and compressible fluid transport - Lectures on the transport of dissolved substances and heat in groundwater. Small scale experimental demonstrations are given during the lectures in which students participate. - Based on the scientific contents of the lectures, practical sessions will be performed to model the fundamental concepts behind successful storage of fluids in the subsurface across scales. These are mostly in the form of weekly computer modelling exercises using, for example, MODFLOW, MT3D, SEAWAT, DARTS, DARSim, or MRST. Python scripts will be used to generate input files, read output files, and analyse the results. - Group experiments in the lab will be performed to quantify physics of the stored fluids, e.g. H₂ and CO₂. Learning Unit 3: Storage design and evaluation project - Project based learning. Students will form multi-disciplinary groups. Each group will work on a specific energy scenario and conceptualise, design and evaluate a storage system based on their scenario. Groups will report findings on a weekly basis to other groups in form of oral presentations. These presentations will focus on the key messages to different stakeholders. These outputs will be formatively assessed. The final designs and evaluations will be communicated in writing (in form of report) and orally (via a presentation).	
Practical Guide	Learning Unit 1: Subsurface storage technology (1 ECTS) Link to learning objectives of the module: CO-SS: LO 1, LO 5 The various technologies related to subsurface storage and the evaluation of subsurface storage are detailed, including the current state of the art and future research directions. Theoretical and quantitative methods to gain insights into technical and environmental performance are taught. Evaluation methodologies, such as exergy analysis and multi-criteria analysis, are outlined. Topics: - Context of subsurface storage: - o global overview of Climate & Energy status - o from production to storage to mitigation. - Overview of the technologies and advancements - o Energy storage in porous media & cavities (reservoirs and salt caverns): - o Borehole thermal energy storage - o Aquifer thermal energy storage - o Hydrogen storage - o Compressed air energy storage - o Supercritical CO₂ energy storage - o Novel storage topics - o CO₂ storage - o Water storage (managed aquifer recharge (MAR)) - Assessment methods - o Exergy Analyses - o Multi-criteria analysis	

- o Environmental performance
- o Geomechanics

Learning Unit 2: Heat, solute and compressible fluid transport (6 ECTS)

Link to learning objectives of the module:

CO-SS: LO 2, LO 3, LO 4

The theoretical aspects of heat, solute and compressible flow in the subsurface are presented, including the flow of heat via conduction, advection and convection, the transport of dissolved solutes and the transport of compressible fluids (e.g. CO₂ and H₂). Reactivity and the effect on transport, leading towards changes in properties or pollution are considered. The coupled behaviour of processes is detailed, e.g. the effect of temperature on physical properties and rate constants.

Topics:

- Solute transport in groundwater
- Heat transport in the surface and groundwater
- Coupled groundwater flow and concentration/heat transport
- o density and viscosity dependency on salinity and temperature
- Reactive transport of contaminants
- o Effect of temperature on rate constants.
- o Mobilization, adsorption and resorption of trace elements due to flow/concentration/temperature
- Compressible fluid flow
- o CO₂
- o H₂
- o Laboratory experiment (geochemistry): measuring CO₂ dissolution, pH change
- Subsurface flow
- o Pore scale to km scale modelling
- Simulation methods
- o Assignments and exercises using various open-source simulators: e.g. MODFLOW, MT3D, SEAWAT, PHT3D and/or PHAST, DARTS, DARSim, MRST
- Applications include seawater intrusion, managed aquifer recharge, aquifer thermal energy storage, CO₂ storage, contaminant plumes.

Learning Unit 3: Storage design and evaluation project (3 ECTS)

Link to learning objectives of the module:

CO-SS: LO 2, LO 3, LO 4, LO 5, LO 6

Students are presented with an energy scenario where they must evaluate the most appropriate storage concept and present a detailed design. The design must include predictions of performance, an evaluation of environmental impact and include analysis of uncertainties. The students must demonstrate the ability to select appropriate theory, modelling concepts and data, perform numerical analysis and critically evaluate the results to form conclusions. They must clearly and precisely communicate their results to their peers.

Topics:

- Scenario evaluation
- Application of theory and tools from other two units
- Critical evaluation of results
- Presentation of project via written technical report and presentation for different stakeholders (focus on non-technical users in the final presentation)

Assessment

Formative assessment (feedback)

Learning Units 1 and 2

During the first two units weekly tasks have to be delivered. Tasks comprise written reports based on laboratory experiments, computer exercises and theoretical knowledge. In addition, class quizzes will be given.

The tasks will be formatively assessed on completeness and quality by involved lecturers. These build towards: (i) the portfolio, and (ii) the activities in Learning Unit 3.

Learning Unit 3

Groups will give weekly presentations on their progress and considering different stakeholders. The presentations will be formatively assessed on completeness and quality by involved lecturers, and peer feedback will be solicited. In addition, formative peer feedback within the group will encourage equal contribution.

Summative assessment (grading)

Portfolio of individual learning activities at the end of Learning Units 1&2

50% of final module grade.

Assessment based on:

Individual cumulative grade. Combination of (approximately) weekly assignments defined during lectures and workshops of Units 1 and 2. The students will have received formative comments after each assignment and can upgrade the assignments for the final grade.

Integrated module assessment at the end of Learning Unit 3

50% of final module grade.

Assessment based on:

Group report

Group presentation at the end of the project

Answering questions after the presentation

A 1-2 page(s) individual report on critical evaluation of all group studies (also to stimulate reading others group reports and actively ask questions during presentations and evaluate based on capacity, efficiency, safety, exergy, public perceptions,).

Both subgrades must be >= 5.8.

Expected prior Knowledge

For all MSc students CE, ENV, AES.

Academic Skills

Analysis of complex (multi-cause) problems, information collection and evaluation, multi-disciplinary and integral problem solving, project management, written and oral communication, teamwork.

Literature & Study Materials

Learning Unit 1: Subsurface storage technology

Books:

- o The city of the future by BNA (Branchevereniging Nederlandse Architectenbureaus), publisher Blauwdruk (2019); and Stad x Klimaat Het gebouw als watermachine by BNA and TU Delft (2020) (in Dutch). The study is interactive available at <https://www.bna.nl/kennis/bna-onderzoek/stad-x-klimaat> which can be automatically translated by Chrome Browser to English, for international students. These books comprise and review a number of practical cases concerning integral ecoengineering solutions for socio-environmental problems occurring in the built environment.
 - o Sustainable energy without the hot air by David Mackay: provides an overview of alternative energy options.
 - o Carbon Dioxide Capture and Storage, by Intergovernmental Panel on Climate Change (UNEP): this open-access report provides guidelines for climate-driven actions related to subsurface storage of greenhouse gases, especially CO₂.
- Specific literature in form of scientific papers and book chapters on diverse subjects of subsurface storage (energy and climate) related to techno-economical, experimental, modelling, computational and analytical techniques will be provided via

	BrightSpace. Experimental material for the CO₂/H₂ storage practical, as well as for the sensor workshop will be provided during the practical. Learning Unit 2: Heat, solute and compressible fluid transport Manuals of the open-source software and the associated packages for the modelling software used. Assignments will be developed in the form of Jupyter Notebooks so that they are fully reproducible and documented. Selected papers on the modeling of ATEs, MAR and storage of compressible fluids. Learning Unit 3: Storage design and evaluation project Resources from Learning Units 1 and 2. Additional specific literature in form of scientific papers and book chapters on multi-criteria analysis and scenario analysis will be provided via BrightSpace
Judgement	N/A
Permitted Materials during Exam	N/A
Collegerama	No

CEGM2007	Resilient Deltas	10
Responsible Instructor	Dr.ir. M.M. Rutten	
Module Manager	Dr.ir. M.M. Rutten	
Instructor	Dr. J.E.A. Storms	
Instructor	Dr. R.E.M. Riva	
Instructor	Dr. D.S. van Maren	
Instructor	Dr. A.P. Siebesma	
Instructor	Prof.dr. C.A. Katsman	
Contact Hours / Week x/x/x/x	approximately 7 hours/week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This topical cross-over deals with the societal challenge of making deltas resilient to climate change, subsidence, unprecedented economic growth and urbanisation. Future increases in sea level rise, droughts, extreme precipitation and flooding can have adverse effects on the physics and dynamics of deltas, salt intrusions and groundwater, which requires new delta interventions to be developed. The focus is on deltas globally, using the Dutch Delta as a nearby example and laboratory. The module is split in three themes: Theme 1 (3EC) - Physics of the coupled delta system under climate change and human interventions - provides students with the scientific understanding of the interaction of atmospheric, fluvial and marine processes that comprise the delta system, their response to human interventions, and their sensitivity to climate change (LO 1,2,3) Theme 2 (3EC) - Climate change and scenarios gives students hands-on experience on the use of statistical and dynamical downscaling techniques and long-term observational datasets, which they need to analyze to understand the impact of natural and man-made climate change on the delta system. Students will study and discuss different methods to construct climate scenarios, contrasting approaches developed in various countries (for example Switzerland, UK and the Netherlands) and learn how to apply them to different regions/ Focus will be on the specific climate variables: (extreme) precipitation/ droughts and sea level rise. (LO 1, 2, 4, 5) Theme 3 (4EC) - Adaptation pathways for deltas - guides students to develop pathways towards new delta technologies (flood defence and water treatment) based on climate model/data analysis using the Dynamic Adaptive Policy Pathway method among other analytical tools, and guided by SCRUM project management (LO 2, 4, 5, 6, 7, 8) Weekly Syllabus The module takes an inquiry-based approach, in which students start in week 1 by exploring a delta system of their choice and interest. They will create a conceptual map of their current understanding of that delta system including the interlinkages or feedbacks between processes (a concept map). They will do this as part of a multidisciplinary team. In week 1, one of the final deliverables of this module a proposal for an adaptation pathway for a hypothetical client/stakeholder - will be explained and discussed with the students. Student teams will serve as advisors for one delta and stakeholders for another delta. The stakeholders are responsible for climate adaptation of a delta, the advisors are hired to develop pathways. Together they will agree on a product that the advisors will deliver to the stakeholders. This product should at least contain an analysis of the local climate change effects on the main delta systems, and proposals for solutions and adaptation pathways. As examples, they might focus on flooding, salt intrusion of beach nourishments in their chosen delta system under climate change. During Weeks 28, the students will take an iterative approach towards their deliverable, enriching their conceptual understanding of the delta system in each iteration with knowledge and skills from Unit 1 & 2, which they put in practise in weekly computer labs. In the computer labs, students may develop python notebooks in which climate projected data obtained through dynamical downscaling with high resolution models are evaluated against observations. Alternatively, students can develop statistical downscaling techniques by combining global climate model data with local observations. They will also be trained in obtaining useful trends from long observational data sets of the present climate and try to attribute these to global warming. Teamwork in Unit 3 towards the deliverable will be supported with SCRUM project management. That product will be developed in three iterative cycles ("sprints"), ending in week 4, week 8 and week 10 (the final assessment).	
Study Goals	1. Explain the interaction of physical processes that take place in different components of the (coupled) delta system (ocean, rivers, land, atmosphere) 2. Explain the influence of natural variability, climate change and socio-economic development on the coupled delta system 3. Explain mechanisms and adaptation timescales associated with historic human interventions and implications for adapting to climate change 4. Construct and interpret climate change scenarios including the use of dynamical and statistical downscaling techniques based on climate model output. 5. Analyse observed and modelled trends in sea level rise, rainfall and flooding, droughts, sediment transport, subsidence and socio-economic development 6. Acquire new knowledge on green-gray solutions for mitigating impacts of human interventions and climate change, e.g. beach nourishment, river training, building with nature. 7. Develop pathways towards engineering solutions for future delta challenges using model scenarios, including uncertainties 8. Conduct a project in a multidisciplinary team	
Education Method	Unit 1: Physics of the coupled delta system under climate change and human interventions Series of lectures on fundamentals and case studies (3 hrs per week) Mini-lectures (1 hr per week) Quizzes / theory test in week 1/2 Unit 2: Climate change and scenarios Series of lectures (2 hrs per week) Computer labs (2 hrs per week) Quizzes / theory test in week 1/2 Unit 3: Adaptation pathways for deltas Discussions, supervised team work, and in-class paper reviews (2 hrs per week) Workshops in system thinking and stakeholder analysis Computer labs (2 hrs per week) Field trip to the Rhine Meuse Delta to be considered	
Literature and Study	Unit 1: Physics of the coupled delta system under climate change and human interventions	

Materials

Hoitink, A. J., Nittrouer, J. A., Passalacqua, P., Shaw, J. B., Langendoen, E. J., Huismans, Y., & van Maren, D. S. (2020). Resilience of river deltas in the Anthropocene. *Journal of Geophysical Research: Earth Surface*, 125(3), e2019JF005201.

Leuven, J. R., Pierik, H. J., Vegt, M. V. D., Bouma, T. J., & Kleinhans, M. G. (2019). Sea-level-rise-induced threats depend on the size of tide-influenced estuaries worldwide. *Nature climate change*, 9(12), 986-992.

Kirwan, M. L., Temmerman, S., Skeehean, E. E., Guntenspergen, G. R., & Fagherazzi, S. (2016). Overestimation of marsh vulnerability to sea level rise. *Nature Climate Change*, 6(3), 253-260.

Van Maren, D. S., Oost, A. P., Wang, Z. B., & Vos, P. C. (2016). The effect of land reclamations and sediment extraction on the suspended sediment concentration in the Ems Estuary. *Marine Geology*, 376, 147-157.

van Maren, D.S., J.G.W. Beemster, Z.B. Wang, Z.H. Khan, R.A Schrijvershof, A.J.F. Hoitink. Tidal amplification and river capture in response to land reclamation in the Ganges-Brahmaputra delta. *Catena*, 2022

Syvitski, J. P., Kettner, A. J., Overeem, I., Hutton, E. W., Hannon, M. T., Brakenridge, G. R., & Nicholls, R. J. (2009). Sinking deltas due to human activities. *Nature Geoscience*, 2(10), 681-686.

Syvitski, J. P., Vorosmarty, C. J., Kettner, A. J., & Green, P. (2005). Impact of humans on the flux of terrestrial sediment to the global coastal ocean. *science*, 308(5720), 376-380.

Talke, S.A. and Jay, D.A. Changing Tides: The Role of Natural and Anthropogenic Factors. *Annu. Rev.Mar. Sci.* 2020. 12:12151

Wang, Z. B., Elias, E. P., van der Spek, A. J., & Lodder, Q. J. (2018). Sediment budget and morphological development of the Dutch Wadden Sea: impact of accelerated sea-level rise and subsidence until 2100. *Netherlands Journal of Geosciences*, 97(3), 183-214.

Winterwerp JC, Wang ZB, van Braeckel A, van Holland G, Kösters F. Man-induced regime shifts in small estuariesII: a comparison of rivers. *Ocean Dyn.* 63:1293306, 2013.

Unit 2: Climate change and scenarios

van den Hurk, B. et al. Drivers of mean climate change around the Netherlands derived from CMIP5. *Clim. Dynam.* 42, 16831697 (2013).

Swiss Climate Change Scenarios CH2011 (C2SM, MeteoSwiss, ETH, NCCR Climate, and OcCC, 2011).

van den Hurk, B. et al. New climate change scenarios for the Netherlands. *Wat. Sci. Technol.* 56, 2733 (2007).

Jenkins, G. J. et al. UK Climate Projections: Briefing Report (Met Office Hadley Centre, 2009).

Frederikse, T., et al. (2020). The causes of sea-level rise since 1900. *Nature*, 584(7821), 393-397. <https://doi.org/10.1038/s41586-020-2591-3>

Tamisiea, M.E., and J.X. Mitrovica (2011). The moving boundaries of sea level change: Understanding the origins of geographic variability. *Oceanography* 24(2):2439, <https://doi.org/10.5670/oceanog.2011.25>

Slangen, A. B. A., Katsman, C. A., Van de Wal, R. S. W., Vermeersen, L. L. A., & Riva, R. E. M. (2012). Towards regional projections of twenty-first century sea-level change based on IPCC SRES scenarios. *Climate dynamics*, 38(5), 1191-1209. <https://doi.org/10.1007/s00382-011-1057-6>

Slangen, A. B. A., Carson, M., Katsman, C. A., Van de Wal, R. S. W., Köhl, A., Vermeersen, L. L. A., & Stammer, D. (2014). Projecting twenty-first century regional sea-level changes. *Climatic Change*, 124(1), 317-332. <https://link.springer.com/article/10.1007/s10584-014-1080-9>

IPCC 5AR Ch 13.1/13.6 [to be replaced by appropriate 6AR chapter this summer]

Katsman, C. A., Sterl, A., Beersma, J. J., Van den Brink, H. W., Church, J. A., Hazeleger, W., ... & Weisse, R. (2011). Exploring high-end scenarios for local sea level rise to develop flood protection strategies for a low-lying delta the Netherlands as an example. *Climatic change*, 109(3), 617-645. <https://link.springer.com/article/10.1007/s10584-011-0037-5>

Practical Guide

Week/Unit

- Physics of Delta System / Climate
- Climate Change / Scenarios
- Adaptation pathways

1

- Deltas in the Anthropocene. An introduction to the range of systems and their response impacted by climate change and human interventions
- An introduction to climate science: climate sensitivity
- Diverse team formation, SCRUM Introduction.
 - Exploration and discussion of generic adaptation pathways and deliverables

2

- Sediment supply from rivers and from coastal seas; sediment type; changes in sediment supply (role of climate, land use & reservoirs)
- Our future climate - Representative Concentration Pathway (RCP) scenarios, climate modelling and downscaling techniques
- Selection of deltas and roles (advisor / stakeholder)
 - Systems thinking workshop & stakeholder Analysis workshop

3

- Tidal amplification response to human interventions (closure dams, channel deepening) b) Sea level rise: causes, processes + observations (altimetry / tide gauges Dutch coast)
- Discussion product deliverables and planning Sprint 1
 - Presentations Conceptual Model and Stakeholder Analysis

4

- Changes in salt intrusion and fresh water availability in response to channel deepening, reservoir construction, and channel reorganization
- Uncertainty in climate model and scenarios, climate change versus natural variability, and attribution of climate change
- Resilience debates

5

- Decline of water quality and ecological value in response to climate change and human interventions
- Atmospheric influences on deltas: Storms, surges, precipitation and river discharge
- Review and Retrospective Sprint 1 and Planning Sprint 2

6

- Subsidence and compaction of deltas
- Atmospheric influences on deltas: Projections of Extremes : Flooding and Droughts
- Discussion on evaluation of strategies

7

- Coastal erosion by natural processes, human interventions, and climate change
- Sea level rise: global and regional projections. High-end scenarios with skewed ice sheet contributions, links to extreme events / coastal safety.

8

- Morphologic response to sea level rise
- Compound effects of sea level rise, storm surges + precipitations / river discharge c) Review and Retrospective Sprint 2

	10 a) Adaptation to sea level rise c) Presentations of adaptation pathways to client/stakeholder, supported by data products consisting of e.g. downscaled climate model output for different scenarios, observational data analyses.
Assessment	Formative assessment (feedback): The formative assessment for Theme 1 and 2 is arranged in the form of quizzes and assignments with feedback. For Theme 3 the conceptual model and stakeholder analysis are formative assessed in week 3. In week 5 and 8 the SCRUM artefacts Review and Retrospective are used to give students feedback on content and process. Summative assessment: The summative assessment consists of: Written exam on content Unit 1 - 3 (50%) Proposal for an adaptation pathway / solution for a delta including analysis of the local climate change effects on the delta system, delivered in either a written/oral/(other form) presentation (50%)
Expected prior Knowledge	Students that enter the module are expected to have in-depth understanding of one of the subsystems of deltas (e.g. coastal area, rivers, the subsurface, the atmosphere etc.). All Master tracks within the programs Civil Engineering, Environmental Engineering and Applied Earth Sciences can participate. Students from other programs need to contact the module manager.
Academic Skills	The professional and personal skills are included in the learning objectives. Specifically students work on their metacognitive skills (learning objective 6) and their collaboration, management and communication skills (learning objective 8)
Literature & Study Materials	(Continuation) Unit 3: Adaptation pathways for deltas For pathway systems thinking and rapid modelling the following material (or comparable resources) will be used together will dedicated material on Brightspace: Haasnoot, M., Brown, S., Scussolini, P., Jimenez, J. A., Vafeidis, A. T., & Nicholls, R. J. (2019). Generic adaptation pathways for coastal archetypes under uncertain sea-level rise. <i>Environmental Research Communications</i>, 1(7), 071006. Wright, D., & Meadows, D. H. (2008). <i>Thinking in systems</i>. Earthscan. https://ocw.tudelft.nl/courses/ngi-2/subjects/8-bonus-module-problem-demarcation-actor-analysis-skills-2/, (accessed 2-4-2021) Goldreich, P., Mahajan, S., & Phinney, S. (1999). Order-of-magnitude physics: Understanding the world with dimensional analysis, educated guesswork, and white lies. University of Cambridge. Relevant part available via https://web.stanford.edu/class/ee353/SanjayMahajanIntro-01-1.pdf (accessed 2-4-2021) Van Berchum, E. C., van Ledden, M., Timmermans, J. S., Kwakkel, J. H., & Jonkman, S. N. (2020). Rapid flood risk screening model for compound flood events in Beira, Mozambique. <i>Natural Hazards and Earth System Sciences</i>, 20(10), 2633-2646. Jafino, B. A., Kwakkel, J. H., & Taebi, B. (2021). Enabling assessment of distributive justice through models for climate change planning: A review of recent advances and a research agenda. <i>Wiley Interdisciplinary Reviews: Climate Change</i>, e721. For teamwork and project process the following material (or comparable resources) will be used together will dedicated material on Brightspace: · Schwartz, S. H. (2014). National culture as value orientations: Consequences of value differences and cultural distance. In <i>Handbook of the Economics of Art and Culture</i> (Vol. 2, pp. 547-586). Elsevier. · https://www.mbtionline.com/ · https://scrumguides.org/ (All accessed 2-4-2021)
Judgement	N/A
Permitted Materials during Exam	N/A
Collegerama	No

CEGM2008	Monitoring of Structural Health and Geohazards	10
Responsible Instructor	Dr. E. Lourens	
Module Manager	Dr. E. Lourens	
Instructor	Ir. C. Kasbergen	
Instructor	Dr.ir. C.C.J.M. Tiberius	
Instructor	Dr.ir. M.J.M.M. Steenbergen	
Instructor	Dr.ir. D.S. Draganov	
Instructor	Ir. F.J. van Leijen	
Instructor	Dr. Y. Yang	
Instructor	Dr. A.A. Nunez Vicencio	
Instructor	Dr. A. Barnhoorn	
Instructor	Dr. K. Anupam	
Instructor	C. Shen	
Instructor	Dr. H. Wang	
Contact Hours / Week x/x/x/x	12-14 hours per week (3-4 lectures, 3-4 workshops of 2*45 min)	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The cross-over will be of the methodological type, oriented around processes and techniques that are shared across multiple domains. Specifically, students from CE and AES will apply predominantly new methods related to the monitoring of civil structures, rock- and soil masses to problems originating in their own as well as other fields of study. The module consists of two units: 1. Theory & Techniques 6 EC 2. Case Studies: Advanced monitoring of structural health and geohazards 4 EC Unit 1: Theory & Techniques [LO1, LO2, LO3, LO4, LO5] The first unit provides the students with knowledge on the theory and techniques required to answer the questions what, when, how, why and where to monitor?. Parts of this unit will form natural extensions to what is taught in the MUDE, including brief recapitulations. Weekly syllabus Week 1: Introduction Degradation & failure mechanisms Relation loading &#8596; materials &#8596; mechanisms &#8596; time- & spatial scale Sensors and data acquisition Advanced signal processing (beyond MUDE) Week 2: Inversion Inversion and quality assessment (including recap MUDE) Prediction (Best Linear Unbiased Prediction, Kriging, Gaussian process Regression) Inversion for sub-surface (layered) structures Week 3: System identification and model updating System identification: EMA/OMA Deterministic (FE) model updating Bayesian inference in the context of structural health and geohazards monitoring (Bayesian inversion) Week 4: Machine learning perspectives Feature engineering Introduction to image segmentation, deep learning and convolutional neural networks (including recap MUDE) Anomaly detection: clustering techniques extended from MUDE, hypothesis testing (time series analysis) Week 5: Exam Written exam (30%) Unit 2: Case studies: Advanced monitoring of structural health and geohazards [LO3, LO4, LO5, LO6] Case studies covering a wide range of application domains will be presented, where a division is made between CE applications (bridge, railway, pavement, and wind turbine monitoring, stability of dykes and embankments), AES applications (soil mass movements and landslides, land subsidence, failure & fault reactivation due to gas extraction, short-term cyclic storage of fluids or resource mining), and applications relating to both fields of study (imaging of structures, rocks and the near surface, monitoring of acoustic emission and seismic activities). Each case study will be developed & supervised by the relevant, contributing research group. Students will work in groups of 2 on these case studies, 1 from CE and 1 from AES. Each case study comprises 2 ECTS. Insofar possible, each group will have representatives from both MSc programmes. The last 4 weeks of the module are reserved solely for the case studies. Weekly syllabus Weeks 6-10: Introduction of the case studies If required: theory specific to the case study & monitoring approach used State-of-the-art review per case study	
Study Goals	[LO1] Identify mechanisms of failure and functionality loss for applications in the geo- and civil engineering domain. [LO2] Discern techniques to inspect and measure the health or integrity of civil structures, rocks, and soil masses. [LO3] Design monitoring systems to detect patterns and anomalies in civil structures, rocks and soil masses and assess the value of information obtained with these systems. [LO4] Post-process monitoring data to extract structural health information. [LO5] Apply numerical and statistical inverse analysis to identify unknowns and update numerical and/or analytical models based on the monitoring data.	

Education Method	[LO6] Integrate monitoring data and physical modelling to support decision-making. Unit 1: The content in unit 1 will be delivered through theoretical teaching sessions, flipped classrooms, live demonstrations in the classroom/laboratory, and computer workshops. Unit 2: For the case studies, the students will be split into sets of groups, where the aim is to have multiple groups working on the same case study, and directly supervised by the relevant research group. For each case study, there will first be theoretical teaching sessions to introduce the case study and point the students towards literature/articles relevant to the case study. This will be followed by regular, supervised practical sessions (per case study), where the groups can work on their assignment and ask questions.
Assessment	Formative assessment (feedback): Unit 1: Small group assignments related to the computer workshops. Unit 2: Short presentations to expert panels, partly peer-graded. During the presentations, societal aspects related to assessment decision-making should also be addressed. Summative assessment: 1. Written exam in week 5 (after Unit 1), before students start the case studies (30%) 2. Assignment portfolio based on group projects (70%) - Case study 1 (35%) - Case study 2 (35%)
Special Information	Case study example 1 [CE]: Structural health monitoring of bridges Students receive a data set containing accelerations and strains measured on a bridge. They are asked to 1) perform operational modal analysis to identify the dynamic properties of the structure, 2) update a (given) simplified model of the bridge using the identified properties, 3) design features that can be used to monitor the global vibrational response of the bridge, 4) implement an algorithm for detecting anomalies in the response, and 5) suggest improvements to the monitoring system (locations/types of sensors, data acquisition). The last task is included for all case studies where it is not practically feasible for students to collect the data themselves. Case study example 2 [AES]: Monitoring of changes in subsurface structures Students receive seismic reflection data, a subsurface fault model, and seismicity data from a subsurface reservoir (e.g. from Groningen) and are asked to 1) perform an assessment of the seismicity locations, 2) perform fault slip processes modelling, 3) assess what the role of subsurface properties has on the outcomes of these analyses, including the role of possible changes in production during continuous production, and 4) devise an improved monitoring plan to improve the quantification of subsurface seismicity properties.
Expected prior Knowledge	N/A
Connection to other modules	Overlap with other parts of the MSc programmes: CEGM1000: Data acquisition, signal processing, Bayesian inference & time series analysis is taught in the MUDE as well. For all three these topics, the theory will be briefly refreshed, and contextualized for monitoring applications. CIEM5110 Structural Mechanics and Dynamics: Also contains content on data acquisition, signal processing, and experimental modal analysis. AESB403B Induced Seismicity: Possible overlap with case studies (Groningen data may be used in this B module as well. The focus is very different, however. AESB404B Raw Material Exploration and Data Acquisition Techniques: Contains one lecture on the seismic exploration method with active and passive sources.
Academic Skills	Project management, teamwork, written and oral communication, information collection and evaluation.
Literature & Study Materials	The complete list for unit 1 still has to be compiled. For the case studies, relevant readings for the specific case study will be collected.
Judgement	N/A
Permitted Materials during Exam	N/A
Collegerama	No

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Environmental Engineering

ENVE Master Thesis Preparation and Portfolio

ENVM4000	Master Thesis Preparation and Portfolio Environmental Engineering	5
Responsible Instructor	Dr.ir. M.W. Ertsen	
Instructor	Dr.ir. M.M. Rutten	
Contact Hours / Week x/x/x/x	4 hours/week. Remark on scheduling: It is possible to complete the Module in each quarter. The actual teaching format will be adapted to the practicalities of the situation as is (number of students, scheduling options, and others). The nominal schedule of the module is Quarter 2 of year 2. The module is also planned to have scheduled activities in Quarter 3 of year 2. For other quarters, scheduling teaching activities is not foreseen yet individual plans will be made.	
Education Period	2 3	
Start Education	2 3	
Exam Period	2 3	
Course Language	English	
Required for	Students from the program ENVM are required to complete this module.	
Course Contents	This module prepares the final MSc thesis project, building on two products that need to be completed. Portfolio Each student compiles a portfolio to show her academic and professional development (efforts, progress, and achievements in becoming an EnvEng professional). This portfolio process starts in the Program Module ENVM1000. Students are guided to build the portfolio and determine progress in reaching the program Intended Learning Objectives. Part of the Portfolio is an assessment of students expertise on ethical issues. The portfolio is completed with one specific activity of a student to master specific, well-defined research skills specifically useful for the individual MSc thesis (some activities (e.g. types of analysis, methods and/or models) are offered, but students can also propose other options. Thesis proposal A thesis research proposal is developed that includes the scientific, professional and/or societal relevance of the research together with a feasible methodology and planning. After topic and supervisor are known, the student plans research activities that allow to execute the thesis research within 30 credits. Budgeting of the research (in terms of time and/or money) is part of this process. Weekly sessions support students developing their proposals (content and structure of research proposals; formulating research questions and hypothesis). A critical review of relevant publications, and a plan how to continue using scientific texts in the thesis project, is part of (writing) a research proposal.	
Study Goals	On completion of this module, the student will be able to: ENVM-TP.1. Identify relevant environmental problems and translate these problems into a scientific research scope. ENVM-TP.2. Critically assess existing knowledge and existing theories to specify scientific research questions. ENVM-TP.3. Consider relevant ethical issues related to the research topic and/or methodology. ENVM-TP.4. Design a research approach, consisting of activities/procedures to collect and analyse data, to answer the main research question. ENVM-TP.5. Draft a work plan, specifying order of and links between research activities over time, to execute the research approach. ENVM-TP.6. Communicate the research plan effectively in written and oral form. ENVM-TP.7. Reflect on her personal development (academically and professionally) as an environmental engineer. ENVM-TP.8. Communicate options to further her personal development in the scope of life-long learning.	
Education Method	Tentative overview of activities, based on a schedule of weekly sessions (4 hrs) plus homework (4 hrs), leaving about 24 hours of writing time for the final proposal. Week 1. Introduction. What is research, which types can be distinguished. The what, how, when and why of a proposal. Week 2. Research question. Working on the main question in relation to the activities that need to be developed to answer the question. Week 3. Literature Review 1. How to find and assess literature; First reading of a paper; Relevance of literature for own thesis project. Week 4. Methods and Time. Linking activities to time, and to sub-questions. Check on data requirements. Week 5. Literature Review 2. Assessing the relevance of the literature that has been found. Check on data source availability. Week 6. Methods and Questions Check on activities, time, and to sub-questions. Determine data needs. Week 7. Literature Review 3. Check on literature and relevance. Determine data availability and immediate need. Week 8. Final draft proposal. Sharing the draft, presenting the draft. Week 9. Improving draft proposal. Self-study. Week 10. Assessment of Thesis Proposal. Meeting with thesis supervisor, using comments from module manager.	
Prerequisites	It is possible to complete the Module in each quarter. The actual teaching format will be adapted to the practicalities of the situation as is (number of students, scheduling options, and others). A student may start the thesis preparation (ENVM4000) if the student has successfully completed all modules of the first year of the Master programme, as well as all courses of the first quarter of the second year of the Master (as far as assessment results are known), within the minimum amount of study time possible and in accordance with their individual study planning. In case these conditions are not met, the student must have at least 75 EC in accordance with their individual study planning.	
Assessment	Assessment (Formative) Portfolio To be (partially) in groups, as this allows comparing portfolios of students, both by the student(s) and coaches. These group sessions are based on what students (want to) share, as access to the portfolios themselves will be restricted. Thesis proposal The draft research proposal, including research questions, hypothesis and research approach will be regularly assessed during the module sessions by (peer) review and regular presentations and discussions with the peer group and thesis committee. Assessment (summative) Portfolio (2 EC) As the portfolio is a direct bridge between the program and the thesis, the self-assessment element of the portfolio is finalized at the start of the thesis preparation. Students complete one skill-related sub-activity in the module. Thesis proposal (3 EC) The research proposal is accepted as a useful start of the thesis project by the thesis supervisor and module manager. Both assessment will be assessed with a rubric that is based on the study goals. The assessment of the Portfolio is a pass when a student has a satisfactory self-assessment. The research proposal is a pass when it is considered as a useful start (by the thesis supervisor) and is in line with what has been taught in the module (by the module manager).	
Expected prior Knowledge	Next to the official admission requirements, it is expected that students have completed a set of modules that links to their thesis topic. Students need to have identified a topic and main supervisor before the start of the module.	

Academic Skills	Critical thinking, problem-based learning and solving, information collection and synthesis, communication, scientific writing, presentation, time management, application of theory to new situations, drafting research questions, reflection on ethics and learning skills.
Literature & Study Materials	TBD, depending on exact program
Judgement	See assessment
Permitted Materials during Exam	N/A
Collegerama	No

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Environmental Engineering

ENVE Master Thesis

ENVM4010	Master Thesis Environmental Engineering	30
Responsible Instructor	Dr.ir. A.M.J. Coenders	
Module Manager	Dr.ir. A.M.J. Coenders	
Contact Hours / Week x/x/x/x	N/A	
Education Period	1 2 3 4	
Start Education	1 2 3 4 5	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	Research/design work, finalised with: - Report (thesis) and internal defence during green light meeting - Final presentation and public defence	
Study Goals	After completion of this module the students are able to: MT-LO1: Relate their thesis work to state-of-the-art literature. (ILO: 1, 2) MT-LO2: Integrate knowledge gained during studies. (ILO: 1, 2) MT-LO3: Apply experimental/computer/design skills acquired during the program when developing a sound scientific argumentation. (ILO: 5, 6) MT-LO4: Contribute creative, innovative ideas to the project. (ILO: 3) MT-LO5: Reflect in a scientific way about their own results. (ILO: 8) MT-LO6: Take initiatives in their research project. (ILO: 10) MT-LO7: Work independently. (ILO: 10) MT-LO8: Plan and work according to an acceptable time schedule, which is adapted if required (related to 30 EC). (ILO: 10) MT-LO9: Write a structured and consistent scientific report. (ILO: 9) MT-LO10: Present orally and defend their work. (ILO: 9)	
Education Method	Guided project of independent research. The thesis project starts with a research proposal as drafted in ENVM4000 (Thesis Preparation). Students will be guided by a supervisory committee, with at least four meetings: thesis kick-off, mid-term, green light, and presentation. - Kick-off: 3 weeks in the thesis project, the committee and student discuss whether the original research design still holds - Mid-term: after ±50% of work, the committee and student discuss progress and confirm/define the required actions to finalize the thesis project as planned - Green light: 3 weeks before presentation/graduation, based on a thesis text that is 95% complete, the committee assesses whether the student masters the topic (internal defence). Final activities to complete the thesis document are defined. - Presentation: The student presents the results to the committee and a general audience in a public defence. Regular progress meetings to assess the students progress and discuss next steps will be held between student and main thesis supervisor (if applicable with the committee). To allow assessing progress in relation to peers, regular meetings for all thesis students in the cohort are organized.	
Prerequisites	A student can only start with the thesis when ENVM4000 Thesis preparation has been completed.	
Assessment	Formative: Students stay in close contact with their TU Delft supervisor / supervisory committee throughout the graduation project. They will receive formative feedback during the progress meetings and the regular supervision meetings. The role of the daily supervisor (who is also part of the supervisory committee) is more prominent: the dedicated supervisor will provide feedback in all meetings, both with the committee and with the student (and additionally by mail/phone). Summative: During the green-light meeting, the supervisory committee decides whether the student can continue preparing for the final defence (and set a date) or whether the student needs to make significant improvements and an extra green-light meeting is required. In the green-light meeting, the student presents the project (for 15-20 minutes), which is discussed in relation to the report. The student will get final feedback on the draft thesis. A draft thesis is defined as a complete report with all components finalized and the report meeting the standard requirements (e.g., referencing and other technicalities). A decision allowing the student to prepare for the final defence can include suggested minor changes to this draft thesis, which can be completed well before the public defence. Based upon the internal defence during the green light, the final thesis, the behavioural competencies, and the final presentation with public defence, the grade will be determined by the members of the supervisory committee as a weighted average of the following criteria: A. Scientific approach (25%) B. Quality of result/product (25%) C. Behavioural Competencies (20%) D. Quality of written presentation (15%) E. Quality of oral presentation and defence (15%) The final MSc thesis will be assessed with a grade between 1 and 10. The chair of the graduation committee is responsible for the final grade.	
Expected prior Knowledge	The thesis is the final course of the program Environmental Engineering.	
Academic Skills	Critical thinking, problem-based learning and solving, information collection and synthesis, communication, scientific writing, presentation, time management, application of theory to new situations, drafting research questions, reflection on ethics and learning skills.	
Literature & Study Materials	Brightspace MSc Environmental Engineering Graduation Project CEG Student portal Graduating CEG Student portal Forms Master TER Annex, Articles 24-25. Rules and Guidelines of the Board of Examiners (RGBE)	
Judgement	See assessment.	
Permitted Materials during Exam	N/A	
Collegerama	No	

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